

Application technology of fungicides for the management of dry root rot of chickpea



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CERTIFICATE – I

This is to certify that the thesis entitled “**Application technology of fungicides for the management of dry root rot of chickpea**” Submitted in partial fulfillment of the requirements for the degree of **MASTER OF SCIENCE** in **Plant Pathology** of Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya, Gwalior is a record of the bonafide research work carried out by **Mr Gourav Chouhan ID No 16131801** under my guidance and supervision. The subject of the thesis has been approved by the Student’s Advisory Committee and the Director of Instruction.

No part of the thesis has been submitted for any other degree or diploma or has been published. All the assistance and help received during the course of this investigation has been acknowledged by the scholar.

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CERTIFICATE – II

This is to certify that the thesis entitled “**Application technology of fungicides for the management of dry root rot of chickpea**” Submitted by **Mr Gourav Chouhan ID No 16131801** to the Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya, Gwalior in partial fulfillment of the requirements for the degree of Master of Science in **AGRICULTURE** in the Department of **Plant Pathology** has been accepted after evaluation by the External Examiner and approved by the Student's Advisory Committee after an oral examination of the same.

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CHAPTER- I

INTRODUCTION

Chickpea (*Cicer arietinum* L.) is an internationally important pulse crop, ranking third after dry beans. In India, it is grown on 9.63 million ha area with a production of 9.38 million tonnes and productivity of 974 kg/ha, whereas in Madhya Pradesh it is grown on 3.22 million ha and contributes about 3.54 million tones with an average productivity of 1100 kg /ha (Anonymous, 2016-17).

Chickpea is rich source of protein (21%), carbohydrate (61.51%), fat (4.5%), vitamin-C and dietary proteins. The nitrogen fixing nodules via interaction with Rhizobia adds to its uniqueness (Ferguson *et al.*, 2010). Soil borne pathogens like *Sclerotium rolfsii* (collar rot), *Fusarium oxysporum* f. sp. *ciceri* (vascular wilt) and *Rhizoctonia bataticola* (dry root rot) are responsible for causing diseases from seedling to flowering and pod formation stages.

Among these, dry root rot (DRR) caused by *Rhizoctonia bataticola* (Taub) Butler [*Macrophomina phaseolina* (Tassi) Goid] is one of the major production constraints that causes 10-20 % annual loss (Vishwadhar and Chaudhary, 2001).

The DRR has been categorized as an acute-emerging disease that occurs irregularly, both temporally and spatially and may cause massive disruptions in system performances and whose range is expanding to new areas (Savary *et al.* , 2011).

Looking at present expanding area in South Zone of India, cropping system and the changing climate scenario, dry root rot of chickpea is becoming more severe in central and southern parts of India, where the crop is mostly grown under rainfed condition as soybean-chickpea or rice-chickpea cropping system. Survey reports indicated that dry root rot had appeared intermittently at higher levels in major chickpea growing areas in India causing substantial yield losses in farmer's field (Ghosh *et al.*, 2013).

R. bataticola is a soil inhabiting organism capable of infecting chickpea at any crop stage, however, It infects chickpea most commonly at pod filling stage when the temperature exceeds > 30⁰C (Sharma and Pande, 2013).

Till date, recommendations have been made for the seed treatment by various workers but due to it's attack at pre-podding or podding stage of the crop, the effect of seed treatment is reduced.

Hence, in the present investigations, efforts were made on application of fungicides after post emergence stage of crop, targeting appearance of dry root rot. The research work was carried out with the following objectives:

Objectives:

1. Surveys of dry root rot disease in and around Sehore.
2. Isolation of causal organism and it's pathogenicity test.
3. *In vitro* evaluation of fungicides against *Rhizoctonia bataticola*.
4. Effect of application method of fungicides on dry root rot incidence after seed treatment and drenching at flowering stage.
5. Effect of fungicides on growth characters of chickpea.

CHAPTER -II

REVIEW OF LITERATURE

“Studies on application technology of fungicides for the management of dry root rot of chickpea” were undertaken to explore the possibility of management of the disease by adopting the different methods of application of fungicides. Information on these aspects is being reviewed in this chapter.

2.1 Historical

The first mention of a plant disease, which may be referred as *Rhizoctonia*, as a causal agent is an important paper by Du Hamel (1728) read before the Paris Academy. In this paper he gave a careful description of a fungus disease of saffron (*Crocus sativus*) occurring in France and named the fungus *Tuberorides*.

De Candolle (1815) discovered a similar disease on alfalfa. In his taxonomic description he was first person who gave the name for the fungus. Later, after careful study of the characteristics and parasitism of the species, he established the genus *Rhizoctonia* in 1815.

In 1924, Small described a *Rhizoctonia*, which develop black sclerotia of variable size and shape in the diseased roots, and many woody plants from Uganda and named it *Rhizoctonia lamrillifera*. Butler (1925), however considered that the fungus should be regarded as *Rhizoctonia* and made the combination *Rhizoctonia bataticola* (Taub.) Butler to a *Sclerotium* causing a root rot of cotton, french beans and cowpeas.

Dry root rot (DRR) in chickpea was first reported from India by Mitra (1931); later, the disease has been reported from most chickpea-growing areas in India and other countries like Iran (Kaiser et al. 1968), the USA (Westerlund *et al.*, 1974) and several countries in Asia and Africa (Nene et al. 1996). The disease was earlier known as “*Rhizoctonia* wilt” in chickpea; however, later it was named as “dry root rot”. The DRR was not of much significance in chickpea earlier; however, it has become a major threat to chickpea production in recent years due to altered weather conditions, particularly on the account of longer drought spells. Higher temperature and soil moisture depletion during crop growth period particularly at post-harvesting stage are predisposing chickpea to DRR (Sharma and Pande 2013).

Disease symptomatology

Rhizoctonia bataticola is a soil borne fungus/pathogen. Several workers have reported various types of symptoms produced on plants with varying intensity of the

disease in different crops and different regions. Narsimhan (1929) observed that plants infected with *Rhizoctonia bataticola* initially showed drooping of leaves followed by wilting. The main roots of such plants show blackening of finer roots.

Singh (1999) reported that dry root rot of chickpea occurred from flowering to podding stage. Infected plants were suddenly dried in the field. The tap roots turned dark brittle and devoid of lateral roots. Khalid and Ilyas (2000) considered the presence of root lesions and sclerotium plugging in xylem vessels of roots and collar region as symptoms of dry root rot of chickpea for screening of germplasms against this disease. Singh and Agarwal (2002) observed the withering and drying of chickpea plants in the field due to the infestation of *R. bataticola*.

The tap root turns black with sign of rotting and is devoid of most of the lateral and tiner roots. The dead root are quit brittle and show shredding of the bark. The tip of the root is easily broken leaving the lower portion of the tap root in the soil, when plants are uprooted. Dark minute sclerotial bodies can be seen on the roots exposed and inner side of the bark or when split open at the collar region vertically (Sharma *et al.*, 2015).

The seedling stage may be affected by the disease but the old plants are more susceptible towards infection (Sharma and Pande, 2013). The most favorable temperature for incidence of disease is found to be 30 °C. The affected plants dry suddenly with straw coloured leaves, while in few cases the stem and lower leaves show a characteristic brown colour. Drooping of leaflets and petioles are confined at the top of the plant. In some cases, when the rest of the plant gets dried, the topmost leaves become chlorotic.

Pathogen

Rhizoctonia is a genus of anamorphic fungi in the order Cantharellales and family Ceratobasidiaceae. *Rhizoctonia* species do not produce spores, but are composed of hyphae and sclerotia (hyphal propagules), asexual stage of fungi. *Rhizoctonia* species are saprophytic, but some act as facultative plant pathogens causing commercially important crop diseases. The genus *Rhizoctonia* (means “root killer”) was described in 1815 by the French mycologist Augustin Pyramus de Candolle for plant pathogenic fungi that produce both hyphae and sclerotia. Subsequent authors added over 100 additional names to the genus. A comprehensive survey and predisposition of species name “*Rhizoctonia*” was published in 1994 by Anderson and Stalpus. According to this published information, *R. bataticola* is used as a synonym to *M. phaseolina* (Tassi) Goid 1947. Currently, *M.*

phaseolina is officially recognized as the correct taxonomic name (CMI description of pathogenic fungi and bacteria No.275) with the sclerotial phase known as *R.bataticola* (Holliday and Punithalingam, 1970). Due to the presence of only sclerotial phase in chickpea, the pathogen is referred as *R. bataticola* (Taub.) Butler in this article.

Haigh (1930) suggested that *R. bataticola* be used for sclerotial stage and pycnidial strains should be called *M. phaseolina*. Goidanich (1947) examined the original material of Tassi and compared it with *M. phaseoli*, *M. corchori*, *M. cajani*, *M. sesami*, *M. philippinensis*, *Dothiorella cajani* and *D. phaseoli*. He confirmed all of them were identical. He corrected the mistake made by Ashby. According to the International Code of Botanical Nomenclature, the binomial *M. phaseolina* was the valid name for the pycnidial stage of *R. bataticola*. *M. phaseolina* is a soil born pathogen, first reported on chickpea from India (Mitra, 1931).

Under *in vitro* conditions, the fungus grows rapidly on potato dextrose agar (PDA) and produces brown to grey coloured mycelium that become darker with age. The young hyphae are thin, hyaline, aseptate and dichotomously branched and later produce typical black sclerotia. The fungus may produce abundant aerial mycelium, as high as to touch the cover of the culture plates, or mycelium is found to be completely or partially suppressed. The characteristic features of *R. bataticola* are right angle branching of the mycelium and constriction of the branch near the point of origin. The sclerotia formed are black, smooth, varying from spherical through oblong to irregular shapes (Sharma *et al.*, 2012).

Survey of dry root rot of chickpea

The first report of occurrence of root rot in chickpea along with wilt was made by Padwick (1948). It was the worst disease of chickpea wilt complex nature *Fusarium*. *Operculellia padwickii* and other non parasitic causes were later reported with this disease. Kaiser *et al.* (1968) observed *M.phaseolina* associated in periodic surveys on wilt and rot root incidence of gram.

Gupta *et al.* (1983) reported incidence of root rot ranging 3.58 to 20.63 per cent in 30 villages of Northern Madhya Pradesh. Reddy *et al.* (1990) observed 100 per cent mortality due to *R. bataticola* (*M. phaseolina*) in JG-62 variety of chickpea. This is one of the most common diseases in South-East Asian countries and limits chickpea production.

In other countries like Mandalay, Sagaing and Bago, central and lower Myanmar the major disease in chickpea was dry root rot caused by *R. bataticola*

(Reddy *et al.*, 1991). Aung *et al.* (1992) also observed that dry root rot caused by *R. bataticola* (*M. phaseolina*) was major disease in these countries. In Ethiopia, root rot of chickpea was also observed by Beniwal *et al.* (1992). Significant reduction (25-70%) in chickpea production is caused by this disease in India (Ahmad *et al.*, 1986, Pandey and Singh, 1990 and Ahmad and Mouselli, 1990).

Shrivastava *et al.* (2002) reported that *Fusarium oxysporum* was most prevalent (56%) in wilted plants followed by *R. bataticola* (17%) and *Trichoderma sp.* (10%). The incidence of *R. bataticola* also caused withering and drying of plant, which range from 8 to 20 percent.

A survey conducted by Manjunatha *et al.* (2011) and reported that the range of dry root rot incidence from 1 to 19% across the districts. Maximum dry root rot incidence in chickpea noticed in Gulbarga district (9.8%) followed by Raichur (7.6%) and the least (6.18 %) in Bidar (Karnataka). This was mainly attributed due to variation in soil type (black and red soil) and cultivars grown (Annegiri and JG 11) under rain fed condition.

Khan *et al.*, (2012) conducted a survey of Jammu and Kashmir showed highest incidence up to 40% in Shangus and lowest up to 4.11% in Naina. The incidence of dry root rot in chickpea caused by *R. bataticola* was observed in late October to mid November. The intensity of the disease was high in the month of February and March during late flowering and podding stage.

Jagre (2012) reported that that highest percent mortality due to dry root rot was observed in the 10th standard week i.e. 2nd week of march 2012 when the temperature range from 11.6 - 30.8⁰c with 18-76 percent relative humidity however the disease initiated when the temperature was >27⁰c with the decreases in relative humidity.

Recent surveys conducted during 2010-2013 indicated widespread and increased incidence of dry root rot in the central and southern states of India (Ghosh *et al.*, 2013). Disease was found irrespective of soil types, cropping system and cultivars used and incidence ranged from 5 to 50% or more in badly infected soils. This noticeable widespread geographic distribution of dry root rot probably makes it a significant disease in future.

The dry root rot of was not of much significance in chickpea earlier, but now, it has become a major threat to chickpea production in recent years due to altered weather conditions, particularly on the account of longer drought spells. Higher temperature and soil moisture depletion during crop growth period particularly at

post-harvesting stage is predisposing chickpea to dry root rot (Sharma and Pande, 2013).

Srinivas (2016) reported through survey that percent disease incidence was ranged from 6.67 to 15.33 per cent in Karnataka and 0.67 to 3.33 per cent in Andhra Pradesh at different locations.

The survey on the incidence of dry root rot of chickpea was carried out to know the prevalence and distribution of the disease in eight districts of North Karnataka viz., Bidar, Kalaburgi, Yadgir, Raichur, Koppal, Ballari, Dharawad and Vijayapura representing chickpea growing rainfed situations during *rabi*, 2015-16. The per cent dry root rot incidence was recorded at 55–80 days old crop. The results indicated that the incidence was noticed in all the fields wherever the survey was conducted (Deepa *et al.*, 2018).

Kadam *et al.* (2018) conducted a roving survey for recording chickpea dry root rot disease incidence was undertaken in Marathawada region. The highest disease incidence was observed in Latur district (23 per cent) and lowest in Aurangabad district (10.20 per cent).

Collection, Isolation, Purification, Identification and Pathogenicity of the *M. phaseolina*.

Several workers described the fungus isolated from different hosts with some variability in morphology and differentiated it in separate species. Pathogenicity plays an important role in confirming that a particular pathogen isolated from diseased plant part, is able to cause disease on same plant. *R.a bataticola* (Taub.) Butler as a plant pathogen was recognized by Halsted (1890). Taubenhause (1913) gave the name of the genus as *Sclerotium* because of absence of spores and the species named as *bataticola* because it was pathogenic to *Ipomea batatus* (L.) Lam. Britton Jones (1925) transferred the fungus to the genus *Rhizoctonia* based on the identification of cultures by Butler (1918). The Pycnidial stage was described by Maublanc as *M. phaseolina*. (Maublanc, 1905)

Ashby (1927) accepted *M.* and rejected the binomial *M. phaseoli* and proposed a new binomial *M. phaseolina* as the pycnidial stage of *R. bataticola* on the basis that *M. phaseoli* was the earliest applicable binomial. He was not aware that Tassi had earlier described *M. phaseolina*.

Deshkar *et al.* (1973) reported *M. phaseolina* was pathogenic in mungbean by paper towel methods. Hooda and Grover (1988) reported that young inoculum of *M.*

phaseolina was more pathogenic than the old inoculum in *Vigna radiata*. De-Mooy and Burke (1990) reported that drenching the soil with suspension of *M. phaseolina* before planting of cowpea was more pathogenic. *M. phaseolina* is a soil borne pathogen. Mathur and Nath (1970) first time reported *M. phaseolina* on mungbean in India

Dhingra and Sinclair (1977) reported *M. phaseolina* causes root rot in number of crops including groundnut, sesame, mungbean, soybean and kidney bean. The pathogen was also reported to cause leaf blight, seedling blight and charcoal rot of guar, mothbean, urdbean, mungbean, groundnut and sunflower (Shukla and Bhargava, 1976, Udit Narain, 1986 and Grover and Sakhuja, 1981). In India, percentage of plants infected with dry root rot is high from late October to mid November, decrease in December – January and increase again February – March (Tripathi and Sharma, 1983).

Singh and Shrivastava (1988) observed typical symptoms of root rot as sudden death of mothbean seedling. The leaves first started withering and drying and the root rot was followed by killing the whole plant. The root showed brown to black lesions on stem. Patel and Patel (1990) observed that the optimum temperature for growth and sclerotial formation by *M. phaseolina* was 35°C, both declined at temperature below 15°C and above 40°C. Under field conditions disease intensity increased with a progressive rise in temperature and decrease in relative humidity. They observed maximum disease occurrence at 35°C and 76 per cent relative humidity.

Mittal (1997) observed in general, disease incidence was less in the mixed crop than in the sole crop and gradually declined from the first to last sowing date. Sowing on 19 October was most effective in reducing the disease and increasing yield of lentil. The fungus is a facultative parasite capable of living saprophytically on dead organic tissue, particularly on many of its natural hosts producing sclerotial bodies, which produces pycnidia. When atmospheric temperature is above 30°C & the pycniospores remain viable for over a year since the fungus attack wide range of plant species.

Goyal and Jain (1998) reported that *M. phaseolina* caused pre and post emergence mortality and reduced seedling vigour when artificially inoculated in black gram and mungbean.

Singh and Kumar (2002) reported that *M. phaseolina* (*R. bataticola*) is an important seed and soil borne pathogen, causing dry root rot and leaf blight of urd bean (*Vigna mungo*). They observed that the pathogen was invariably present in the seed coat and in the cotyledons of infected seeds, survived in infective state up to twenty months in seeds transmitted to seedlings by local contact.

El-Araby *et al.* (2009) observed symptoms including shrunken, unfilled pods and brown wilted attached to dead petioles and stems along with the stem cortex due to *M. phaseolina* causing charcoal rot in soybean. The fungus is mainly a soil dweller and spreads from plant to plant through irrigation water, food and implements and cultural operation. The sclerotia and pycnidia may also become air borne and cause further spread of the pathogen (Rangaswami and Mahadevan, 2008).

The root rot disease caused by soil and seed borne fungus *M. phaseolina* (Tassi) Goid is major limiting factor in mungbean. It is a serious disease of many crops infecting upto 100% yield losses in mungbean under dry and hot condition. It is known to infect more than 500 plant species worldwide (Wyllie, 1993).

Katariya *et al.* (2012) made studies to determine the cultural and pathogenic variation of *M. phaseolina* isolates collected from infected chickpea from various parts of Rajasthan. Pure cultures of these isolates were found the pathogenic on 40-day-old chickpea cv. C-235 grown in pots.

Chemical control

In vitro

Goyal and Mehrotra (1981) tested 9 fungicides against *R. bataticola* *in vitro* and on (chickpea) Benlate, Bavistin. Thiophanate methyl and Dyrene were promising. *In vitro*, Venomyl and Bavistin inhibited fungal growth most.

Sarwar and Raju (1985) tested 8 fungicides *in vitro* an antibiotic Topsin M-70 was highly effective at 1000 ppm against *R. bataticola*.

Peshney *et al.* (1992) reported that thiram, Campton, PCNB, Mancozeb Iprodione, Carboxin, Carbendazim, Thiobedazole (all 0.2%) and Tridemorph (0.07%) effectively controlled growth on Sclerotial germination of *R. bataticola* *in vitro*. Carbendazim and Thiobendazole were most effective and inhibited growth at the concentration of 0.006%. In successive cultures, the fungus should increase tolerance to Mancozeb and PCNB and to a lesser extent to Thiram, Campton,

Carboxin and Iprodione. There was no reduction in sensitivity towards carbendazim or Thiobendazole.

Lambhate *et al.* (2002) reported that, Carbendazim, Thiophanate methyl @ 0.1, 0.2, 0.3 were found effective in inhibiting the growth of *M. phaseolina*. Nagamma (2012) observed that, Thiram @ 0.3%, Carboxin + Thiram 0.2% and Carbendazim @ 0.1% showed cent percent inhibition of *R. bataticola*.

Prajapati *et al.* (2002) tested the efficacy of 11 fungicides, 8 non- target pesticides and 2 biological control agents against *R. bataticola* (*M. phaseolina*) the causal agent of root rot of chickpea and found that carbendazim, carbendazim +Thiram, carboxin and Topsin- M (Thiophanate methyl) completely inhibited the growth of *R. bataticola*.

Salunke *et al.* (2008) carried an experiment to see the efficacy of seven fungicides viz., companion (0.2%), zineb (0.2%), tricyclazole (0.2%), curzate M8TM (0.2%), copper oxychloride (0.3%), carbendazim (0.1%) and thiram (0.3%) against *R. bataticola* by poisoned food technique. Companion and zineb each at 0.2% concentration showed complete growth inhibition of all isolates. All isolates showed tolerance to copper oxychloride, thereby showing its least efficacy.

In vitro studies conducted by Konde *et al.* (2008) revealed that carbendazim + thiram (0.1+ 0.2%), penconazole (0.1%) and thiophanate-M (0.1%) completely inhibited (100%) the growth of *R. bataticola*.

Ammajamma and Hegde (2009) evaluated efficacies of systemic and non-systemic fungicides against *R. bataticola* at different concentrations. Among systemic fungicides, carboxin (0.05 and 0.1) and hexaconazole, metalaxyl and traidemifon at 0.1 per cent completely (100%) inhibited the growth of *R. bataticola* and among the non-systemic fungicides tested thiram at 0.1, 0.2 and 0.3 per cent concentrations was found effective against *R. bataticola*. Kumar *et al.*, (2011) also reported that bavistin and vitavax inhibit 100 per cent of mycelial growth at 50 ppm.

Khan *et al.* (2012) found that Indofil M-45, Bavistin, companion, copper oxychloride and Benelate were completely inhibited the growth of the fungus *R. bataticola* causing dry root rot of chickpea under laboratory condition on potato dextrose agar medium as compare to check.

Nagamma *et al.* (2012) tested fungicides *in vitro* against *M. phaseolina* causing dry root rot of chickpea. Cent percent inhibition was recorded in

hexaconazole, carbendazin, difenoconazol and propiconazole at 50 ppm among all the systemic fungicides used.

In vitro efficacy of non systemic, systemic and combine fungicides are evaluated against *R. bataticola*. Among contact fungicides tested, Ziram recorded 100 per cent inhibition at all the concentrations (i.e., 0.1, 0.2 and 0.3 %) with the mean inhibition, which was significantly superior over all the treatments whereas, Mancozeb and Thiram showed 100 per cent inhibition at 0.3 per cent concentration. Among systemic fungicides tested, Tebuconazole showed complete inhibition at all the concentration (i.e., 0.05, 0.10 and 0.15), whereas, Propiconazole showed 100 per cent inhibition of *R. bataticola* at 0.10 and 0.15 per cent concentration. Among combine products tested, Carbendazim 12% + Mancozeb 63% WP, Trifloxystrobin 25% + Tebuconazole 50% EC and Carboxin 37.5% + Thiram 37.5% WP showed cent per cent (100%) inhibition at all the concentrations (i.e., 0.10%, 0.20% and 0.30 %) (Maruti *et al.*, 2017).

A lab experiment was conducted by Ravichandran and Hegde (2017) to evaluate six combine products, five contact and four systemic fungicides against *Rhizoctonia bataticola* causing dry root rot in chickpea. Among combine products evaluated, carbendazim 25% + mancozeb 50% (Sprint), carboxin 37.5% + thiram 37.5% (Vitavax power 75% WP) and carbendazim 12% + mancozeb 63% (Saaf) were found to be most effective with complete inhibition of mycelial growth of *R. bataticola* at all the concentrations tested. Among the contact fungicides tested, chlorothalonil and mancozeb at 0.2% were effective in inhibiting cent per cent inhibition. Among the four systemic fungicides evaluated against *R. bataticola*, carbendazim, difenconazole and tebuconazole were best with cent per cent inhibition of mycelial growth at all concentrations tested.

Pawar *et al.*, (2018) tested four fungicides viz., Vitavax+Thiram @ (0.3%), Carbendazim + Mancozeb @ (0.3%), Thiram + Mancozeb @ (0.3%) and Thiophanate methyl @ (0.1%) against *R. bataticola* in *in vitro* and found that all the fungicides inhibited cent percent (100%) radial mycelial growth of *R. bataticola*.

In vivo

Verma (1976) found seed treatment with Bavistin or Carboxin at the rate of 0.25 per cent to protect seedling in field for 30 days or more from root rot of chickpea.

Pardeshi *et al.* (1989) reported that root and shoot lengths and seedling fresh weight were increased most effectively after application of captan followed by thiram, sulphur and mancozeb. Most of the fungicide treatments increased seedling dry weight.

Taya *et al.* (1990) stated that effective management of *R. bataticola* can be done by single application of carbendazim or in combination with thiram as seed treatment, soil- drench and seed treatment plus drenching.

Mathieson and Rush (1991) observed that seed treatment with Triadimenol had significantly greater emergence and dry matter production in chickpea in root rot infested field.

Singh and Sindhan (1998) observed that carbendazim and carboxin (0.2%) used as seed treatment and soil drench or seed treatment + soil drench significantly reduced the incidence of dry root rot of chickpea caused by *R. bataticola* (*M. phaseolina*) with a corresponding increase in phenolics and carbohydrates in the plants. They examined that application of carbendazim and carboxin with seed treatment + soil drenching was the most effective method in reducing disease incidence followed by soil drenching and seed treatment alone.

Vijay-Mohan *et al.* (2006) reported in fungicidal trails on management of dry root rot of chickpea caused by *R. bataticola* carbendazim (0.2%) and Etaconazole (0.1%) used as seed treatment, soil drenching and seed treatment plus soil drenching recorded lowest disease incidence of 15.6 and 18.2 per cent highest grain yield of 192 and 18.9 q/ha respectively, during rabi 2001-02 and 2002-03 crop seasons. The above treatment recorded 57.2 and 52.4% higher yield over control with per rupee return of 12.78 and 11.80 respectively.

Kaushal (2008) reported that seed treatment with thiram and carbendazim at 2 g/kg was effective in control of chickpea dry root rot.

Thawale *et al.* (2010) found that seed treatments with thiram + carboxin and hydration (2 hrs) followed by dry dressing with thiram, recorded maximum germination, seedling length, seedling dry weight and vigour index. All treatments recorded more number of pods/plant, 100 seed weight, and seed yield/plant over control.

Manjunatha, *et al.* (2011) evaluated two biocontrol agents individually and in combinations and in integration with carbendazim seed treatment and hexaconazole seed soaking for management of dry root rot of chickpea. Seed and soil application

with bioagents was effective for the maximum germination with less root rot incidence as compared to seed treatment with carbendazim alone and seed soaking in hexaconazole alone. Integration of seed and soil application of biocontrol agents resulted in higher germination and reduced mortality due to disease as against check. The application of biocontrol agents individually either soil application or seed treatment also showed good germination with reduced mortality due to dry root rot as compared to treatments imposed with fungicides.

The combined use of host resistance with fungicide treatment resulted in better seedling emergence and delayed the onset of root rots. Treating the seeds with captan or thiram is also helpful in reducing the disease (Gurha *et al.* 2003). Practice of seed treatment with fungicide (bavistin and thiram) in farmers' field reduced the DRR incidence in central and southern parts of India (Ghosh *et al.* 2013).

Chilkuri and Giri (2014) also found that seed treatment with Thiram + Carbendazim @ 3 g/kg increases the seed germination percent (85%), shoot length (11.17), root length (9.27) and seedling vigour index (1765.60) in greengram. Sharma and Kumari (2017) tested the efficacy of various fungicides viz. Carbendazim, Thiram, Carboxin, Phenylpyrrole, Thiophnate Methyl and Chlorothalonil as seed treatment and soil application were evaluated against *R. bataticola* causing root rot disease of Chickpea. Management study of root rot disease of chickpea done under both pot and field condition. Seed treatment with Carbendazim was proved most effective in minimizing the disease as well as pathogen population in soil followed by Thiophnate Methyl. Seed treatment with Chlorothalonil was found least effective in minimizing the disease as well as pathogen population in soil.

Lakhran and Ahir (2018) tested efficacy of fungicides against dry root rots of chickpea under *in vivo*. According to the result, all fungicides were found significantly superior over control in reducing per cent disease control at 40 and 60 days after sowing. Minimum percent disease incidence was recorded with carbendazim (13.33 and 15.38 %) followed by carbendazim + mancozeb (16.66 and 20.00%) while the maximum percentage disease incidence was observed in control (46.60 and 62.50 %) at both of the stage.

Pawar *et al.* (2018) reported Thiram + Carbendazim @ (0.3%) was found most effective for increasing the seed germination (81.66%), shoot length (10.77

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cm), root length (9.27 cm) and seedling vigour index (1636.46) followed by T4 Thiophanate methyl @ (0.1%) with seed germination (73.33%) shoot length (9.96 cm) root length (8.45 cm) and seedling vigour index (1350.00).

CHAPETR-III

MATERIAL AND METHODS

(A) Materials

1. General

Corning and Borosil make glasswares were used throughout the course of study. Glasswares were cleaned with chromic acid solution followed by washing with detergent and finally rinsed with tap water. After drying, glasswares were sterilized in hot air oven at 180°C for 2 hours. The metallic equipments like forceps, needle and cork borer were sterilized by dipping in alcohol and heating to red hot over flame of a spirit lamp.

Surface sterilization of plant parts and diseased materials was done by dipping them in 0.1% mercuric chloride solution for 1-2 minutes followed by 3 times, washing by sterilized water.

The culture media were sterilized in autoclave at 15 lbs pressure per square inch (1.05 kg/cm²) for 20 minutes. The soil and sand were sterilized at 30 lbs pressure per square inch (3.1 kg/cm²) for two hours.

1.1 Media

Following media were used in the course of investigation with following ingredients:

1.1.1 Potato Dextrose Agar (Riker and Riker, 1936)

Peeled and Sliced Potato	200 g
Dextrose	20 g
Agar	20 g
Distilled water	1000 ml

1.1.2 Soil maize media

Maize powder	200g
Sand	400g
Distilled water	1000 ml

1.2 Source of seed and other materials

Chickpea seeds were procured from All India Coordinated Research Project on Chickpea, R.A.K. College of Agriculture, Sehore (M.P.). Fungicides, talk based

formulation of *Trichoderma* strain were obtained from the Department of Plant Pathology, R.A.K. College of Agriculture, Sehore (M.P.).

1.3 Survey

Survey for dry root rot was done in the chickpea growing areas of Sehore district. Diseased plant parts were collected in paper envelopes and brought to the laboratory for isolation. The chickpea crop was surveyed at flowering stage. During the survey per cent disease incidence was recorded.

1.4 Experimental site:

All the field experiments were conducted at the experimental field of Plant Pathology Farm of R.A.K. College of Agriculture, Sehore during *rabi* season 2017-18.

(B). METHODS

2.1 Survey : Collection of diseased plant samples and symptomatology

Dry root rot chickpea plants were collected from 23 villages in Sehore district of Madhya Pradesh at adult plant stages. Samples were brought in the laboratory and symptoms were recorded in detail. The tissues of affected plants showing the symptom of dry root rot were examined under the microscope for the presence of the causal agent.

2.2 Isolation of pathogen from diseased samples

The roots of diseased plant showing symptoms were washed thoroughly with tap water, small pieces of infected roots were cut with the help of sterilized blade. These pieces were surface sterilized with 1:1000 mercuric chloride (HgCl_2) solution for one minute followed by three washings with sterilized distilled water to remove traces of HgCl_2 . The pieces were then transferred aseptically to Petri plates containing PDA. Inoculated Petri plates were incubated at $25 \pm 2^\circ\text{C}$ for three to five days and examined at frequent intervals to see the growth of the fungus developing from different pieces. As and when fungal colony appears they were transferred to PDA slant.

2.3 Purification and identification of isolates of *R. bataticola*

The cultures of *R. bataticola* were purified by frequent sub culturing by the hyphal tip method as described by Dingra and Sinclair (1985). Isolated fungus was identified based on their morphological characters according to Barnett and Hunter (1987) and stored at 4°C until use.

2.4 Pathogenicity tests

2.4.1 Pot inoculation method

The test fungus was multiplied on soil maize medium for 7 days. Sterilized soil was used in pot experiment. The soil was collected from the local field and dried thoroughly in sun and sterilized. Pots of 25 cm face diameter were used during experiments.

The pots were filled with 1kg sterilized soil and then 20g of inoculum was added in each pot. One inch of soil was placed over the inoculum. The contents were then mixed and surface sterilized seeds were sown and covered by half inch layer of soil. In control, no inoculum was added and pots were filled with sterilized soil. Ten seeds of variety L 550 were sown in each pot. Observations on disease intensity were also recorded.

2.5 In vitro test

2.5.1 Efficacy of *Trichoderma* spp. against *R. bataticola*

Efficacy of fungicides was tested by poisoned food techniques (Nene and Thapliyal, 1979).

Table-3.1: Fungicide with their coined, trade and chemical name

Common name	Trade name	Formulation	Chemical name
Fosetyl Al	Aliette	80% WP	aluminum tris (O-ethyl phosphonate)
Carbendazim	Bavistin	50% WP	Methyl-1-2-benzimidazole carbamate (MBC)
Mancozeb+carbendazim	Sprint	25%+50% WS	Manganese ethylene bis (dithiocarbamate) polymeric complex + methyl-2-benzimidazole carbamate.
Thiram +carboxin	Vitavax power	37.5% +37.5% WP	5,6-dihydro-2-methyl-1,4-oxathiin-3-carboxanilido + Tetramethyl thiurum disulphide
Thiophanate methyl + Pyraclostrobin	Xelora	45%+5% FS	Dimethyl 4,4'-o-phenylene bis(3-thioallophanate) + Methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl] phenyl} (N-methoxy) carbamate

Three replications were kept for recommended dose. Required quantity of the fungicide was added in PDA and thoroughly mixed at the time of pouring. Twenty ml

of medium was poured in each pre-sterilized 90 mm diameter Petri plate. A five mm disc of *R. bataticola* was placed on the medium in upside down position to maintain a continuous contact of fungus with poisoned medium. The medium without fungicide served as control. The radial growth of the colony was measured after seven days growth under different fungicides and control was calculated.

2.6 Effect of seed treatment on dry root rot incidence and growth promoters of chickpea

Design	:	Randomized block design
Plot size	:	5 x 2.4 m ²
Row to row	:	30 cm
Seed / plot	:	630 seeds
No. of rows	:	9
Plot to plot distance	:	60 cm
Replications	:	4
Treatments	:	6
Crop	:	Chickpea
Variety	:	JG 6

T₁- Carbendazim @ 2g/kg

T₂- Carboxin + Thiram @ 2g/kg

T₃- *Trichoderma* @ 5g/kg

T₄- Carbendazim + Mancozeb @ 2.5g/kg

T₅- Thiophanate methyl+ Pyraclostrobin @ 2ml/kg

T₆- Control

2.7 Effect of drenching of fungicides on dry root rot incidence and growth promoters of chickpea

Design	:	Randomized block design
Plot size	:	5 x 2.4 m ²
Row to row	:	30 cm
Seed / plot	:	630 seeds
No. of rows	:	9
Plot to plot distance	:	60 cm
Replications	:	4
Treatments	:	6

Crop : Chickpea
Variety : JG 6

- T₁- Fosetyl Al 80% WP a.i. 1600 @ 2000 g/ha at 30* and 45** days after sowing
T₂- Fosetyl Al 80% WP a.i. 2000 @ 2500 g/ha at 30 and 45 days after sowing
T₃- Fosetyl Al 80% WP a.i. 2400 @ 3000 g/ha at 30 and 45 days after sowing
T₄- Carbendazim a.i. 500 @ 1000 g/ha at 30 and 45 days after sowing
T₅- Fosetyl Al 80% WP a.i. 4800 @ 6000 g/ha at 30 and 45 days after sowing
T₆- Control at 30 and 45 days after sowing

* 30 days after sowing (Mid crop stage)

** 45 days after sowing (Pod initiation stage)

Observations:

(a) Effect on germination

Germination per cent of chickpea was recorded by counting the complete plant population of each plot after thirty days of sowing and percentage emergence was calculated on the basis of seeds sown/plot. The data were transformed angularly and analyzed.

(b) Plant vigour

Plant growth promoting activity of *Trichoderma* spp. was assessed based on vigour index and it was calculated by using the formula of Baki and Anderson (1973).

Vigour index = (Mean shoot length + Mean root length) × germination (%)

(c) Incidence of dry root rot

Incidence of dry root rot were taken at 40 and 60 days after sowing

(d) Shoot and root length

Shoot and root length of each plant were recorded in cm. The shoot and root length of plant were recorded after 40 and 60 days of sowing and mean of 10 plants was calculated.

(e) Fresh and dry weight of shoot

Fresh and dry weight of each plant was recorded in g. The fresh and dry weight of plant was recorded after 40 and 60 days of sowing and mean of 10 plants was calculated. The sample was weighed with the help of electronic balance.

(f) Fresh and dry weight of root

Fresh and dry weight of root was recorded in g. The fresh and dry weight of root was recorded after 40 and 60 days of sowing and mean of 10 plants was calculated. The sample was weighed with the help of electronic balance.

(g) Number of nodules, fresh and dry weight of nodules

Numbers of nodule, fresh and dry weight (mg) of each plant were recorded after 40 and 60 days of sowing and mean of 10 plants was calculated.

(h) Effect on yield components

The yield contributing characters like, number of pods, number of seeds per pod, seed yield per plant and seed yield per plot were recorded on 10 plants tagged in treatment of the experiment.

Statistical analysis

The data were subjected to statistical analysis after transformation. The data converted into percentage were transformed to angular values.

The difference between the two means was subjected to further testing by computing critical difference at 5% probability level.

(1) Standard error for treatment mean:

$$S. Em. \pm = \sqrt{\frac{Em.s}{r}}$$

(2) Critical difference:

$$C.D. = S. Em. \times \sqrt{2} \times t$$

Where,

Ems = Error means sum of square

r = Number of replication

t = 't' value at 5% probability levels

CHAPTER IV

EXPERIMENTAL FINDINGS

4.1 Survey of chickpea crop for dry root rot incidence

Survey of chickpea crop fields was conducted in the second and third week of February 2017 at four blocks of Sehore district in 23 locations. Percent disease index (PDI) was recorded and presented in table-4.1 and fig.-1.

Table-4.1: Root rot incidence in chickpea fields in different blocks of Sehore district

S.No .	Block	Village	Date of sowing	Date of observation	Variety	Mean PDI (%)
1	Sehore	Badnagar	16.11.2017	10.02.2018	JG 130	27.50
2		Dondi	17.11.2017	10.02.2018	KAK 2	19.40
3		Amla	16.10.2017	10.02.2018	Vishal	30.50
4		Kachnariya	15.11.2017	10.02.2018	Kripa	27.60
5		Hirapur	17.11.2017	10.02.2018	Dollar	18.80
6		Jamni	19.11.2017	10.02.2018	Desi(katila)	36.30
	Mean PDI (%)					26.88
7	Ichhawar	Kheri	18.11.2017	12.02.2018	Dollar	22.70
8		Kaneriya	17.11.2017	12.02.2018	Vishal	35.60
9		Mogra	19.11.2017	12.02.2018	JG 130	25.40
10		Nagli	15.10.2017	12.02.2018	JAKI 9218	19.60
11		Arwaliya	21.11.2017	12.02.2018	Kripa	27.40
12		Bijori	17.11.2017	12.02.2018	Desi(katila)	35.90
	Mean PDI (%)					27.77
13	Rehti	Ambajhir	19.11.2017	14.02.2018	KAK 2	26.10
14		Dongri	17.10.2017	14.02.2018	JAKI 9218	29.60
15		Barkheda	20.10.2017	14.02.2018	Dollar	27.50
16		Ramgarha	17.11.2017	14.02.2018	Kripa	37.50
17		Khajuri	18.11.2017	14.02.2018	Dollar	27.70
18		Kosmi	18.11.2017	14.02.2018	Desi(katila)	39.60
Mean PDI (%)						31.33
19	Budni	Nander	19.11.2017	16.02.2018	Desi(Katila)	38.30
20		Sagpur	19.11.2017	16.02.2018	Kripa	32.80
21		Rampura	18.11.2017	16.02.2018	KAK 2	23.60
22		Khatpura	17.11.2017	16.02.2018	JAKI 9218	28.20
23		Hirani	16.11.2017	16.02.2018	Vishal	30.40
Mean PDI (%)						30.66

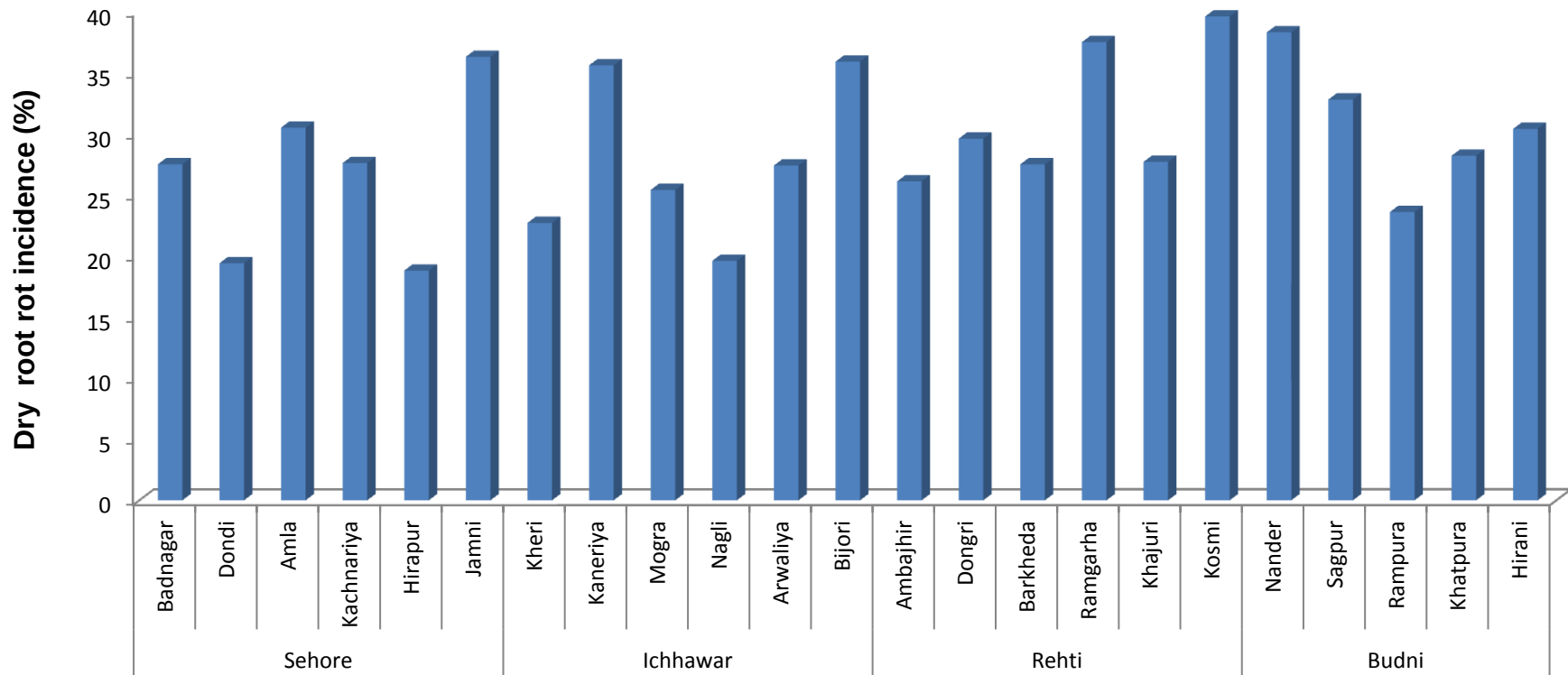


Fig.-1: Dry root rot incidence in chickpea fields in different blocks of Sehore district during *rabi* 2017-18

The data presented in the table-4.1 and fig.-1 reveals that the per cent disease incidence of dry root rot varied from location to location. Status of incidence of dry root rot (DRR) at maturity stage of chickpea around Sehore district varied from 18.80-39.60 per cent. The highly prone area of dry root rot of chickpea was Kosmi (39.60%) village followed by Nander (38.30%), Ramgarha (37.50%), Jamni (36.30%), Bijori (35.90%) and Kaneriya (35.60%) villages. The least incidence of disease was noticed in Hirapur village (18.80%) as compared to other locations of the survey. Block-wise incidence of dry root rot in the four blocks surveyed ranged from 26.88 to 31.33%. In Rehti block, DRR incidence was maximum 31.33 % followed by the blocks viz. Budni (30.66%) and Ichhawar (27.77%). Minimum 26.88% DRR incidence was found in Sehore block.

Out of six villages of Sehore block, highest per cent DRR incidence was noticed in village Jamni (36.30%) followed by 30.50% in Amla (table-4.1). However, it was lowest 19.40% and 18.80% in villages Dondi and Hirapur, respectively.

In Ichhawar block, out of six villages, two villages viz., Kaneriya (35.60%) and Bijori (35.90%) showed DRR incidence above 30% whereas , it was lower than 30% in Arwaliya (27.40%), Mogra (25.40%), Kheri (22.70%) and Nagli (19.60%).

Six villages of Rehti block viz., Ambajhir, Dongri, Barkheda, Ramgarha, Khajuri and Kosmi were surveyed. Highest per cent disease incidence was observed in Kosmi (39.60%) followed by Ramgarha (37.50%) and lower than 30% per cent in rest of the villages viz., Khajuri (27.70%), Barkheda (27.50%) and Ambajhir (26.10%).

In Budni block, out of five villages, the highest per cent DRR was noticed in village Nander (38.30%) followed by 32.80% in Sagpur and 30.40% in Hirani village (table-4.1). However, it was lowest 28.20% and 23.60% in villages Khatpura and Rampura, respectively.

4.2 Isolation and Identification of associated fungi

The incidence of dry root rot was mostly observed during flowering to podding stage of chickpea. The first symptom of the disease is yellowing of the leaves, within a day or two such leaves dry and while plant die off in the next two or three days. The leaves and stem of the affected plants become straw coloured. The tap root seems dark and quite brittle in dry soil. The dark black sclerotia can be observed on and within host tissue of the stem. Dry root rot affected plants can be pulled easily

and showed blackening of roots (Plate-1). The tissues were weakened and break off easily. In advanced cases, sclerotial bodies were seen scattered on the affected tissue. Isolation made from roots of infected chickpea samples yielded growth of *Rhizoctonia bataticola*. The *R.bataticola* isolate was purified by hyphal tip method, sub cultured and maintained on PDA slants for further studies.

The fungal isolates from roots of DRR infected chickpea plants were identified as *R. bataticola* on the basis of morphological characters, viz. initial radial hyaline colonies which later became carbonaceous brown to black, mycelium septate and dark brown in colour, typical right angled branching of mycelium and production of sclerotia.

4.3 Effect of seed treatment

4.3.1 Effect on germination and plant vigour of chickpea

Effect of seed treatment with *T. viride* and fungicides on germination of chickpea seeds was studied and data are presented in table-4.2 and fig.-2.

The data from table-4.2 reveals that germination percentage in the seed treated with *T. viride* and fungicides was significantly higher than control (56.24%). Among the treatments, maximum seed germination (82.34%) was recorded in *T. viride* @ 5g/kg which was at par with Carboxin +Thiram @ 2g/kg (80.50%), Carbendazim + Mancozeb@2.5g/kg (78.35%) , Carbendazim @2g/kg(77.00%) but it was superior to Thiophanate methyl+ Pyraclostrobin @ 2ml/kg (76.20%).

Table-4.2: Germination percentage of chickpea in *T. viride* and fungicides treatments

Treatment	Germination percentage*	Plant vigour*
Carbendazim @ 2g/kg	77.00 (61.59)	5872.27
Carboxin + Thiram @ 2g/kg	80.50 (64.18)	5592.99
<i>Trichoderma viride</i> @ 5g/kg	82.34 (65.64)	6519.20
Carbendazim + Mancozeb @ 2.5g/kg	78.35 (62.57)	5682.48
Thiophanate methyl+ Pyraclostrobin @ 2ml/kg	76.20 (61.03)	4917.97
Control	56.24 (48.59)	2696.45
Sem±	2.82	251.45
CD at 5%	6.02	535.84

*Average of four replications.

** Data in parenthesis are angular transformed value

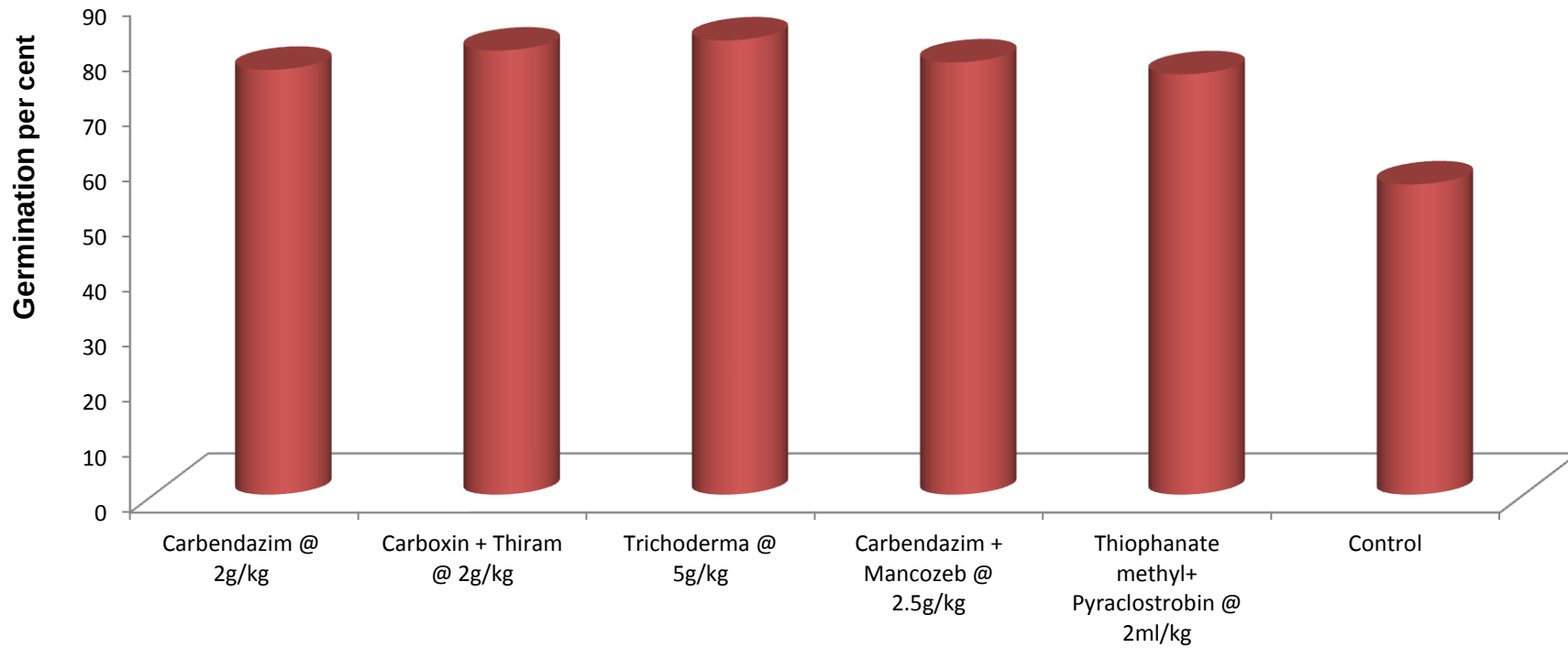


Fig.-2: Germination percentage of chickpea in seed treated with *Trichoderma viride* and fungicides

The data in table-4.2 and fig.-3 indicate that the seed treatments with *T. viride* and fungicides significantly increased the plant vigour as compared to control. Among the treatments, maximum plant vigour (6519.20) was recorded in *T. viride* @ 5g/kg which was superior to rest of the treatments. Carbendazim @ 2g/kg (5872.27) was at par with Carbendazim + Mancozeb @ 2.5g/kg (5682.48), Carboxin + Thiram @ 2g/kg (5592.99). However, seed treatment with Carbendazim was significantly superior to Thiophanate methyl+ Pyraclostrobin @ 2ml/kg (4917.97). In control least plant vigour 2696.45 was recorded.

4.4 Effect of seed treatment on Growth Promoting Ability (GPA) of chickpea

4.4.1 Effect on length, fresh and dry weight of chickpea shoot

The data presented in table-4.3 shows the effect of seed dressing with *T. viride* and fungicides on length, fresh weight and dry weight of chickpea shoot.

Table-4.3: Length, fresh and dry weight of chickpea shoots in different seed treatments

Treatment	Chickpea shoot					
	Length/plant (cm)		Fresh weight/plant (g)		Dry weight/plant (g)	
	40 DAS	60 DAS	40 DAS	60 DAS	40 DAS	60 DAS
Carbendazim @ 2g/kg	33.50	56.62	27.08	62.15	10.86	21.09
Carboxin + Thiram @ 2g/kg	31.25	51.97	24.32	53.53	9.34	17.74
<i>Trichoderma viride</i> @ 5g/kg	34.75	58.73	28.44	65.77	11.87	23.3
Carbendazim + Mancozeb @ 2.5g/kg	32.00	54.08	25.45	55.70	9.81	19.35
Thiophanate methyl+ Pyraclostrobin @ 2ml/kg	30.75	48.65	23.37	50.11	8.06	16.26
Control	27.25	37.20	21.10	38.31	4.31	13.48
Sem $\pm$	1.02	1.77	0.95	1.86	0.53	1.15
CD at 5%	2.17	3.78	2.03	3.96	1.14	2.46

*Average of four replications.

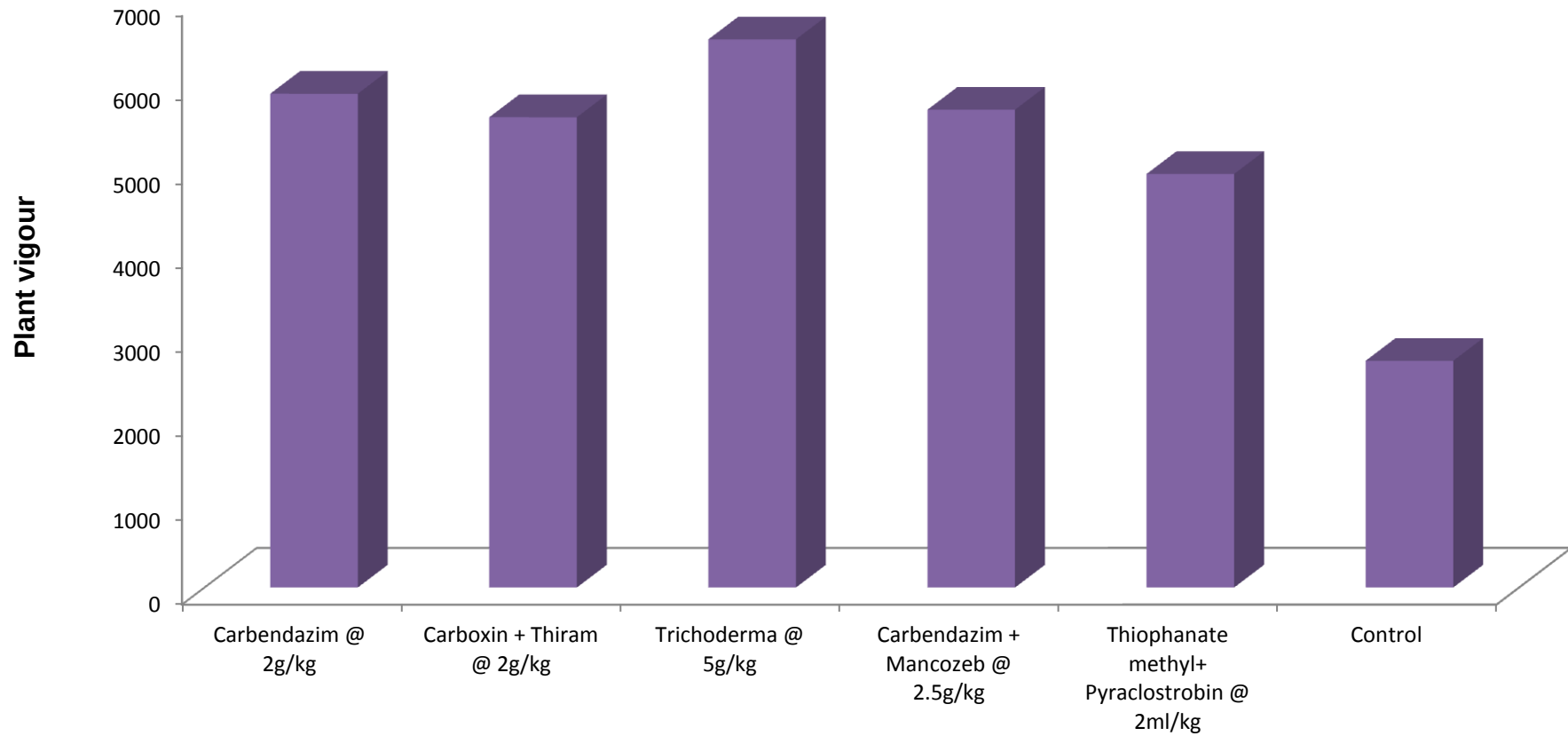


Fig.-3: Plant vigour of chickpea in seed treated with *Trichoderma viride* and fungicides

The data presented in table-4.3 and fig.-4 indicate that after 40 days of sowing significant increase in shoot length/plant was observed in seed dressing with *T. viride* and fungicides as compared to control. Among the treatments, *T. viride* @ 5g/kg recorded maximum shoot length/plant (34.75cm) at par with Carbendazim @2g/kg (33.50cm) and significantly superior to rest of the treatments. Carbendazim+ Mancozeb @2.5g/kg (32.00 cm), Carboxin + Thiram @2g/kg (31.25 cm) and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (30.75 cm) were equally effective in increasing the shoot length/plant of chickpea. The least shoot length/plant 27.25 cm was recorded in control.

After 60 days of sowing, significant increase in shoot length/ plant was observed in seed dressing with *T. viride* and fungicides as compared to control. Among the treatments, *T. viride* @ 5g/kg recorded maximum shoot length/plant (58.73cm) at par with Carbendazim @ 2g/kg (56.62cm) and significantly superior to rest of the treatments. Carbendazim+ Mancozeb @ 2.5g/kg (54.08 cm), Carboxin + Thiram @2g/kg (51.97 cm) and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (48.65 cm) were same as in increasing the shoot length of chickpea. The least shoot length/plant 37.20 cm was recorded in control.

Data in the table-4.3 and fig.-5 shows that 40 days after sowing seed treatment with *T.viride* and fungicides significantly increased in the fresh shoot weight of chickpea as compared to control. Among the treatments, maximum fresh shoot weight/plant (28.44 g) was in *T. viride* @ 5g/kg at par with Carbendazim @ 2g/kg (27.08g) and significantly superior to rest of the treatments .Fresh shoot weight of chickpea was at par in Carbendazim+ Mancozeb @2.5g/kg (25.45 g), Carboxin + Thiram @2g/kg (24.32 g) and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (23.37 g) after 40 of sowing. In control the fresh shoot weight was lowest 21.10 g/plant.

Similarly, 60 days after sowing seed treatment with *T. viride* and fungicides significantly increased the fresh shoot weight of chickpea as compared to control. Among the treatments, maximum fresh shoot weight (65.77 g) was recorded in *T. viride* @ 5g/kg at par with Carbendazim @ 2g/kg (62.15 g) and significantly superior to rest of the treatments. Carbendazim+ Mancozeb @2.5g/kg (55.70 g), Carboxin + Thiram @2g/kg (53.53 g) and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (50.11 g) were equally effective in increasing fresh shoot weight of chickpea. In control the fresh shoot weight was lowest 38.31 g/plant.

After 40 days of sowing, seed treatment with *T. viride* and fungicides significantly increased the shoot dry weight of chickpea as compared to control. Among the treatments, maximum shoot dry weight/plant (11.87 g) was obtained in *T. viride* @ 5g/kg at par with Carbendazim @ 2g/kg (**10.86g**) and significantly superior to rest of the treatments. Dry shoot weight of chickpea was at par in Carbendazim+ Mancozeb @2.5g/kg (9.81 g), Carboxin + Thiram @2g/kg (9.34 g) and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (8.06 g). Least shoot dry weight 4.31 g/plant was recorded in control.

After 60 days of sowing, seed treatment with *T. viride* and fungicides significantly increased the shoot dry weight of chickpea as compared to control. Among the treatments, maximum shoot dry weight/plant(23.30 g) was obtained in *T. viride* @ 5g/kg at par with Carbendazim @ 2g/kg (**21.09 g**) and significantly superior to rest of the treatments. Carbendazim+ Mancozeb @2.5g/kg (19.35 g), Carboxin + Thiram @2g/kg (17.74 g) and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (16.26 g) were same as in increasing shoot dry weight of chickpea. Least shoot dry weight 13.48 g/plant was recorded in control. **(Table-4.3 and fig.-6).**

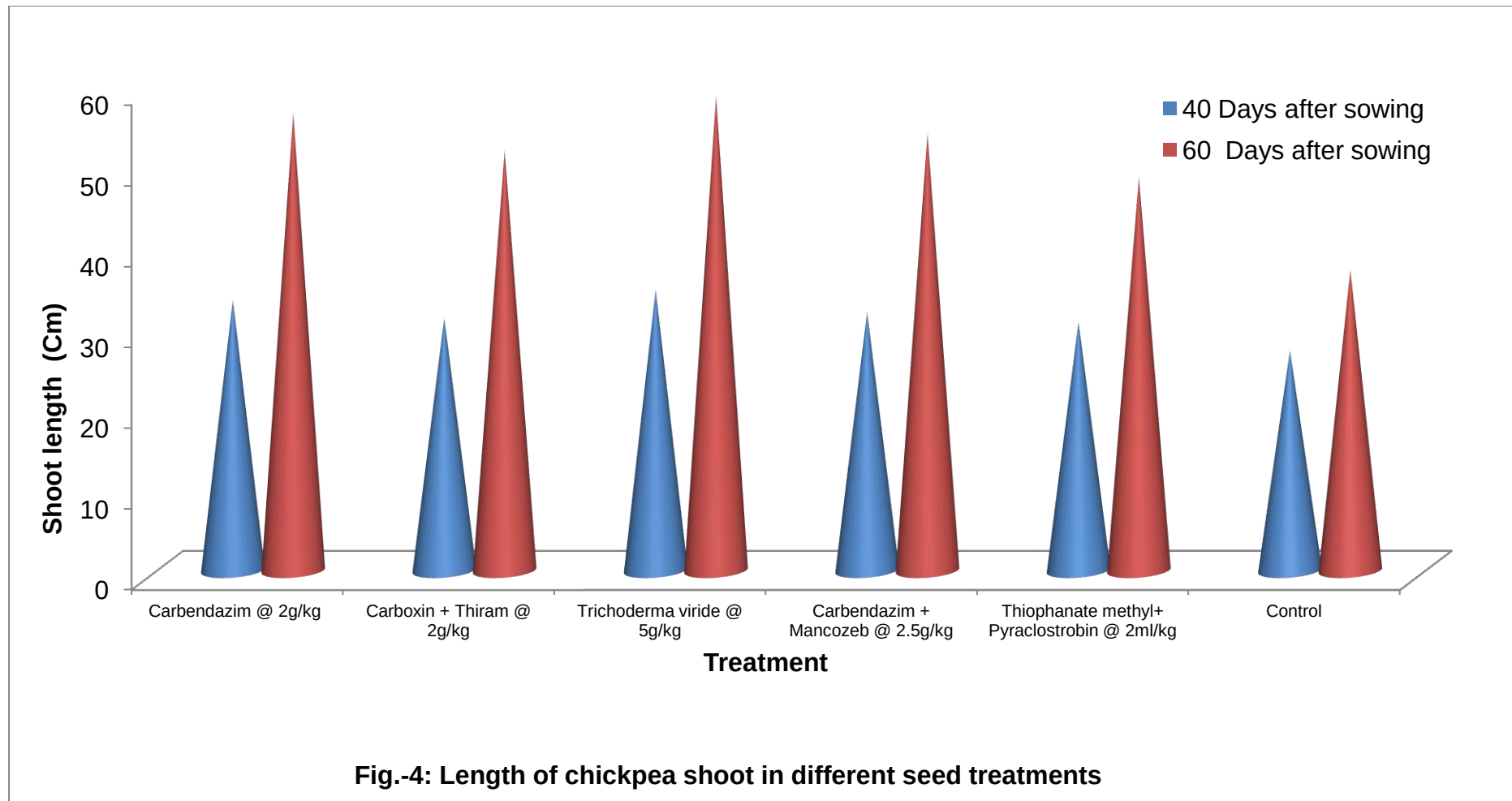
4.4.2 Effect on length, fresh and dry weight of chickpea root

Data presented in table-4.4 and fig-7 revealed that due to seed treatments root length increased significantly ranged from 10.20 cm to 13.23 cm against 7.75 cm in control at 40 days after sowing. Among the treatments, maximum root length/plant (13.23 cm) was obtained in *T. viride* @ 5g/kg at par with Carbendazim @ 2g/kg(12.50 cm), Carbendazim + Mancozeb @ 2.5g/kg (11.75 cm) and significantly superior to rest of the treatments. Carboxin + Thiram @ 2g/kg (11.13 cm) and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (10.20 cm) were equally effective in increasing root length of chickpea.

Table-4.4: Length, fresh and dry weight of chickpea root in different seed treatments

Treatment	Chickpea root					
	Length/plant (cm)*		Fresh weight/plant (g)		Dry weight/plant (g)	
	40 DAS	60 DAS	40 DAS	60 DAS	40 DAS	60 DAS
Carbendazim @ 2g/kg	12.50	19.63	13.63	24.55	5.86	12.65
Carboxin + Thiram @ 2g/kg	11.13	17.47	12.13	21.83	5.22	11.21
<i>Trichoderma viride</i> @ 5g/kg	13.23	20.76	14.42	25.96	6.21	13.38
Carbendazim + Mancozeb @ 2.5g/kg	11.75	18.45	12.81	23.06	5.54	11.91
Thiophanate methyl+ Pyraclostrobin @ 2 ml /kg	10.20	16.01	11.12	19.97	4.81	10.32
Control	7.75	10.85	7.91	14.25	2.64	7.27
SEm $\pm$	0.70	1.09	0.76	1.38	0.31	0.68
CD at 5%	1.50	2.32	1.62	2.94	0.67	1.46

*Average of four replications.



All the seed treatments increased root length significantly ranging from 16.01 to 20.76 cm against 10.85 cm in control at 60 days after sowing. Among the treatments, maximum root length/plant (20.76 cm) was obtained in *T. viride* @ 5g/kg at par with Carbendazim @ 2g/kg (19.63 cm), Carbendazim + Mancozeb @ 2.5g/kg (18.45 cm) and significantly superior to rest of the treatments. Root length of chickpea was at par in Carboxin + Thiram @ 2g/kg (17.47 cm) and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (16.01 cm).

Seed treatment with *T. viride* and fungicides significantly increased the fresh root weight of chickpea 40 days after sowing as compared to control. Among the treatments, maximum fresh root weight/plant (14.42 g) was recorded in *T. viride* @ 5g/kg at par with Carbendazim @ 2g/kg(13.63 g), Carbendazim + Mancozeb @ 2.5g/kg (12.81g) and significantly superior to rest of the treatments. Carboxin + Thiram @ 2g/kg (12.13 g) and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (11.12 g) were same as in increasing fresh root weight of chickpea. In control the fresh root weight was lowest 7.91 g.

Similarly, Seed treatment with *T. viride* and fungicides significantly increased the fresh root weight of chickpea 60 days after sowing as compared to control. Among the treatments, maximum fresh root weight/plant (25.96 g) was recorded in *T. viride* @ 5g/kg at par with Carbendazim @ 2g/kg(24.55 g), Carbendazim + Mancozeb @ 2.5g/kg (23.06 g) and significantly superior to rest of the treatments. Fresh root weight of chickpea was at par in Carboxin + Thiram @ 2g/kg (21.83 g) and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (19.97 g). In control the fresh root weight was lowest 14.25 g.

Seed treatment with *T. viride* and fungicides significantly increased the root dry weight of chickpea and were superior over control after 40 days of sowing. The results thus obtained are presented in table-4.4 and fig.-9 indicate that among the treatments, seed treated with *T. viride* @ 5g/kg gave maximum root dry weight/plant (6.21 g) at par with Carbendazim @ 2g/kg (5.86 g), Carbendazim + Mancozeb @ 2.5g/kg (5.54 g) and significantly superior to rest of the treatments. Carboxin + Thiram @ 2g/kg (5.22 g) and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (4.81 g) were same as in increasing dry root weight of chickpea. Lowest root dry weight 2.64 g was recorded in control.

Similarly, after 60 days of sowing, significant increase in root dry weight of chickpea was observed in all seed treatments as compared to control. Among the treatments, maximum root dry weight/plant (13.38 g) was obtained in *T. viride* @ 5g/kg at par

with Carbendazim @ 2g/kg (12.65 g), Carbendazim + Mancozeb @ 2.5g/kg (11.91 g) and significantly superior to rest of the treatments. Carboxin + Thiram @ 2g/kg (11.21 g) and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (10.32 g) were equally effective in increasing dry root weight of chickpea. Lowest root dry weight 7.27 g was recorded in control. **(Table-4.4).**

4.4.3 Number of nodules and nodule fresh and dry weight

The data on effect of seed treatment with *T. viride* and fungicides on number of nodules/plant, nodule fresh and dry weight/plant are presented in table-4.5 and fig.-10.

After 40 days of sowing, significant increase in number of nodules/plant was observed in all seed treatments as compared to control. Among the treatments, maximum number of nodules/plant (30.00) was obtained in *T. viride* @ 5g/kg at par with Carbendazim @ 2g/kg (28.50) and significantly superior to rest of the treatments. Number of nodules/plant was at par in Carbendazim + Mancozeb @ 2.5g/kg (27.00), Carboxin + Thiram @ 2g/kg (26.25) and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (25.50). In control least number of nodules/plant 14.13 was recorded.

Similarly, after 60 days of sowing, significant increase in number of nodules/plant was observed in all seed treatments as compared to control. Among the treatments, maximum number of nodules/plant (50.25) was obtained in *T. viride* @ 5g/kg at par with Carbendazim @ 2g/kg (48.75) and significantly superior to rest of the treatments. Carbendazim + Mancozeb @ 2.5g/kg (47.00), Carboxin + Thiram @ 2g/kg (46.25) and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (43.50) were same as in increasing number of nodules per plant in chickpea. In control least number of nodules/plant 19.25 was recorded. **(fig.-10).**

Table-4.5 and **fig.-11** shows that after 40 days of sowing, significant increase in nodule fresh weight was observed in all seed treatments as compared to control. Among the treatments, maximum nodule fresh weight/plant (0.039 g) was obtained in *T. viride* @ 5g/kg at par with Carbendazim @ 2g/kg (0.036g) and significantly superior to rest of the treatments. Carbendazim + Mancozeb @ 2.5g/kg (0.034 g), Carboxin + Thiram @ 2g/kg (0.031 g) and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (0.029 g) were equally effective in increasing nodule fresh weight of chickpea. Lowest nodule fresh weight 0.015 g was recorded in control.

Table-4.5: Number of nodules, nodule fresh and dry weight of chickpea in different seed treatments

Treatment	Chickpea Nodules					
	Number of nodules/plant*		Fresh weight/plant (g)		Dry weight/plant (g)	
	40 DAS	60 DAS	40 DAS	60 DAS	40 DAS	60 DAS
Carbendazim @ 2g/kg	28.50	48.75	0.036	0.145	0.012	0.050
Carboxin + Thiram @ 2g/kg	26.25	46.25	0.031	0.125	0.010	0.043
<i>Trichoderma viride</i> @ 5g/kg	30.00	50.25	0.039	0.158	0.013	0.054
Carbendazim + Mancozeb @ 2.5g/kg	27.00	47.00	0.034	0.136	0.011	0.046
Thiophanate methyl+ Pyraclostrobin @ 2ml/kg	25.50	43.50	0.029	0.119	0.009	0.039
Control	14.13	19.25	0.015	0.079	0.004	0.022
Sem±	1.15	1.51	0.002	0.008	0.0007	0.003
CD at 5%	2.44	3.22	0.004	0.019	0.0015	0.006

*Average of four replications.

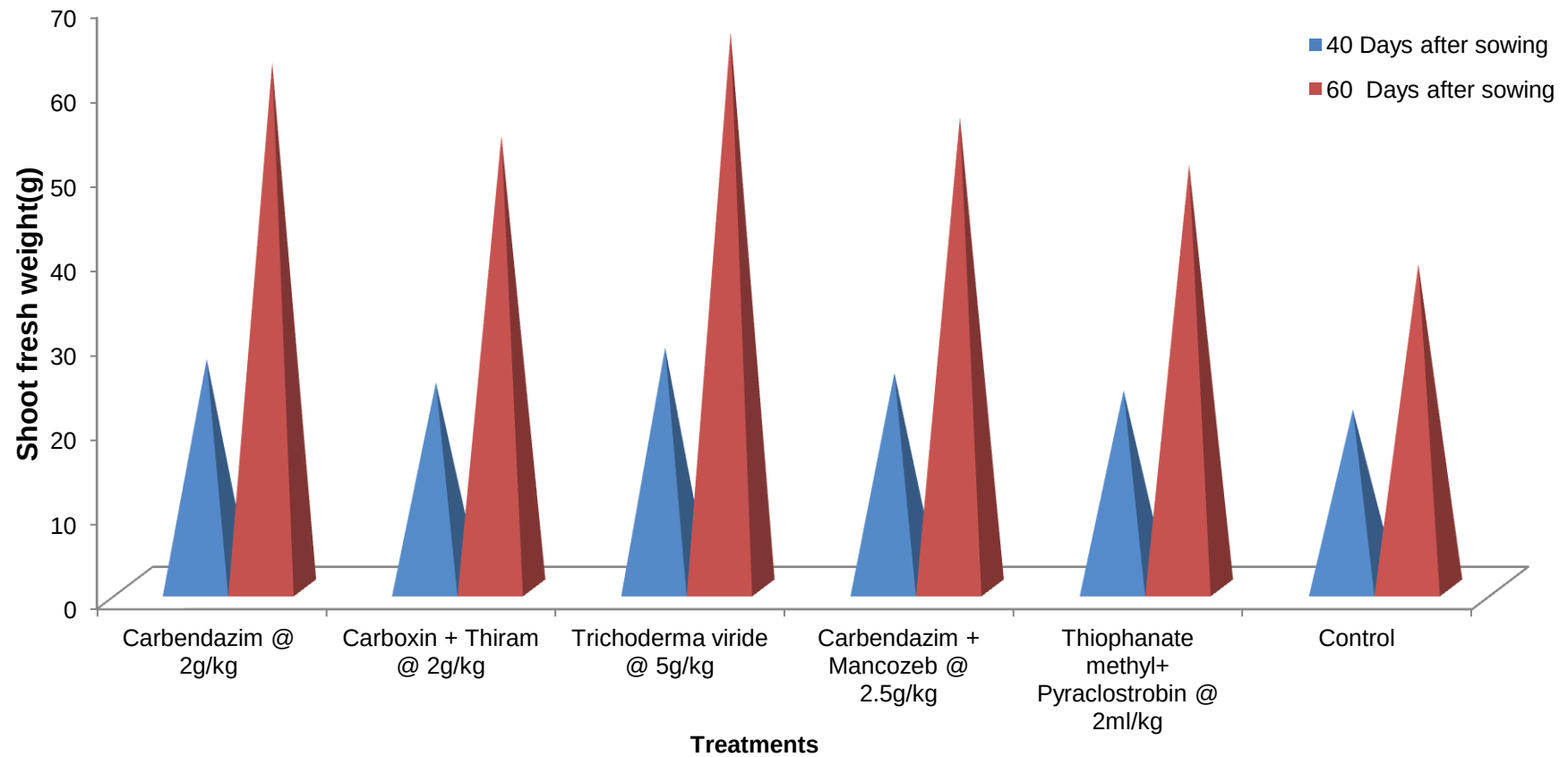


Fig.-5: Fresh weight of chickpea shoot in different seed treatments

Similarly, after 60 days of sowing, significant increase in nodule fresh weight was observed in all seed treatments as compared to control. Among the treatments, maximum nodule fresh weight/plant (0.158 g) was obtained in *T. viride* @ 5g/kg at par with Carbendazim @ 2g/kg (0.145 g) and significantly superior to rest of the treatments. Nodule fresh weight of chickpea was at par in Carbendazim+Mancozeb @ 2.5g/kg (0.136 g), Carboxin + Thiram @ 2g/kg (0.125 g) and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (0.119 g). Lowest nodule fresh weight 0.079 g was recorded in control.

The nodule dry weight varied among different treatments (Table-4.5 and **fig.-12**). After 40 days of sowing, significant increase in nodule dry weight was observed in all seed treatments as compared to control. Among the treatments, maximum nodule dry weight/plant (0.013 g) was obtained in *T. viride* @ 5g/kg at par with Carbendazim @ 2g/kg (0.012 g) and significantly superior to rest of the treatments. Carbendazim + Mancozeb @ 2.5g/kg (0.011 g), Carboxin + Thiram @ 2g/kg (0.010 g) and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (0.009 g) were same as in increasing nodule dry weight of chickpea. In control least nodule dry weight/plant 0.004 g was recorded. Similarly, after 60 days of sowing, significant increase in nodule dry weight was observed in all seed treatments as compared to control. Among the treatments, maximum nodule dry weight/plant (0.054 g) was obtained in *T. viride* @ 5g/kg at par with Carbendazim @ 2g/kg (0.050g) and significantly superior to rest of the treatments. Carbendazim + Mancozeb @ 2.5g/kg (0.046 g), Carboxin + Thiram @ 2g/kg (0.043 g) and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (0.039 g) were equally effective in increasing nodule dry weight of chickpea. In control least nodule dry weight/plant 0.022 g was recorded.

4.4.4 Post harvest observations

(i) Number of pods per plant

Data (Table- 4.6 and Fig.-13) revealed that seed dressing with *T. viride* and fungicides significantly increased **number of pods/plant** of chickpea. **The average number of pods/plant ranged between 29.58 to 46.18. All the seed treatments significantly increased the number of pods/plant as compared to control. Among the treatments, maximum average number of pods/plant (46.18) were recorded in *T. viride* @ 5g/kg at par with Carbendazim @ 2g/kg (43.91) and significantly superior to rest of the treatments. Number of pods/plant was at par in Carbendazim + Mancozeb @ 2.5g/kg (40.50), Carboxin + Thiram @ 2g/kg (37.99)**

and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (35.49). Lowest number of pods/plant 29.58 was recorded in control.

(ii) Number of seeds per plant

It is evident from data in table-4.6 and fig-13 that seed treatment with *T.viride* and fungicides significantly increased number of seeds/plant as compared to control. Among the treatments, significantly maximum number of seeds/plant (43.41) was found in *T. viride* @ 5g/kg at par with Carbendazim @2g/kg (41.27) and significantly superior to rest of the treatments .Carbendazim + Mancozeb @ 2.5g/kg (38.07), Carboxin + Thiram @ 2g/kg (35.71) and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (33.36) were equally effective in increasing number of seeds/plant of chickpea. Lowest number of seeds/plant 27.21 was recorded in control.

(iii) Yield (kg/ha)

The yield (kg/ha) varied among different treatments (Table-4.6 and Fig.-14). Seed treatment with *T. viride* and fungicides significantly increased yield (kg/ha) as compared to control. Among the treatments, significantly maximum yield (kg/ha) (1914.94 kg) was noted in *T. viride* @ 5g/kg at par with Carbendazim @ 2g/kg (1820.61 kg) and significantly superior to rest of the treatments. Carbendazim + Mancozeb @ 2.5g/kg (1679.11 kg), Carboxin + Thiram @ 2g/kg (1575.35 kg) and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (1471.58 kg) were same as in increasing yield(kg/ha) of chickpea. In control least yield (kg/ha) 845.66 was recorded.

Table-4.6: Post harvest observations of chickpea in different seed treatments under field condition

Treatment	Number of Pods/ plant*	Number of Seeds/ plant*	Yield (kg/ha)*
Carbendazim @ 2g/kg	43.91	41.27	1820.61
Carboxin + Thiram @ 2g/kg	37.99	35.71	1575.35
<i>Trichoderma viride</i> @ 5g/kg	46.18	43.41	1914.94
Carbendazim +Mancozeb @ 2.5 g/kg	40.50	38.07	1679.11
Thiophanate methyl+ Pyraclostrobin @ 2ml /kg	35.49	33.36	1471.58
Control	29.58	27.21	845.66
Sem $\pm$	1.17	1.10	47.64
CD at 5%	2.49	2.34	101.52

*Average of four replications.

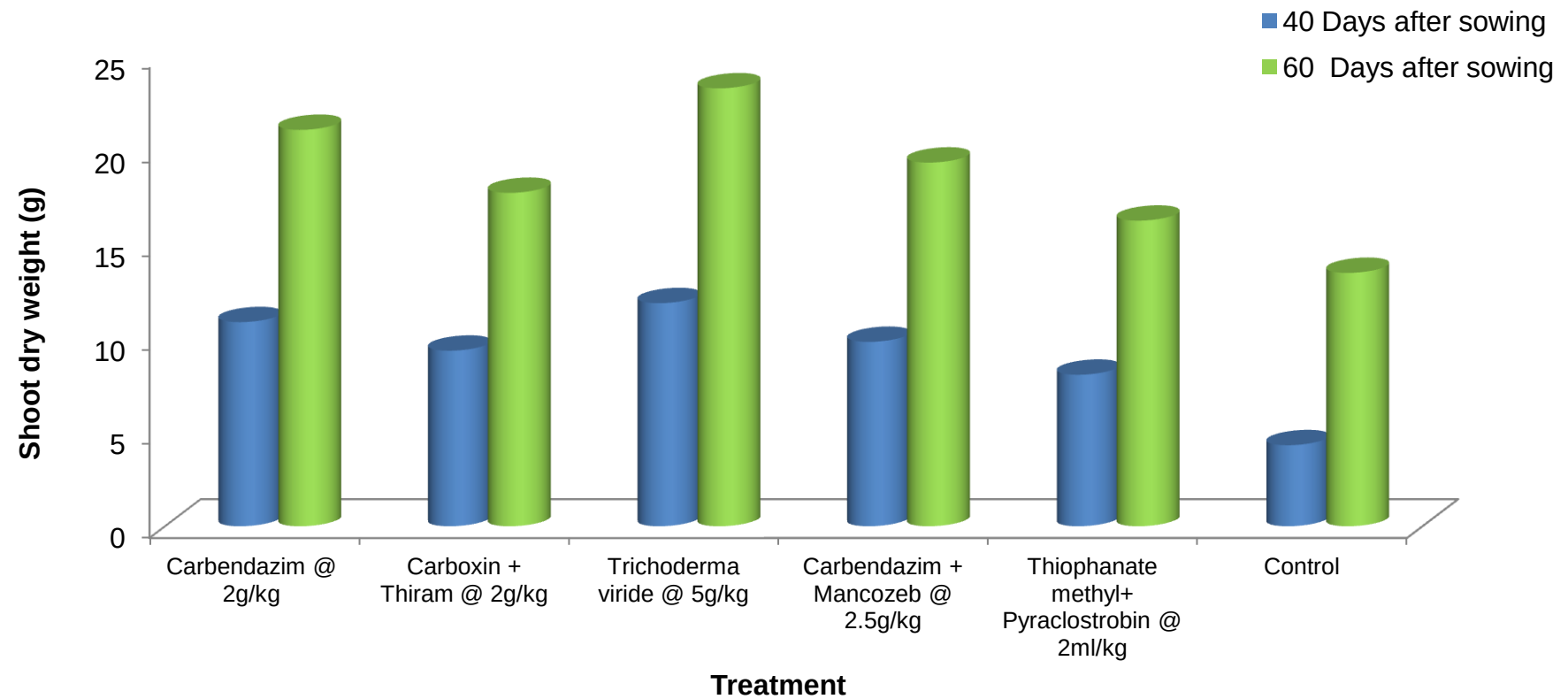


Fig.-6: Shoot dry weight of chickpea in different seed treatments

4.4.5 Evaluation of biological agent and chemicals for management of dry root rot of chickpea

The results pertaining to management of dry root rot of chickpea conducted during the year 2017-18 *rabi* season are presented in the table-4.7 and fig.-15. Incidence of dry root rot caused by *R. bataticola* was significantly minimum in all the treatments as compared to control.

Upon field trials, seed treatment with *T. viride* @ 5g/kg(5.91%) was found to be best treatment for the control of dry root rot of chickpea at par with Carbendazim @ 2g/kg (6.38%), Carbendazim + Mancozeb @ 2.5g/kg (7.03%) and significantly superior to rest of the treatments. Carboxin + Thiram @ 2g/kg (10.27%) and Thiophanate methyl + Pyraclostrobin @ 2ml/kg (11.19%) were equally effective for the control of dry root rot of chickpea. In control, where no treatment was done, the incidence of root rot was significantly maximum (33.57%) .

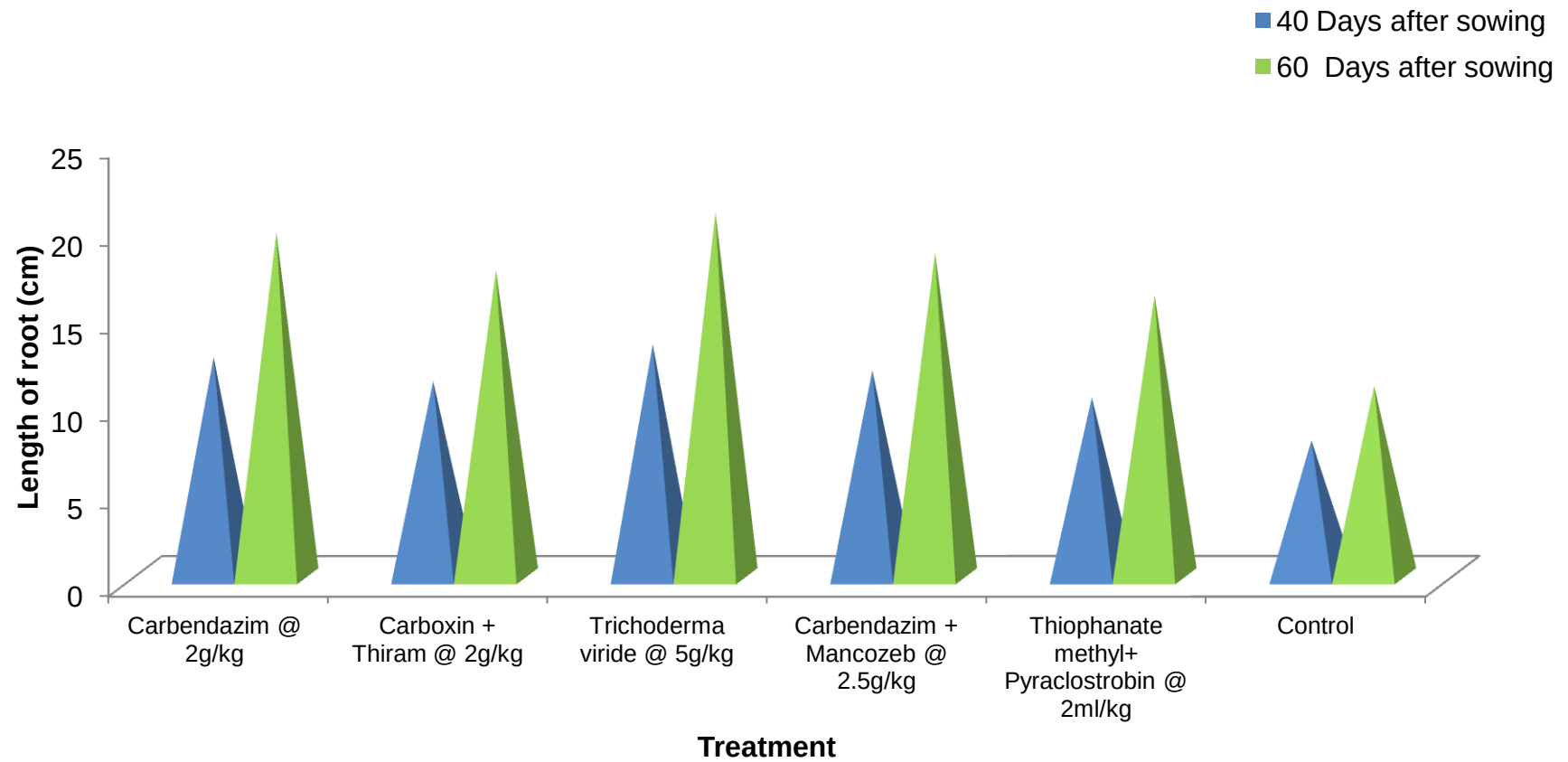
Table-4.7: Incidence of dry root rot of chickpea in different seed treatments

Treatment	DRR incidence (percentage)*
Carbendazim @ 2g/kg	6.38 (14.61)
Carboxin + Thiram @ 2g/kg	10.27 (18.67)
<i>Trichoderma viride</i> @ 5g/kg	5.91 (14.05)
Carbendazim + Mancozeb @ 2.5g/kg	7.03 (15.35)
Thiophanate methyl+ Pyraclostrobin @ 2ml/kg	11.19 (19.53)
Control	33.57 (35.38)
Sem±	0.72
CD at 5%	1.53

*mean of four replications

** Data in parentheses are arcsine transformed values

Fig.-7: Length of root of chickpea in different seed treatments



4.5 Evaluation of fungicides against *R. bataticola* *in vitro*

Seven fungicides including new one in different doses were evaluated in laboratory for their toxicity against *R. bataticola* by poison food technique.

Table-4.8: Radial growth of *R. bataticola* in fungicides amended medium

Fungicides	Mycelial radial growth (mm)*
Fosetyl Al @ 0.20%	16.67
Fosetyl Al @ 0.25%	0.00
Fosetyl Al @ 0.30%	0.00
Carbendazim 50% WP	0.00
Mancozeb 50% WS +carbendazim 25%	0.00
Thiram 37.5% WS+carboxin 37.5%	0.00
Thiophanate methyl 45%+ Pyraclostrobin 5%	0.00
Control	89.17
S.E_m±	0.45
C.D. at 5%	0.96

* Mean of four replications.

In vitro evaluation of fungicides and their readymade combi products was carried out by poison food technique. Observations on radial growth of *R. bataticola* were recorded after seven days of inoculation (Plate-2). Data in table-4.8 showed that all fungicides completely inhibited mycelial growth of *R. bataticola* except Fosetyl Al @ 0.20% (16.67 mm).

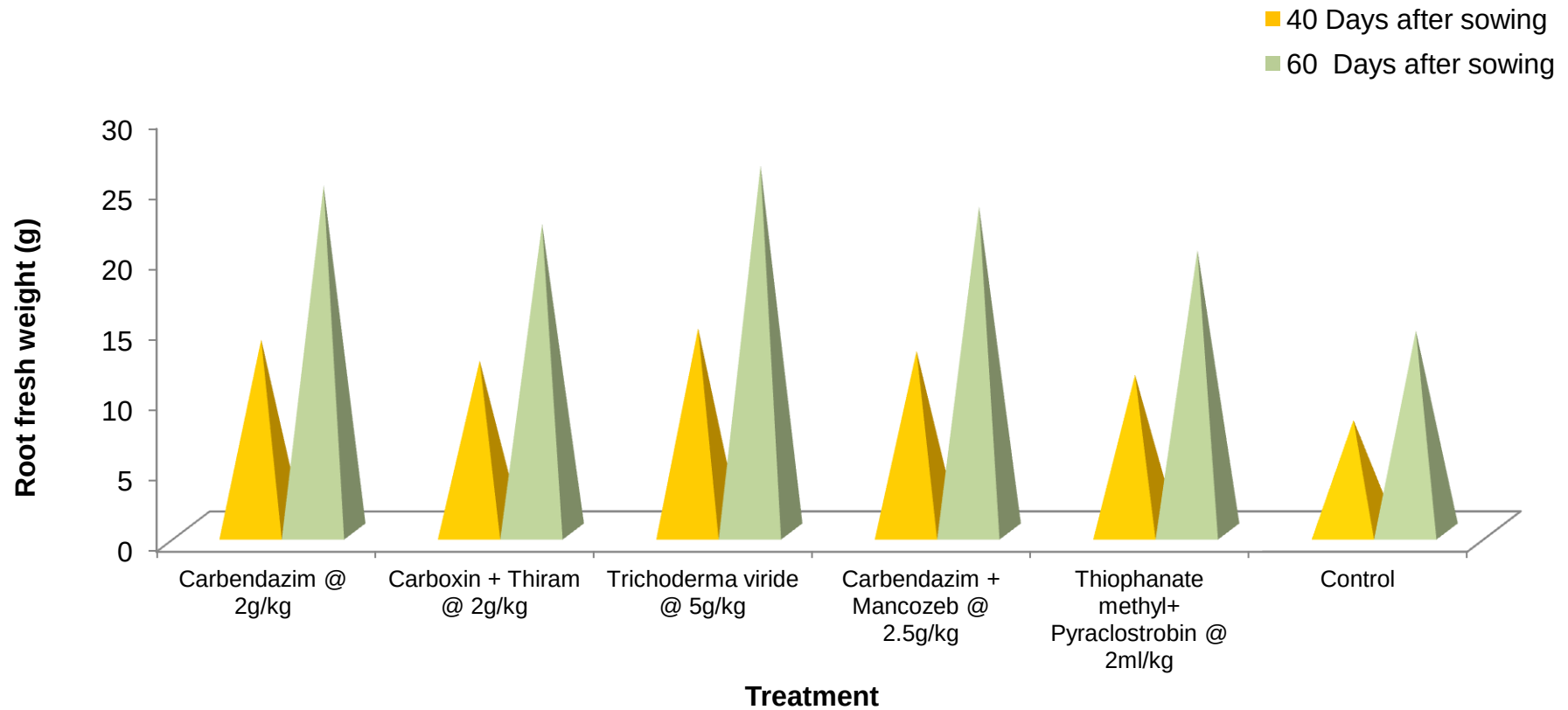


Fig.-8: Root fresh weight of chickpea in different seed treatments

4.6 Evaluation of drenching of fungicides and their doses for management of dry root rot of chickpea

The results pertaining to management of dry root rot of chickpea conducted during the year 2017-18 *rabi* season are presented in the table-4.9 and fig.-16. Incidence of dry root rot caused by *R. bataticola* was significantly minimum in all the treatments as compared to control.

Table-4.9: Incidence of dry root rot of chickpea in different treatments with drenching of fungicides and their doses

Treatment	DRR incidence (percentage)*
Fosetyl Al 80% WP a.i. 1600 @ 2000 g/ha	12.35 (20.55)
Fosetyl Al 80% WP a.i. 2000 @ 2500 g/ha	10.69 (19.06)
Fosetyl Al 80% WP a.i. 2400 @ 3000 g/ha	7.95 (16.37)
Fosetyl Al 80% WP a.i. 4800 @ 6000 g/ha	6.15 (14.35)
Carbendazim 50% WP @ 1000 g/ha	5.57 (13.63)
Control	32.27 (34.59)
Sem $\pm$	0.74
CD at 5%	1.57

*mean of four replications

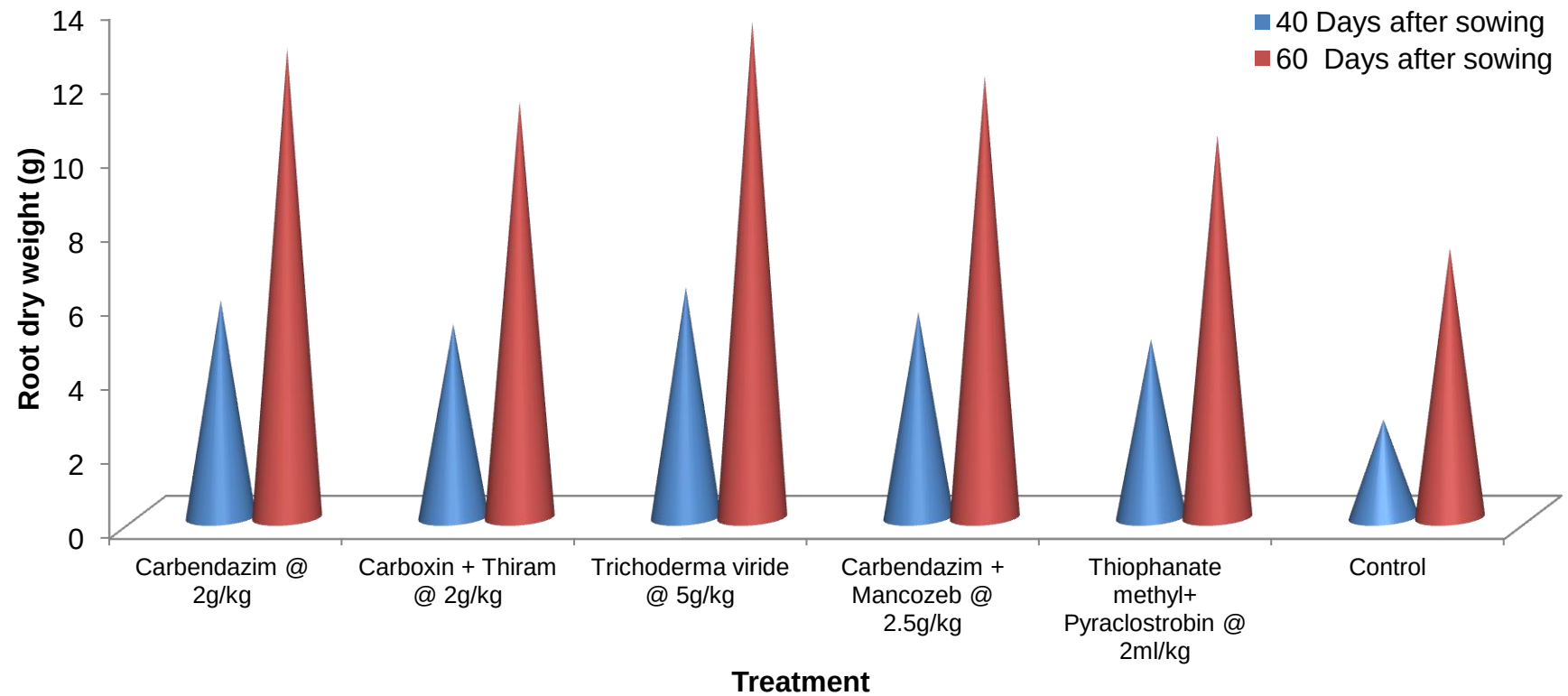


Fig.-9: Root dry weight of chickpea in different seed treatments

** Data in parentheses are arcsine transformed values

In field experiments, drenching with Carbendazim 50% WP (5.57%) at 30 and 45 days after sowing was the best treatment for the control of dry root rot of chickpea and it was at par with drenching with Fosetyl AI 80% WP a.i. 4800 (6.15%). Fosetyl AI 80% WP a.i. 2400, Fosetyl AI 80% WP a.i. 2000 and Fosetyl AI 80% WP a.i. 1600 (7.95%, 10.69% and 12.35%, respectively) were equally effective for the control of dry root rot of chickpea. In control, where no treatment was done, the incidence of root rot was significantly maximum (32.27%).

4.7 Effect of drenching of fungicides and their doses on dry root rot incidence

4.7.1 Effect on length of shoot, fresh and dry weight of chickpea shoot

The data presented in table-4.10 shows the effect of drenching with fungicides on shoot length, fresh weight and shoot dry weight of chickpea.

Table-4.10: Length of shoot, fresh and dry weight of chickpea shoot in drenching of different fungicides and their doses

Treatment	Chickpea shoot					
	Shoot length/plant (cm)		Fresh weight/plant (g)		Dry weight/plant (g)	
	40 DAS	60 DAS	40 DAS	60 DAS	40 DAS	60 DAS
Fosetyl AI 80% WP a.i. 1600 @ 2000 g/ha	30.68	51.96	23.86	51.34	8.57	17.06
Fosetyl AI 80% WP a.i. 2000 @ 2500 g/ha	32.72	54.65	24.50	53.89	8.93	17.21
Fosetyl AI 80% WP a.i. 2400 @ 3000 g/ha	32.74	55.00	25.58	55.81	9.23	18.21
Fosetyl AI 80% WP a.i. 4800 @ 6000 g/ha	33.29	56.26	26.03	59.73	10.55	20.50
Carbendazim 50% WP @ 1000 g/ha	33.94	57.08	27.03	62.51	11.38	22.34
Control	26.96	36.80	20.63	35.73	4.21	13.18
Sem±	1.41	3.31	1.14	3.39	0.57	1.15
CD at 5%	3.01	7.05	2.42	7.22	1.21	2.45

*Average of four replication

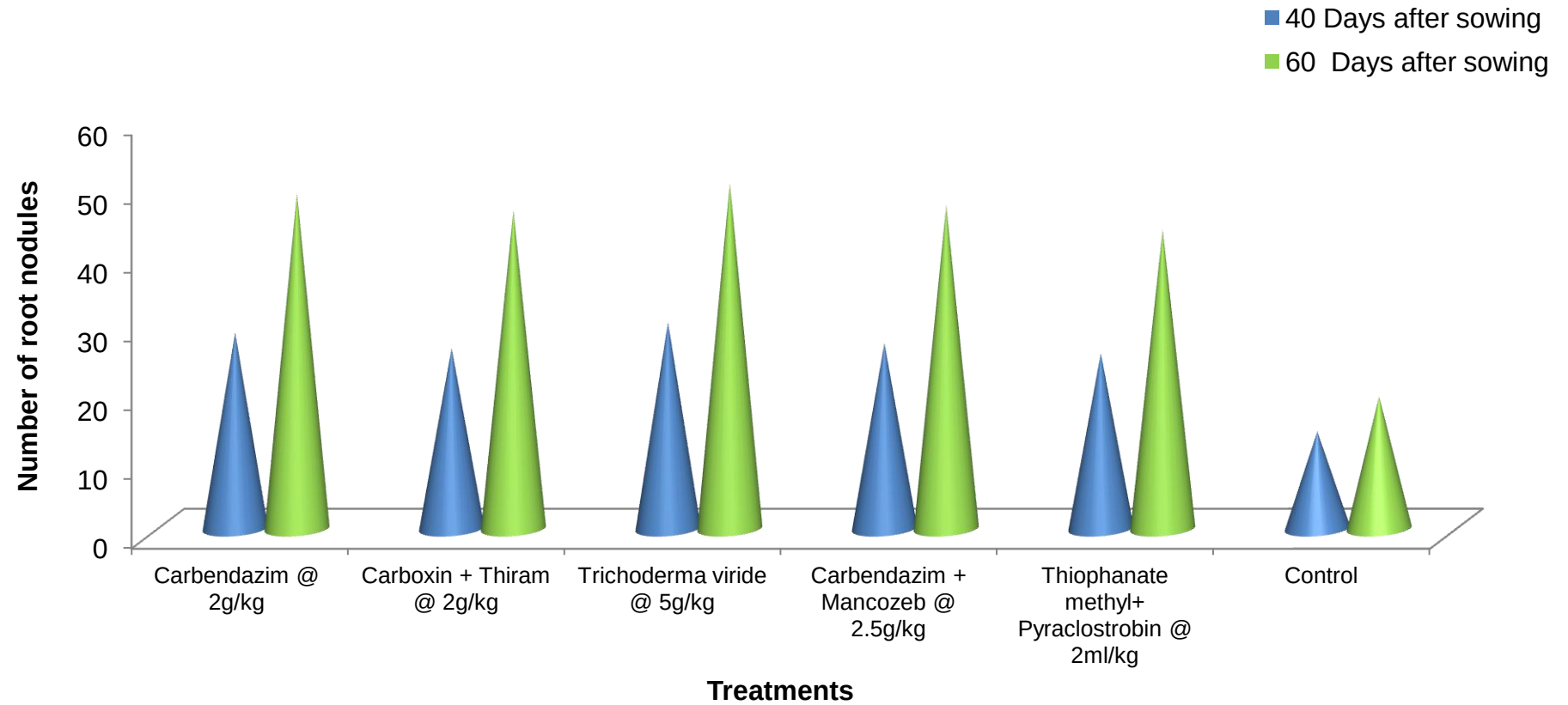


Fig.-10: Number of nodules of chickpea in different seed treatments

The data presented in table-4.10 and fig.-17 indicate that after 40 days of sowing significant increase in shoot length/plant was observed in drenching with fungicides as compared to control. Among the treatments, maximum shoot length/plant (33.94 cm) was obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 (33.29 cm), Fosetyl Al 80% WP a.i. 2400 (32.74 cm), Fosetyl Al 80% WP a.i. 2000 (32.72 cm) but it was significantly superior to Fosetyl Al 80% WP a.i. 1600 (30.68 cm) . In control least shoot length/plant 26.96 cm was recorded.

Similarly, after 60 days of sowing, all the drenched fungicides significantly increased shoot length/plant and hence showed their superiority over control. Among the treatments, Carbendazim 50% WP recorded maximum shoot length/plant (57.08 cm) at par with Fosetyl Al 80% WP a.i. 4800 (56.26 cm), Fosetyl Al 80% WP a.i. 2400 (55.00 cm), Fosetyl Al 80% WP a.i. 2000 (54.65 cm) Fosetyl Al 80% WP a.i. 1600 (51.96 cm). In control least shoot length/plant 36.80 cm was recorded.

Data in the table-4.10 and fig.-18 shows that after 40 days of sowing, all the drenching with fungicides significantly increased fresh shoot weight of chickpea and were significantly superior to control. **Among the treatments,** Carbendazim 50% WP **significantly recorded maximum fresh shoot weight/plant (27.03 g)** at par with Fosetyl Al 80% WP a.i. 4800 (26.03 g) and Fosetyl Al 80% WP a.i. 2400 (25.58 g) and significantly superior to rest of the treatments. Fosetyl Al 80% WP a.i. 2000 (24.50 g) and Fosetyl Al 80% WP a.i. 1600 (23.86 g) were equally effective in increasing fresh shoot weight/plant of chickpea. In control least fresh shoot weight/plant 20.63 g was recorded. **Similarly, after 60 days of sowing,** all the drenching with fungicides significantly enhanced fresh shoot weight of chickpea and were superior over control. **Among the treatments, significantly maximum shoot fresh weight/plant (62.51 g) was observed in** Carbendazim 50% WP **at par with** Fosetyl Al 80% WP a.i. 4800 (59.73 g) and Fosetyl Al 80% WP a.i. 2400 (55.81 g) and significantly superior to rest of the treatments. Fosetyl Al 80% WP a.i. 2000 (53.89 g) and Fosetyl Al 80% WP a.i. 1600 (51.34 g) were same as in increasing fresh shoot weight of chickpea. In control least fresh shoot weight/plant 35.73 g was recorded.

Drenching with Carbendazim and Fosetyl in different doses significantly increased the shoot dry weight of chickpea as compared to control after 40 days of sowing. Among the treatments ,significantly maximum shoot dry weight/plant (11.38 g) was obtained in Carbendazim 50% WP **at par with** Fosetyl Al 80% WP a.i. 4800 **(10.55 g)** and significantly superior to rest of the treatments.

Dry root weight of chickpea was at par in Fosetyl Al 80% WP a.i. 2400 (9.23 g), Fosetyl Al 80% WP a.i. 2000 (8.93 g) and Fosetyl Al 80% WP a.i. 1600 (8.57 g). Lowest **shoot dry weight/plant 4.21 g was recorded in control.**

Similarly, after 60 days of sowing, **drenching with Carbendazim and Fosetyl in different doses significantly increased the shoot dry weight of chickpea as compared to control. Among the treatments, significantly maximum shoot dry weight/plant (22.34 g) was obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 (20.50g)** and significantly superior to rest of the treatments. Dry root weight of chickpea was at par in Fosetyl Al 80% WP a.i. 2400 (18.21 g), Fosetyl Al 80% WP a.i. 2000 (17.21 g) and Fosetyl Al 80% WP a.i. 1600 (17.06 g). Lowest **shoot dry weight/plant 13.18 g was recorded in control. (table-4.10 and fig.-19).**

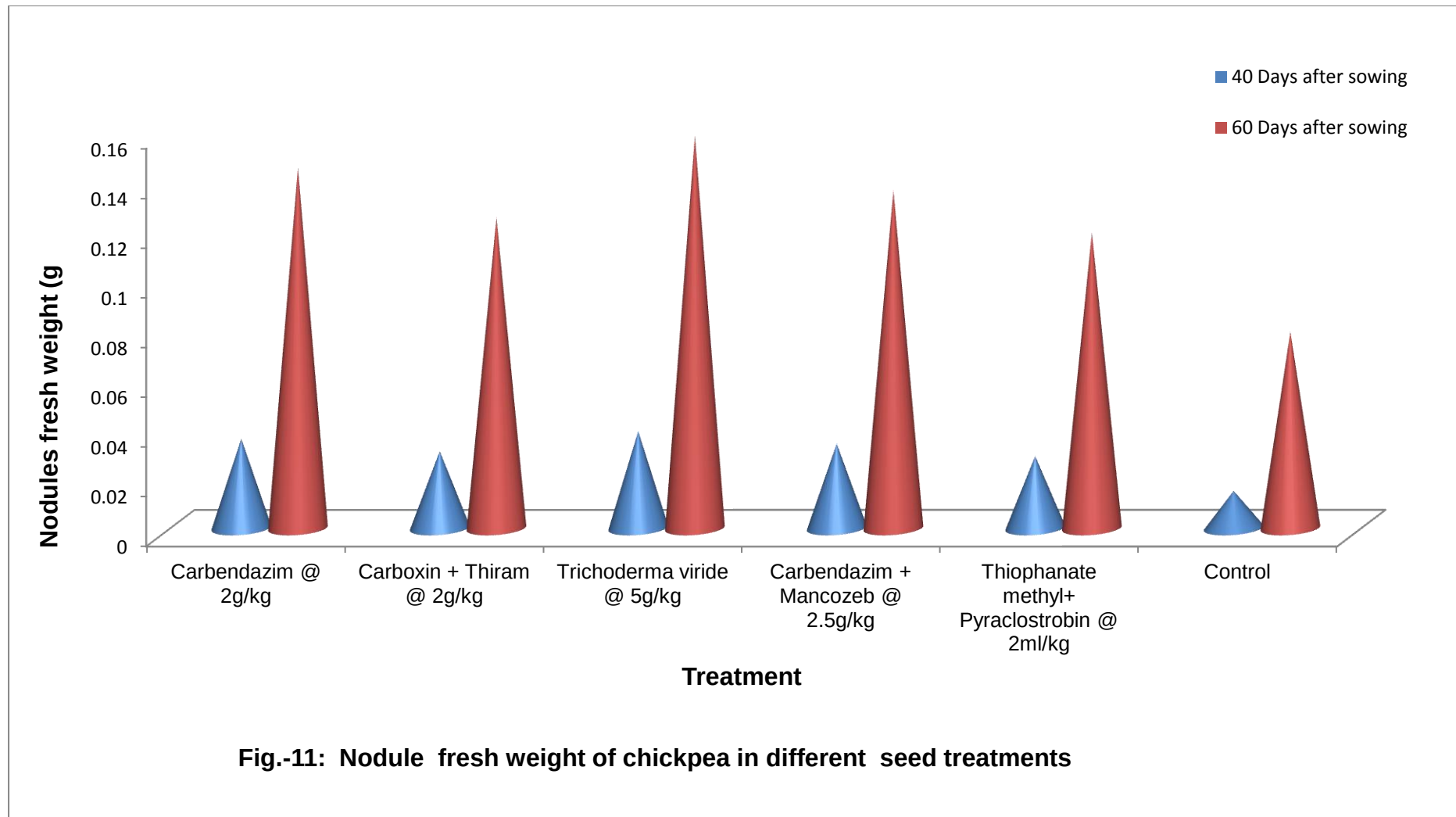
4.7.2 Effect on length of root, fresh and dry weight of chickpea root

It is obvious from the data presented in table-4.11 and fig-20 that drenching with all the fungicides increased root length significantly in chickpea ranging from 9.90 cm to 13.14 cm against 9.06 cm in control at 40 days after sowing. Among the treatments, maximum root length/plant (13.14 cm) was obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 (12.56 cm) and significantly superior to rest of the treatments. Fosetyl Al 80% WP a.i. 2400 (11.31 cm), Fosetyl Al 80% WP a.i. 2000 (10.76 cm) and Fosetyl Al 80% WP a.i. 1600 (9.90 cm) were at par in increasing root length/plant of chickpea. **Similarly** after 60 days of sowing, drenching with all the fungicides increased root length significantly ranging from 15.53 cm to 20.63 cm against 12.68 cm in control. Among the treatments, maximum root length (20.63 cm) was obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 (19.71 cm) and significantly superior to rest of the treatments. Root length of chickpea was at par in Fosetyl Al 80% WP a.i. 2400 (17.75 cm), Fosetyl Al 80% WP a.i. 2000 (16.90 cm) and Fosetyl Al 80% WP a.i. 1600 (15.53 cm) at 60 days after sowing.

Table-4.11: Length of root, fresh and dry weight of chickpea root in drenching with different fungicides and their doses

Treatment	Chickpea root					
	Root length/plant (cm)		Fresh weight/plant (g)		Dry weight/plant (g)	
	40 DAS	60 DAS	40 DAS	60 DAS	40 DAS	60 DAS
Fosetyl Al 80% WP a.i. 1600 @ 2000 g/ha	9.90	15.53	10.79	15.92	4.67	11.10
Fosetyl Al 80% WP a.i. 2000 @ 2500 g/ha	10.76	16.90	11.73	17.24	5.05	11.98
Fosetyl Al 80% WP a.i. 2400 @ 3000 g/ha	11.31	17.75	12.33	18.21	5.32	12.57
Fosetyl Al 80% WP a.i. 4800 @ 6000 g/ha	12.56	19.71	13.68	20.28	5.89	13.52
Carbendazim 50% WP @ 1000 g/ha	13.14	20.63	14.32	21.84	6.17	14.30
Control	9.06	12.68	8.90	13.14	3.08	7.85
Sem±	0.71	1.09	0.76	1.13	0.32	0.75
CD at 5%	1.50	2.33	1.62	2.41	0.68	1.60

*Average of four replications.



Data in the table-4.11 and fig.-21 reveals that drenching with all the fungicides increased fresh root weight significantly ranging from 10.79 g to 14.32 g against 8.90 g in control at 40 days after sowing. Among the treatments, maximum fresh root weight/plant (14.32 g) was obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 (13.68g) and significantly superior to rest of the treatments. Fosetyl Al 80% WP a.i. 2400 (12.33 g), Fosetyl Al 80% WP a.i. 2000 (11.73 g) and Fosetyl Al 80% WP a.i. 1600 (10.79 g) were same as in increasing fresh root weight of chickpea after 40 days of sowing. **Similarly,** after 60 days of sowing, drenching with all the fungicides increased fresh root weight significantly ranging from 15.92 g to 21.84 g against 13.14 g in control. Among the treatments, maximum fresh root weight/plant (21.84g) was obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 (20.28 g) and significantly superior to rest of the treatments. Fresh root weight of chickpea was at par in Fosetyl Al 80% WP a.i. 2400 (18.21 g), Fosetyl Al 80% WP a.i. 2000 (17.24 g) and Fosetyl Al 80% WP a.i. 1600 (15.92 g) after 60 days of sowing.

Drenching with all the fungicides significantly increased the root dry weight of chickpea. The results thus obtained are presented in table-4.11 and fig.-22 indicate that drenching with all the fungicides increased dry root weight significantly ranging from 4.67 g to 6.17 g against 3.08 g in control at 40 days after sowing. Among the treatments, maximum dry root weight/plant (6.17 g) was obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 (5.89g) and significantly superior to rest of the treatments. Fosetyl Al 80% WP a.i. 2400 (5.32 g), Fosetyl Al 80% WP a.i. 2000 (5.05 g) and Fosetyl Al 80% WP a.i. 1600 (4.67 g) were equally effective in increasing dry root weight of chickpea after 40 days of sowing. **Similarly** after 60 days of sowing, drenching with all the fungicides increased dry root weight significantly ranging from 11.10 g to 14.30 g against 7.85 g in control. Among the treatments, maximum dry root weight/plant (14.30 g) was obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 (13.52 g) and significantly superior to rest of the treatments. Dry root weight of chickpea was at par in Fosetyl Al 80% WP a.i. 2400 (12.57 g), Fosetyl Al 80% WP a.i. 2000 (11.98 g) and Fosetyl Al 80% WP a.i. 1600 (11.10 g) after 60 days of sowing.

4.6.3 Number of nodules and nodule fresh and dry weight

The data on effect of drenching with fungicides on number of nodules/plant, nodule fresh and dry weight/ plant are presented in table-4.12.

It is clear from the data presented in table-4.12 and fig.-24 that drenching with all the fungicides increased number of nodules/plant significantly as compared to control at 40 days after sowing. Among the treatments, maximum number of nodules/plant (25.92) was obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 (24.63) and Fosetyl Al 80% WP a.i. 2400 (23.09) and significantly superior to rest of the treatments. Fosetyl Al 80% WP a.i. 2000 (22.58) and Fosetyl Al 80% WP a.i. 1600 (22.24) were same as in increasing number of nodules/plant in chickpea after 40 days of sowing. In control least number of nodules/plant 15.93 was recorded. **Similarly** after 60 days of sowing, drenching with all the fungicides increased number of nodules/plant significantly as compared to control. Among the treatments, maximum number of nodules/plant (43.42) was obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 (42.12), Fosetyl Al 80% WP a.i. 2400 (40.19) , Fosetyl Al 80% WP a.i. 2000 (39.78) but it was significantly superior to Fosetyl Al 80% WP a.i.1600 (37.93). In control least number of nodules/plant 21.48 was recorded.

Table-4.12 and fig.-25 shows that drenching with all the fungicides increased nodule fresh weight significantly as compared to control at 40 days after sowing. Among the treatments, maximum nodule fresh weight/plant (34.13 g) was obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 (31.32 g) and significantly superior to rest of the treatments. Nodule fresh weight/plant of chickpea was at par in Fosetyl Al 80% WP a.i. 2400 (29.08 g), Fosetyl Al 80% WP a.i. 2000 (26.88 g) and Fosetyl Al 80% WP a.i. 1600 (25.94 g). In control least nodule fresh weight/plant 17.91 g was recorded after 40 days of sowing. **Similarly** after 60 days of sowing, drenching with all the fungicides increased nodule fresh weight significantly as compared to control. Among the treatments, maximum nodule fresh weight/plant (136.51 g) was obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 (124.86 g) and significantly superior to rest of the treatments. Fosetyl Al 80% WP a.i. 2400 (118.59 g), Fosetyl Al 80% WP a.i. 2000 (107.50 g) and Fosetyl Al 80% WP a.i. 1600 (102.82 g) were equally effective in increasing nodule fresh weight/plant of chickpea. In control least nodule fresh weight/plant 87.63 g was recorded after 60 days of sowing.

Table-4.12: Number of nodules and nodule fresh and dry weight of chickpea in different drenching of fungicides and their doses

Treatment	Chickpea nodule					
	Number of nodules/plant*		Fresh weight/plant (g)		Dry weight/plant (g)	
	40 DAS	60 DAS	40 DAS	60 DAS	40 DAS	60 DAS
Fosetyl AI 80% WP a.i. 1600 @ 2000 g/ha	22.24	37.93	0.025	0.102	0.008	0.034
Fosetyl AI 80% WP a.i. 2000 @ 2500 g/ha	22.58	39.78	0.026	0.107	0.009	0.037
Fosetyl AI 80% WP a.i. 2400 @ 3000 g/ha	23.09	40.19	0.029	0.118	0.010	0.040
Fosetyl AI 80% WP a.i. 4800 @ 6000 g/ha	24.63	42.12	0.031	0.124	0.010	0.043
Carbendazim a.i. 500 @ 1000 g/ha	25.92	43.42	0.034	0.136	0.011	0.047
Control	15.93	21.48	0.017	0.087	0.004	0.024
Sem $\pm$	1.42	2.40	0.001	0.007	0.0005	0.002
CD at 5%	3.04	5.11	0.003	0.015	0.0001	0.005

*Average of four replications.

The nodule dry weight varied among different treatments (table-4.12 and fig.-26). Drenching with all the fungicides increased nodule dry weight significantly as compared to control at 40 days after sowing. Among the treatments, maximum nodule dry weight/plant (0.011 g) was obtained in Carbendazim 50% WP at par with Fosetyl AI 80% WP a.i. 4800 (0.010 g) and significantly superior to rest of the treatments. Nodule dry weight/plant of chickpea was at par in Fosetyl AI 80% WP a.i. 2400 (0.010 g), Fosetyl AI 80% WP a.i. 2000 (0.009 g) and Fosetyl AI 80% WP a.i. 1600 (0.008 g). In control least nodule dry weight/plant 0.004 g was recorded after 40 days of sowing. **Similarly** after 60 days of sowing, drenching with all the fungicides increased nodule dry weight significantly as compared to control. Among the treatments, maximum nodule dry weight/plant (0.047 g) was obtained in Carbendazim 50% WP at par with Fosetyl AI 80% WP a.i. 4800 (0.043 g) and significantly superior to rest of the treatments. Nodule dry weight/plant of chickpea was equally effective in Fosetyl AI 80% WP a.i. 2400 (0.040 g), Fosetyl AI 80% WP a.i. 2000 (0.037 g) and Fosetyl AI 80% WP a.i. 1600 (0.034 g). In control least nodule dry weight/plant 0.024 g was recorded after 60 days of sowing.

4.6.4 Post harvest observations

(i) Number of pods/plant

Data in table-4.13 and fig.-27 revealed that drenching with all the fungicides increased **number of pods/plant** of chickpea significantly **as compared to control**. Among the treatments, significantly, maximum average numbers of pods/plant (**44.02**) were recorded in Carbendazim 50% WP at par with Fosetyl AI 80% WP a.i. 4800 (**42.05**) and significantly superior to rest of the treatments. Fosetyl AI 80% WP a.i. 2400 (37.96), Fosetyl AI 80% WP a.i. 2000 (36.64) and Fosetyl AI 80% WP a.i. 1600 (34.46) were at par in increasing number of pods/plant of chickpea. In control least number of pods/plant 28.04 was recorded.

Table-4.13: Post harvest observations of chickpea in different treatments with drenching of fungicides and their doses under field condition

Treatment	Number of pods/plant*	Number of seeds/plant*	Yield(kg/ha)*
Fosetyl AI 80% WP a.i. 1600 @ 2000 g/ha	34.46	32.32	1250.86
Fosetyl AI 80% WP a.i. 2000 @ 2500 g/ha	36.64	34.24	1339.06
Fosetyl AI 80% WP a.i. 2400 @ 3000 g/ha	37.96	35.33	1455.37
Fosetyl AI 80% WP a.i. 4800 @ 6000 g/ha	42.05	39.15	1611.97
Carbendazim a.i. 500 @ 1000 g/ha	44.02	41.98	1729.60
Control	28.04	26.86	837.76
Sem±	2.36	2.22	87.15
CD at 5%	5.03	4.74	185.72

*Average of four replications.

(ii) Number of seeds/plant

It is evident from data in table-4.13 and fig-27 that drenching with all the fungicides increased number of seeds/plant of chickpea significantly as compared to control. Among the treatments, significantly, maximum number of seeds/plant (41.98) were found in Carbendazim 50% WP at par with Fosetyl AI 80% WP a.i. 4800 (39.15) and significantly superior to rest of the treatments. Fosetyl AI 80% WP a.i. 2400 (35.33), Fosetyl AI 80% WP a.i. 2000 (34.24) and Fosetyl AI 80% WP a.i. 1600 (32.32) were equally effective in increasing number of seeds/plant of chickpea. In control least number of seeds/plant 26.86 was recorded.

(iii) Yield (kg/ha)

The yield(kg/ha) varied among different treatments (table-4.13 and fig.-28). Drenching with all the fungicides increased yield (kg/ha) of chickpea significantly as compared to control. Among the treatments, significantly, maximum yield (kg/ha)

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1729.60 kg was obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 (1611.97 kg) and significantly superior to rest of the treatments. The yield(kg/ha) of chickpea was at par in Fosetyl Al 80% WP a.i. 2400 (1455.37 kg), Fosetyl Al 80% WP a.i. 2000 (1339.06 kg) and Fosetyl Al 80% WP a.i. 1600 (1250.86 kg). In control least yield (kg/ha) 837.76 kg was recorded.

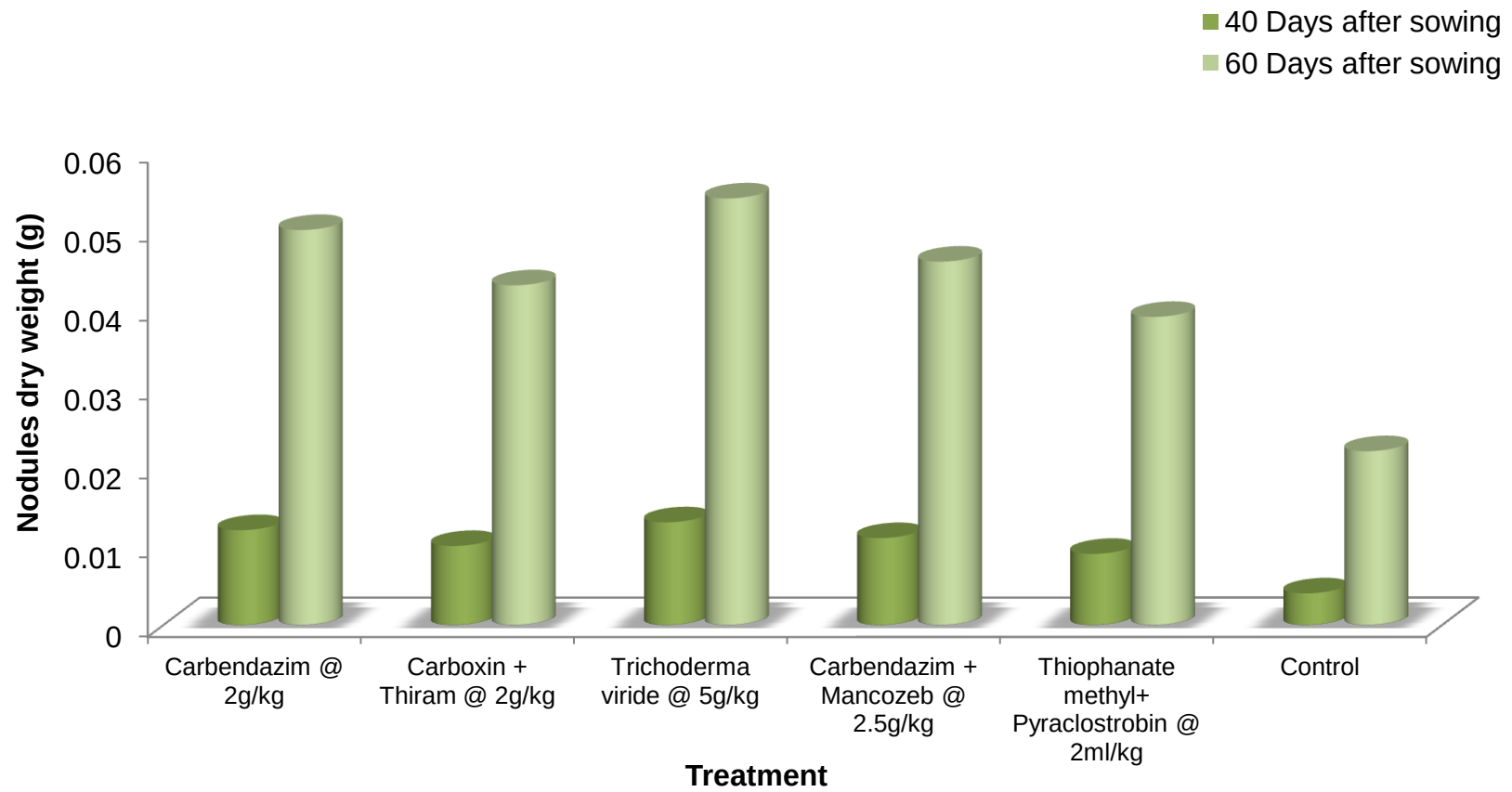


Fig.-12: Nodules dry weight of chickpea in different seed treatments

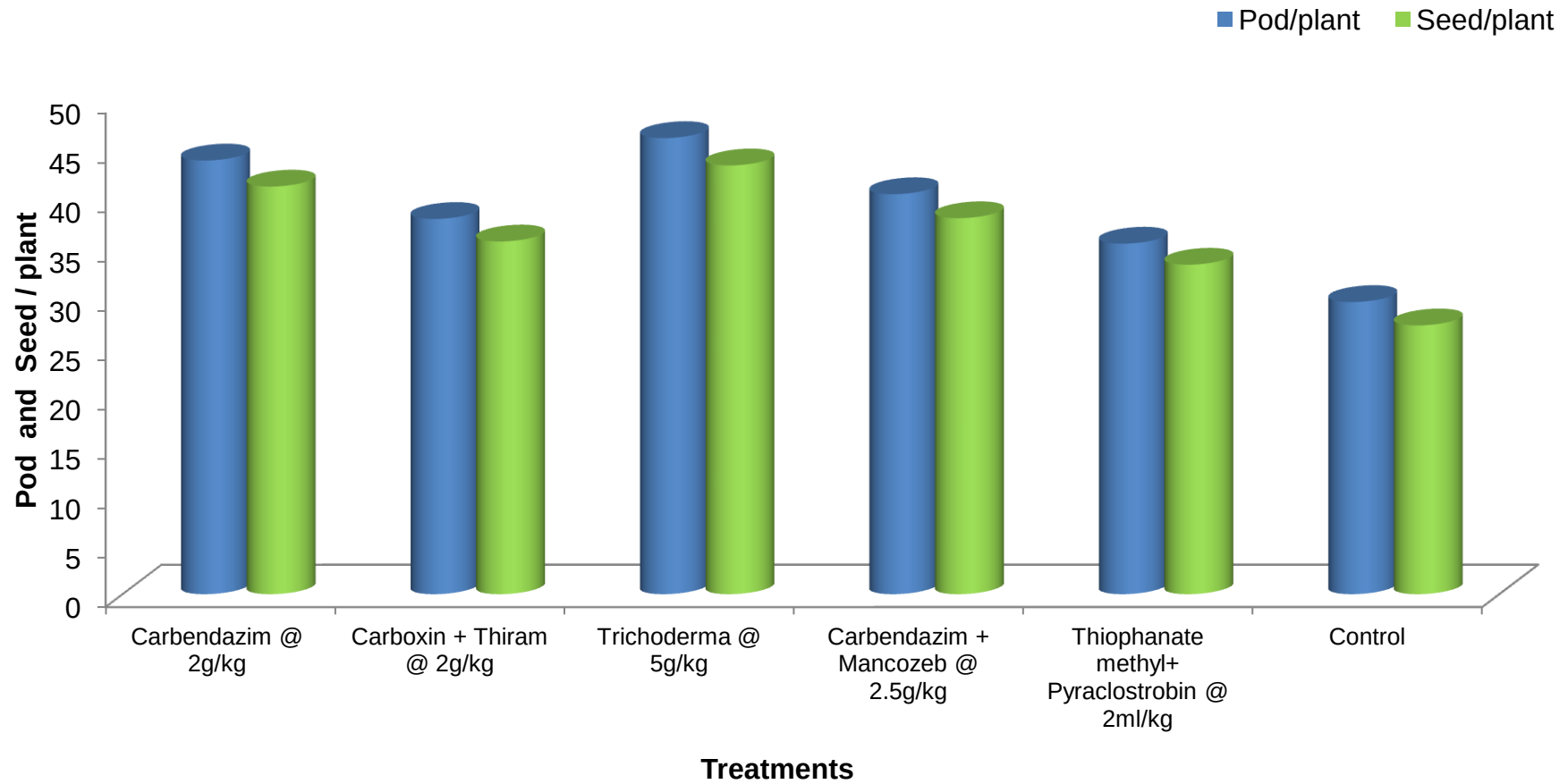


Fig.-13: Pods/plant, and seeds/plant of chickpea in different seed treatments under field condition

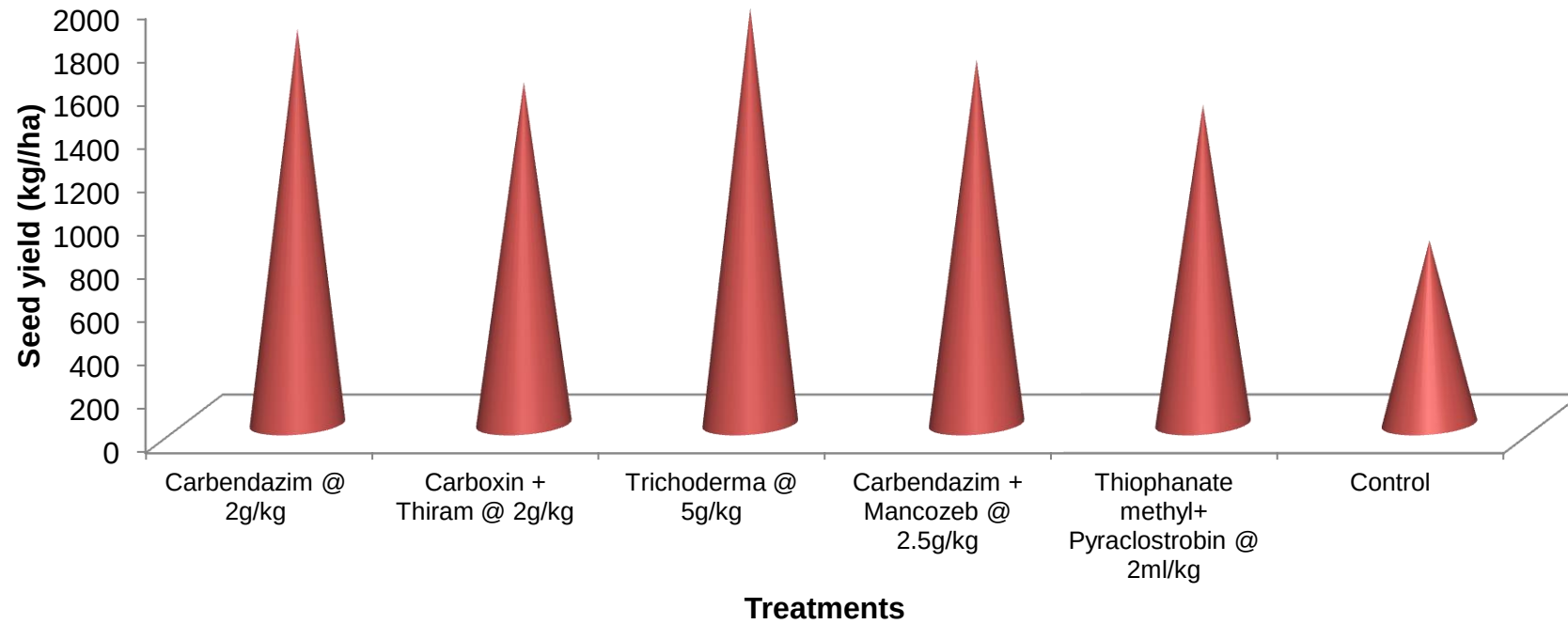


Fig.-14: Seed yield (kg/ha) of chickpea in different seed treatments under field condition

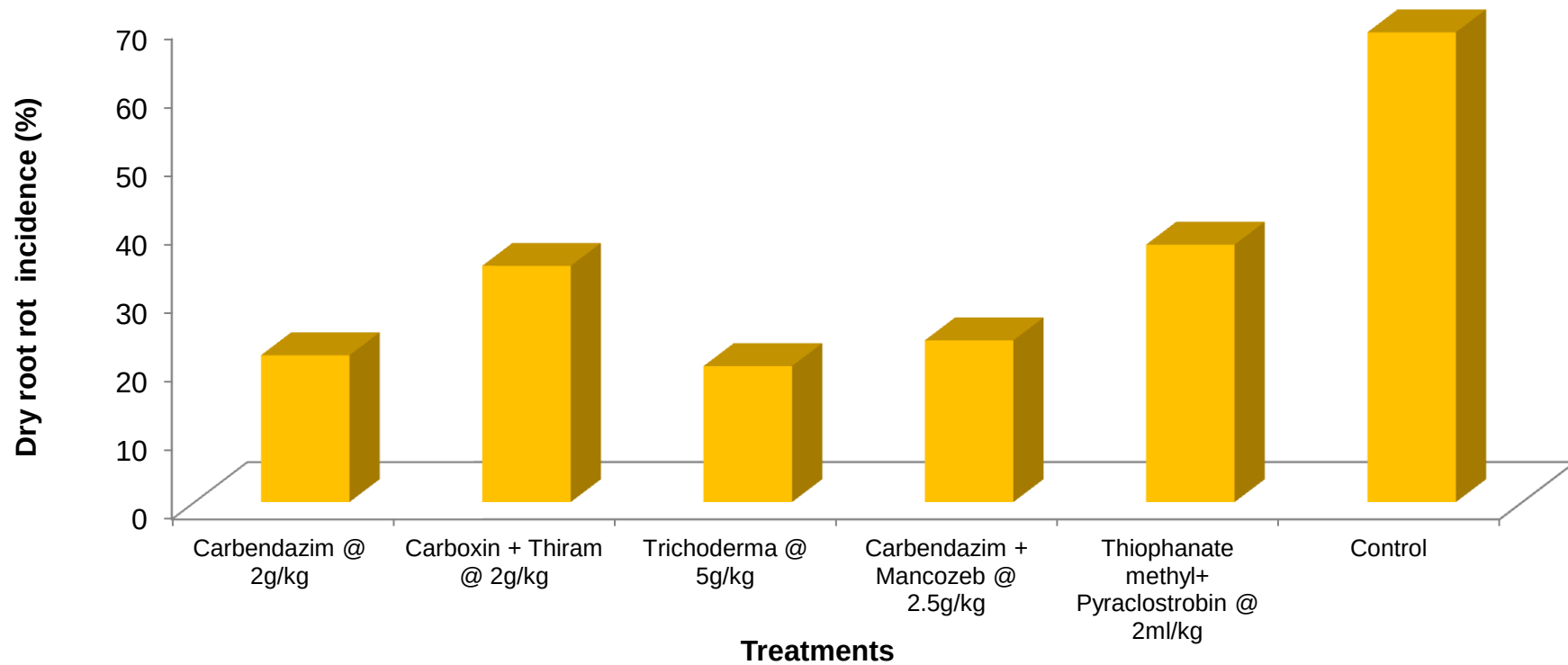
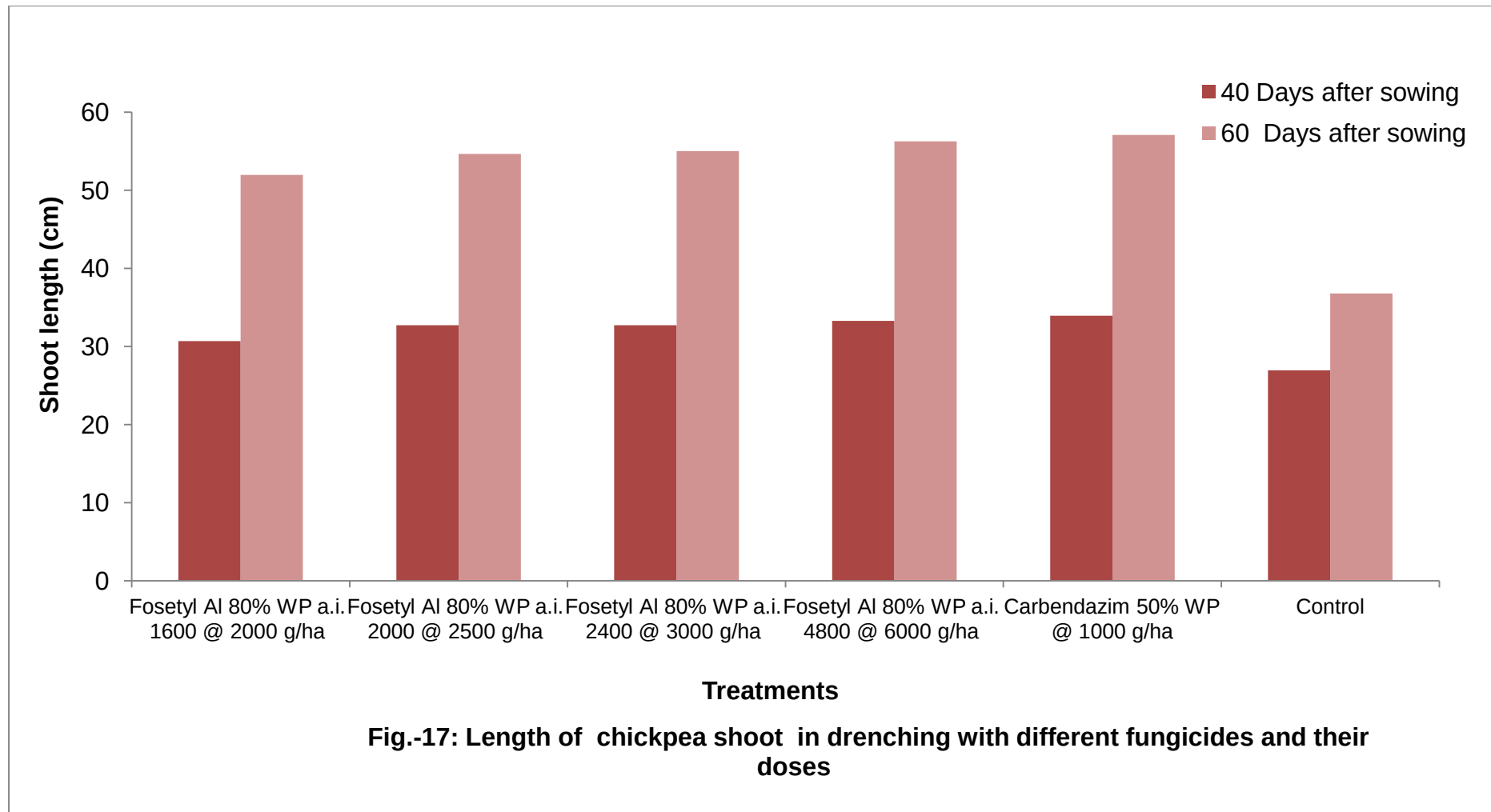


Fig.-15: Incidence of dry root rot of chickpea in different seed treatments



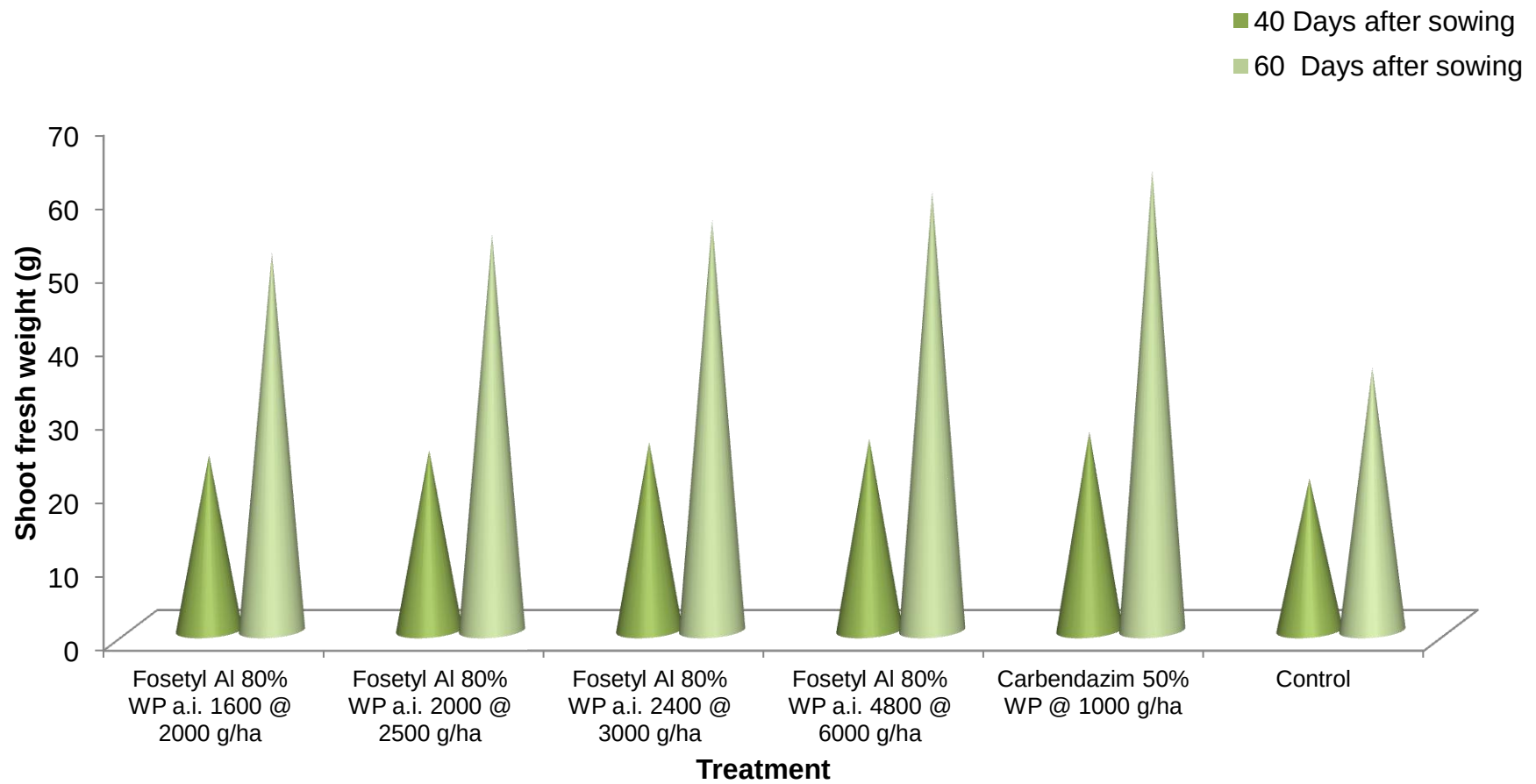


Fig.-18: Fresh weight of chickpea shoot in drenching with different fungicides and their doses

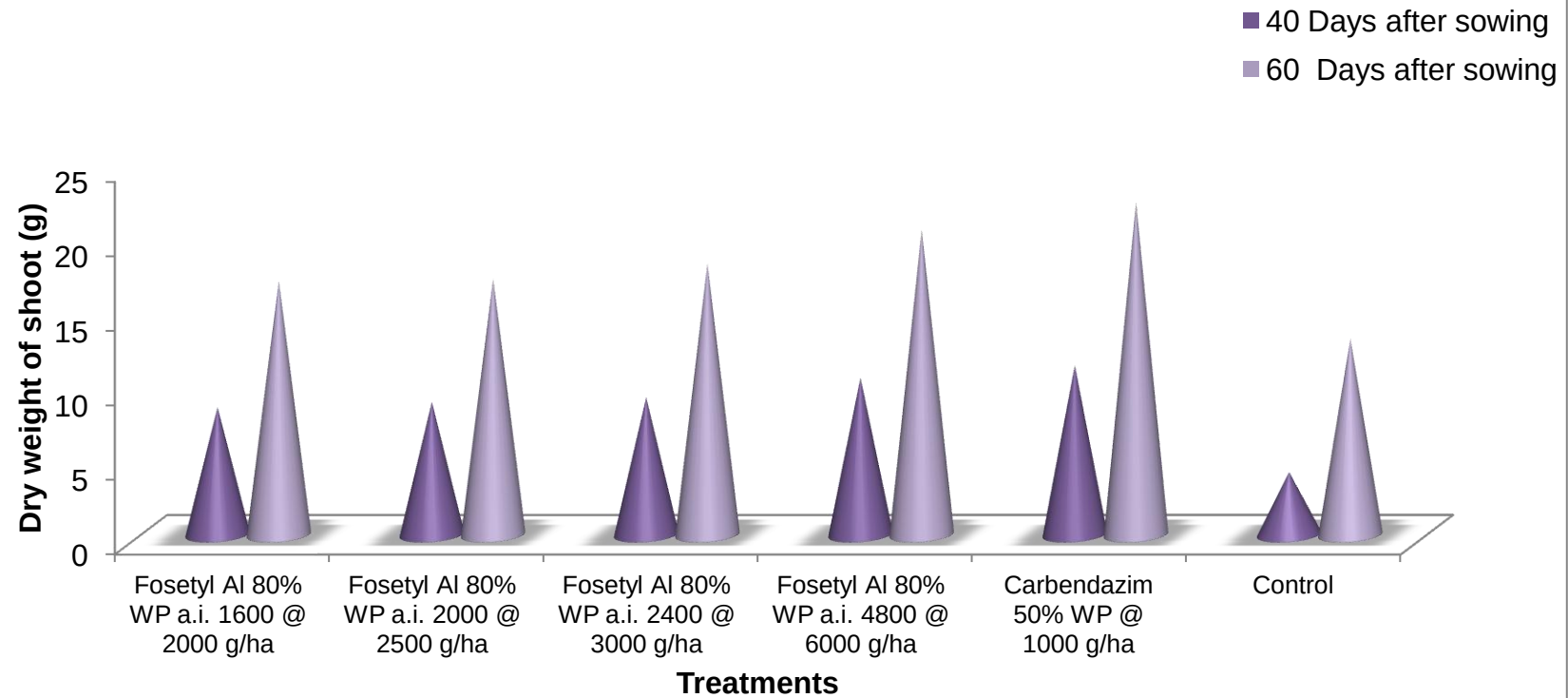


Fig.-19: Dry weight of chickpea shoot in drenching with different fungicides and their doses

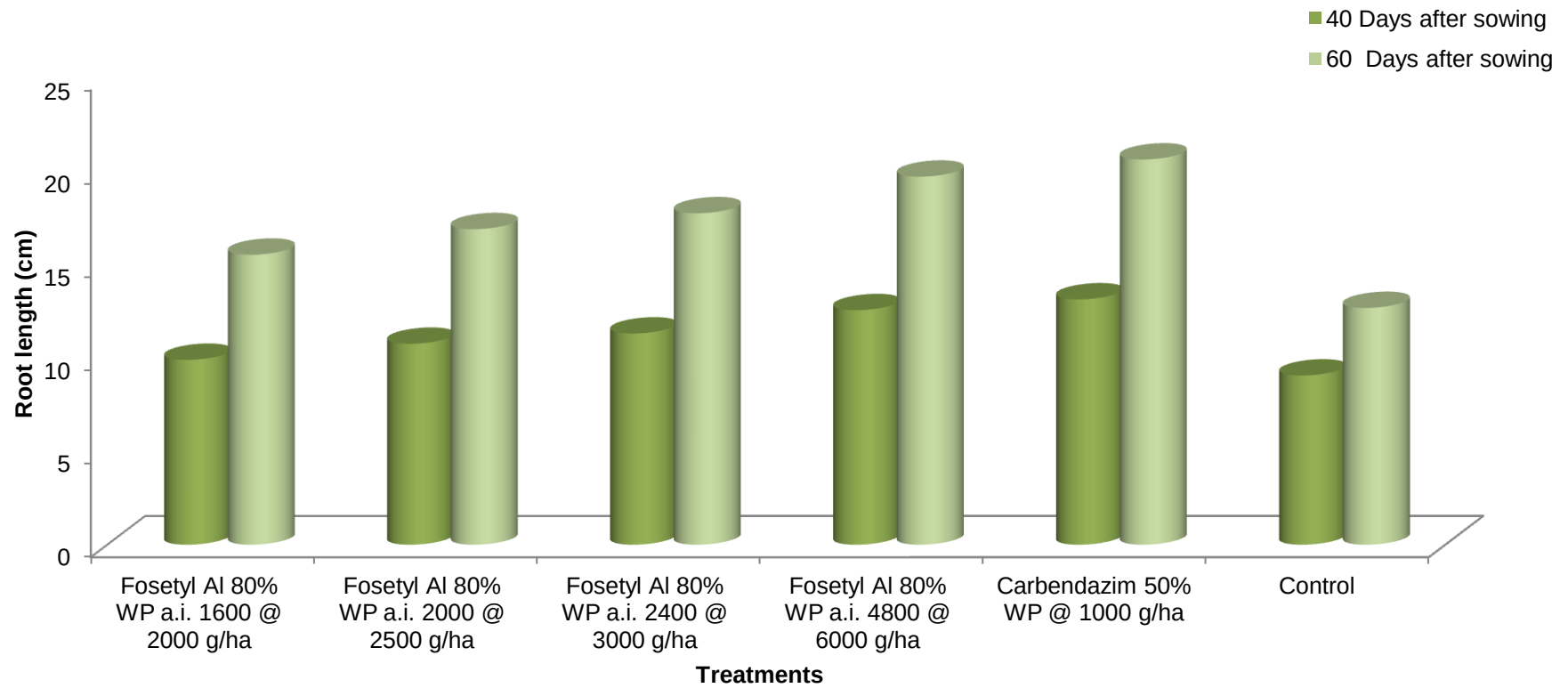


Fig.-20: Length of root of chickpea in drenching with different fungicides and their doses

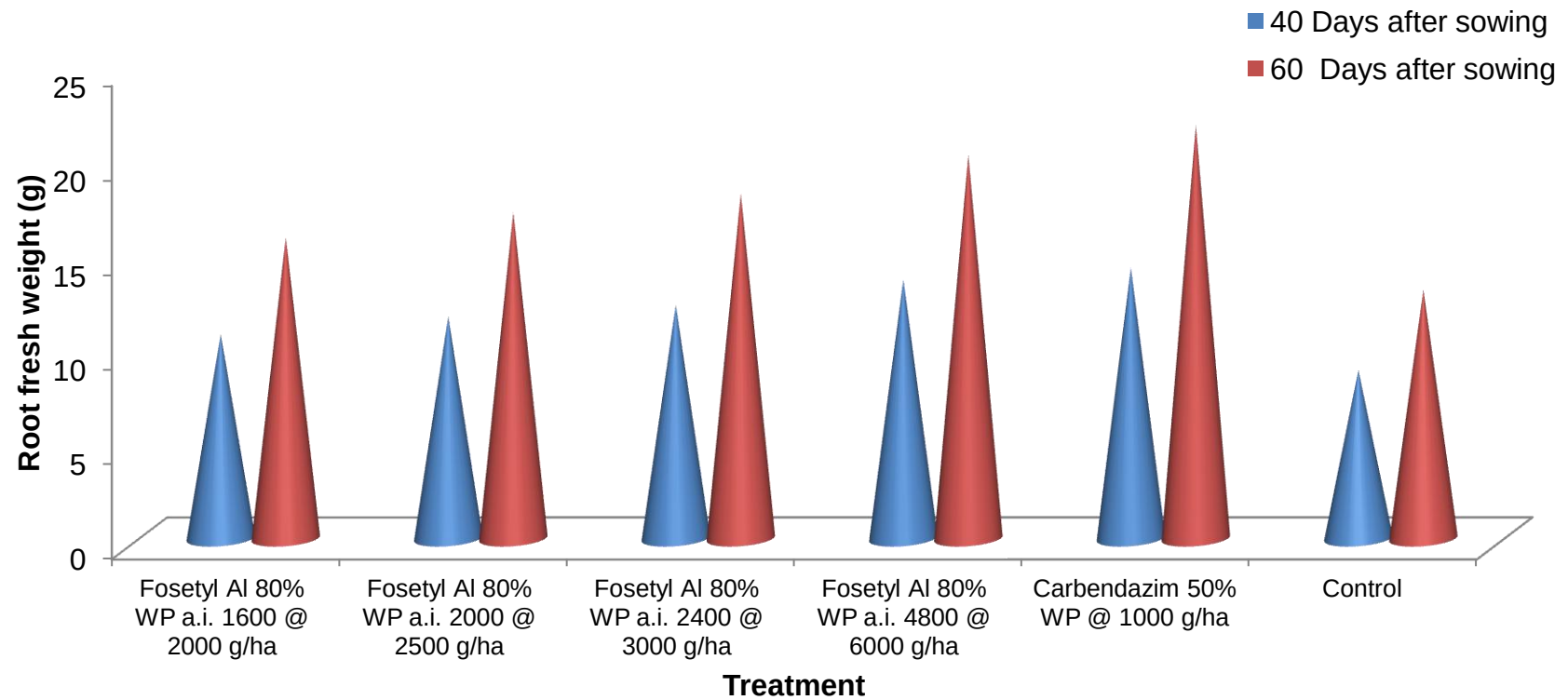


Fig.-21: Fresh weight of root of chickpea in drenching with different fungicides and their doses

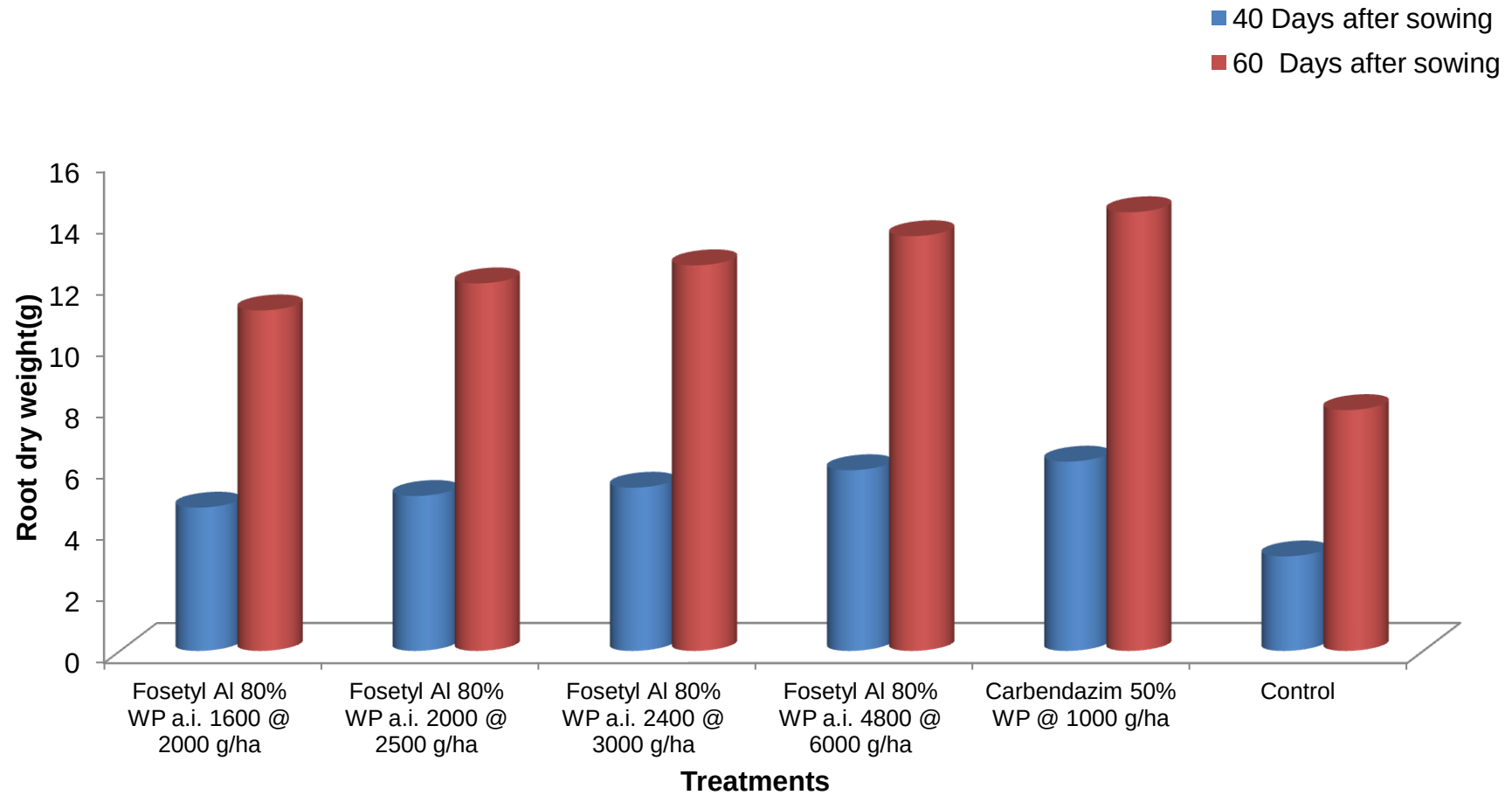


Fig.-22: Dry weight of root of chickpea in drenching with different fungicides and their doses

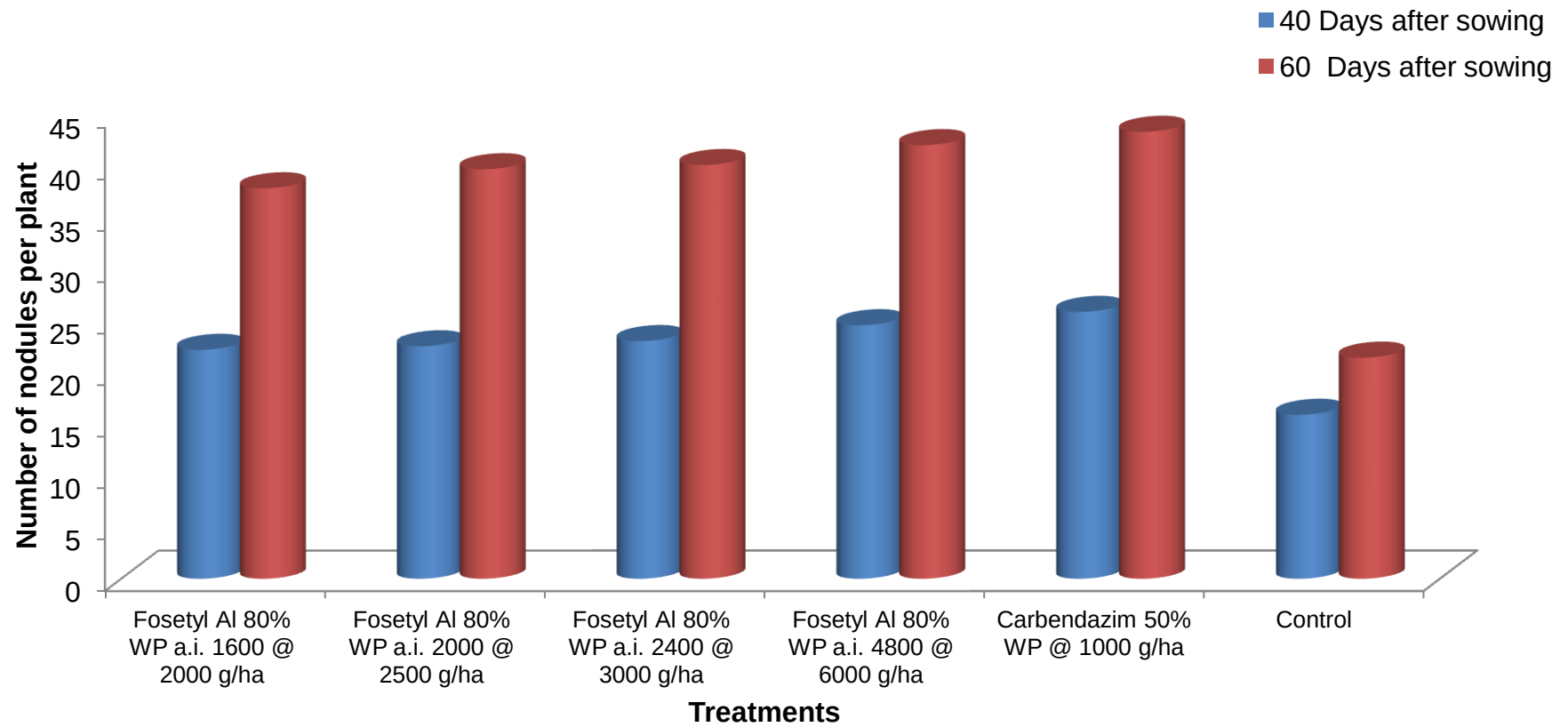


Fig.-23: Number of nodules of chickpea in drenching with different fungicides and their doses

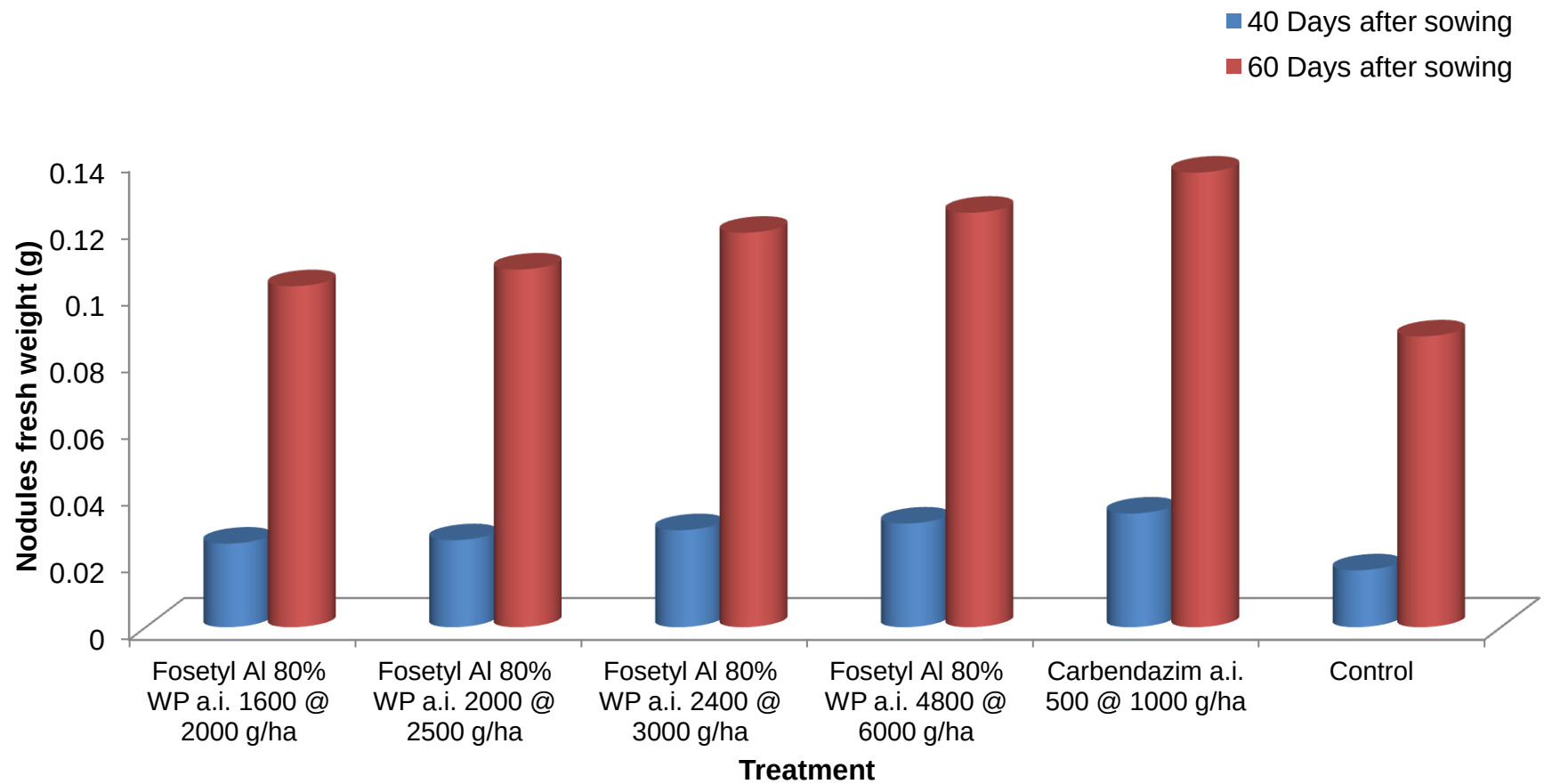


Fig.-24: Nodules fresh weight of chickpea in drenching with different fungicide and their doses

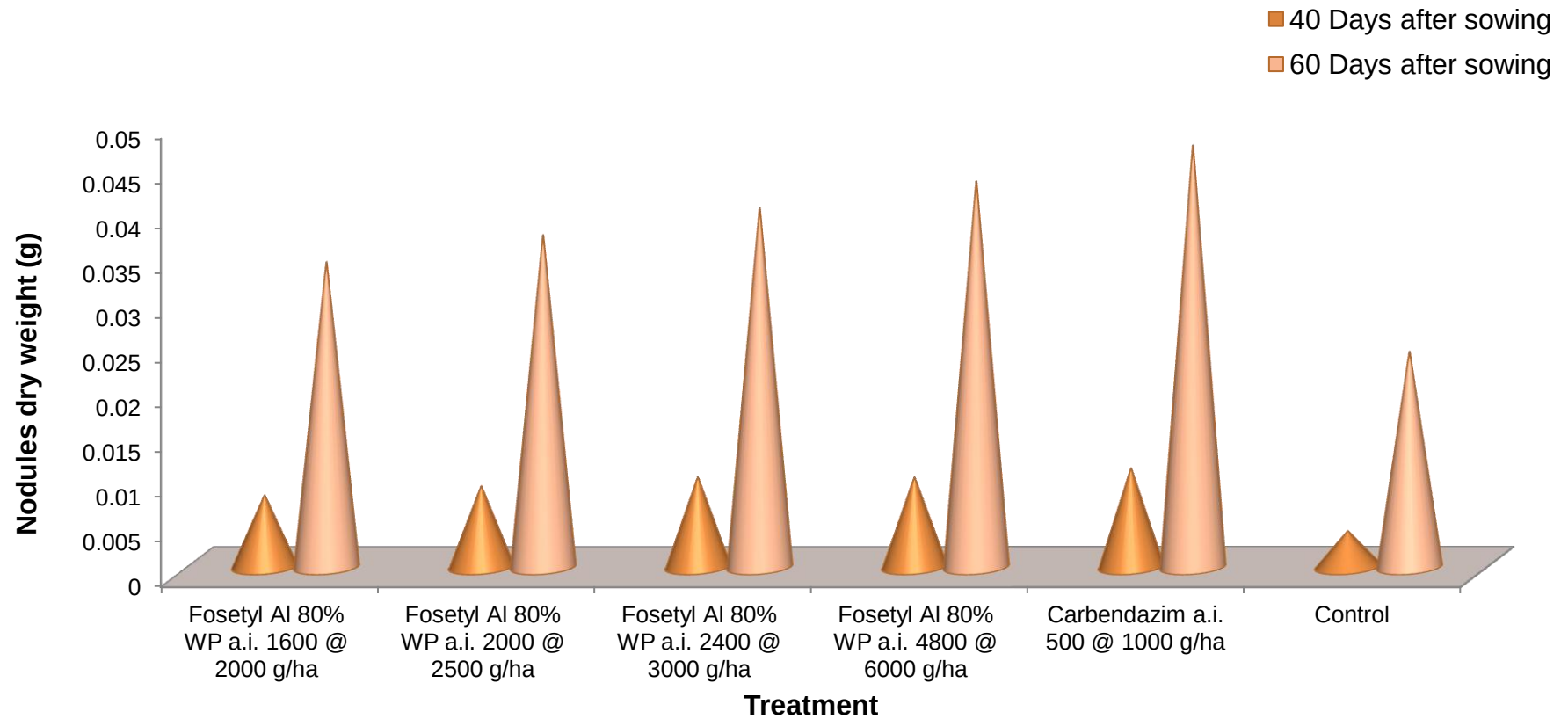
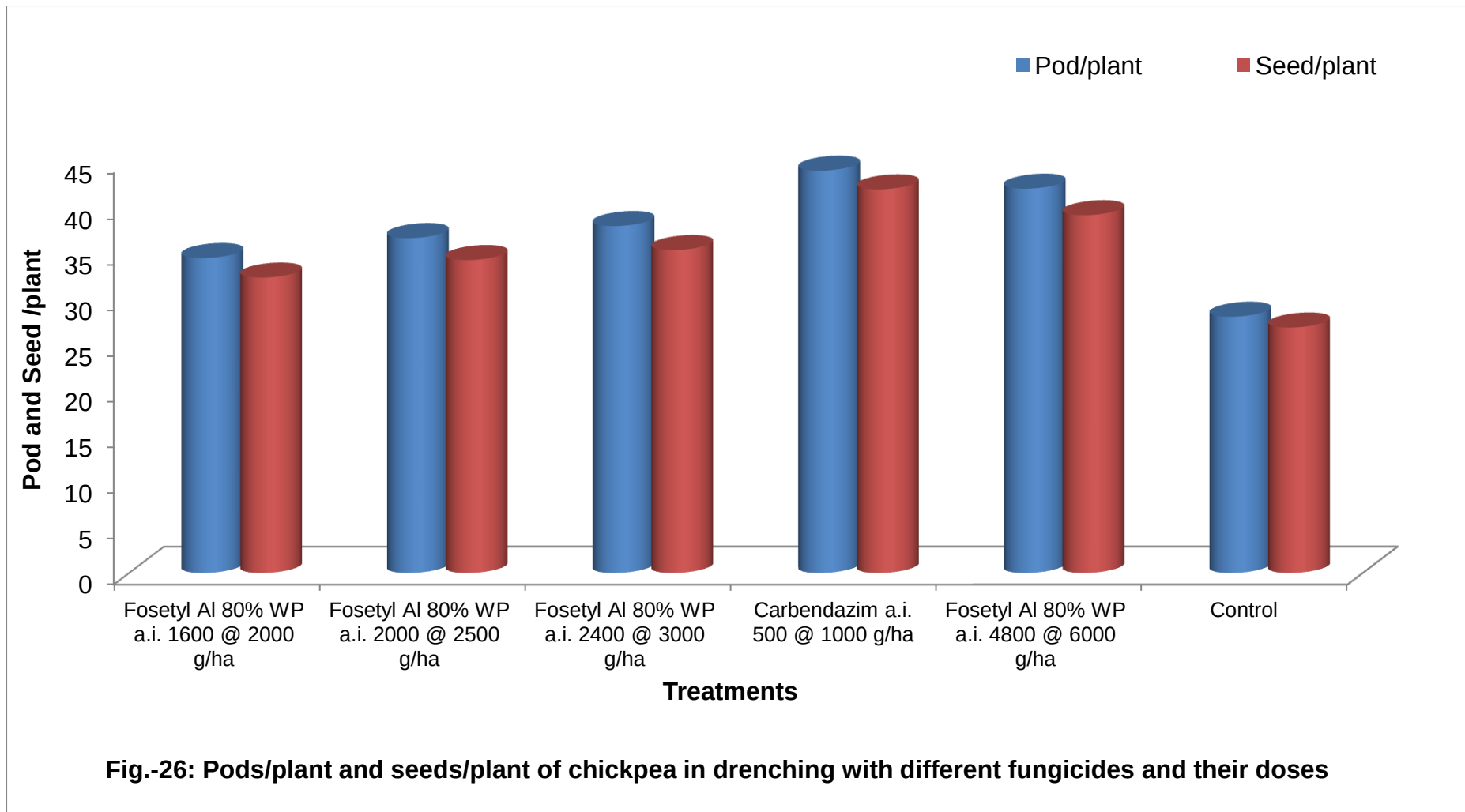


Fig.-25: Nodules dry weight of chickpea in drenching with different fungicides and their doses



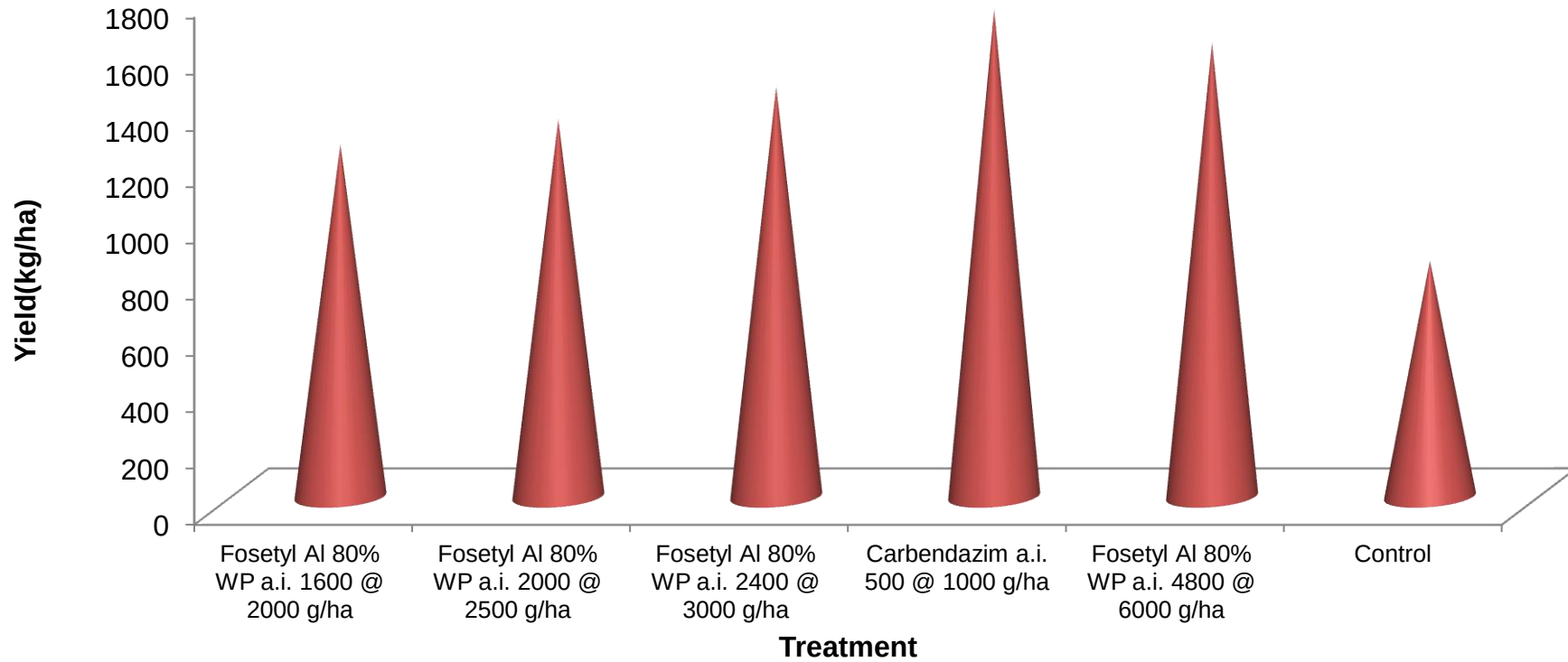


Fig.-27: Yield(kg/ha) of chickpea in drenching with different fungicides and their doses

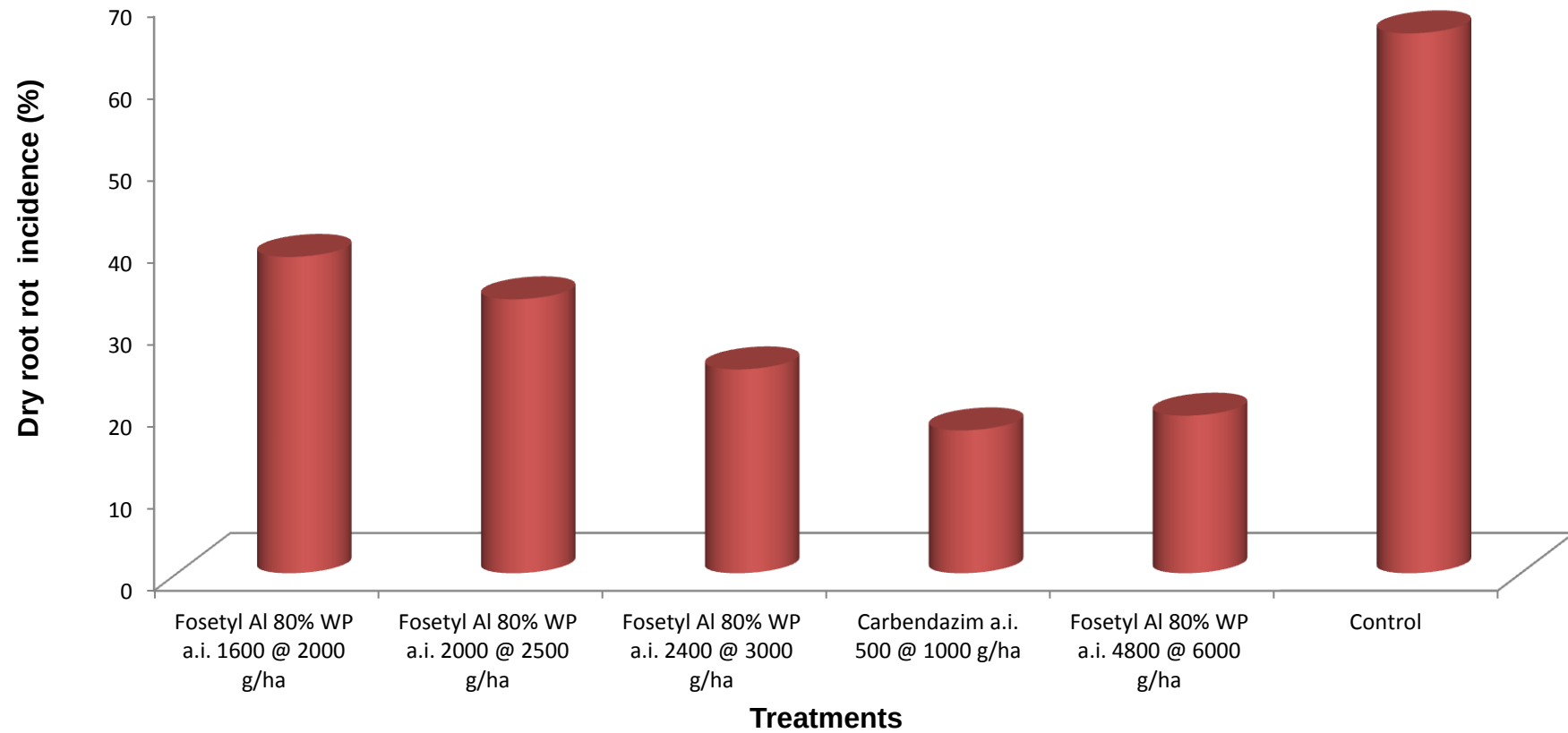


Fig.-16: Incidence of dry root rot of chickpea in drenching with fungicides and their doses

CHAPTER V

DISCUSSION

In Indian agriculture, pulses play an important role from economical and nutritional point of view. Chickpea is an important pulse crop of the Indian subcontinent.

Recent reports indicate that among the major fungal diseases infecting chickpea, dry root rot is one of the most destructive and wide spread disease emerging as a potential threat to chickpea production (Gupta and Sharma, 2015). Several workers have reported the pathogen associated with dry root rot of chickpea. A roving survey was conducted during *rabi*, 2017-18 around Sehore district of Madhya Pradesh. The dry root rot incidence varied from 18.80 to 39.60 per cent. The results indicate that the incidence was noticed in all the fields, wherever, the survey was conducted. The highly prone areas of dry root rot of chickpea were Kosmi followed by Nander, Ramgarha, Jamni, Bijori and Kaneriya. The least incidence of disease was noticed in Hirapur as compared to survey of other locations. In block-wise seasonal incidence, average of dry root rot incidence in the four blocks surveyed ranged from 26.88 (Sehore) to 31.33 % (Rehti). However, the chickpea crops grown in the block of Rehti were found to be infected with maximum per cent of disease incidence followed by the other blocks viz., Ichhawar and Budni. Among the soil borne diseases, dry root rot caused by *R. bataticola*, is the major limiting factor in chickpea production (Deepa *et al.*, 2018). The disease has also been reported in Marathawada region ranging from 10.20 to 23 per cent. (Kadam *et al.*, 2018).

Dry root rot affected plants were identified based on symptoms like drying and withering of plants and, when these plants were pulled out, showed blackening of tap root and were devoid of lateral and finer roots and shredding of bark which comes cut in the form of flakes. These symptoms were similar to earlier reports of Malviya (2018).

The pathogen was isolated from the root region and purified by single hyphal tip method and maintained on PDA for further studies. Based on morphological characters, the pathogen was identified as *R. bataticola* Butler. Similar reports were observed by Barnett and Hunter (1972) and Nagar (2017).

Pathogenicity of the fungus was established by inoculation and reisolation from artificially infected plants proving Koch's Postulates. In the present study, soil

inoculation methods were applied to establish the disease. Several workers found soil inoculation method as the most suitable in establishing the disease caused by *R. bataticola* (Manjunatha, et al. 2011).

Seven fungicides were evaluated in laboratory for their toxicity against *R. bataticola* by poison food technique. Observations on mycelial radial growth of *R. bataticola* were recorded after seven days of inoculation. All fungicides completely inhibited the mycelial growth of *R. bataticola* except Fosetyl Al @ 0.20%. Pawar et al., (2018) tested four fungicides viz., Vitavax + Thiram @ (0.3%), Carbendazim + Mancozeb @ (0.3%), Thiram + Mancozeb @ (0.3%) and Thiophanate methyl @ (0.1%) against *R. bataticola in vitro* and found that all the fungicides inhibited cent percent (100%) radial mycelial growth of *R. bataticola*.

Effect of seed treatment with *T. viride* and fungicides on germination percentage in the seed treated with *T. viride* and fungicides was significantly higher than control (56.24%). Among the treatments, maximum seed germination (82.34 %) was recorded in *T. viride* @ 5g/kg which was at par with Carboxin + Thiram @ 2g/kg (80.50%), Carbendazim + Mancozeb @ 2.5g/kg (78.35%) , Carbendazim @ 2 g/kg (77.00%) but it was superior over to Thiophanate methyl+ Pyraclostrobin @ 2ml/kg (76.20%). Ingle et al. (2018) results revealed that in comparison with bio-agent (*Pseudomonas fluorescens* and *Trichoderma viride*) seed dressing with Carboxin 37.5% + Thiram 37.5% (combi product) @ 2g/kg seed recorded maximum seed germination (92.96%), least per cent incidence of root rot (101, 2.79 and 6.22%) at 15, 30 and 45 days after germination respectively.

Seed treatments with *T. viride* and fungicides significantly increased the plant vigour as compared to control (2696.45). Among the treatments, maximum plant vigour (6519.20) was recorded in *T. viride* @ 5g/kg which was superior to rest of the treatments. Carbendazim @ 2g/kg (5872.27) was at par with Carbendazim + Mancozeb @ 2.5g/kg (5682.48), Carboxin + Thiram @ 2g/kg (5592.99). However, seed treatment with Carbendazim was significantly superior to Thiophanate methyl+ Pyraclostrobin @ 2ml/kg (4917.97). Tancic et al. (2013) also observed increase in vigour index of soybean treated with four isolates of *Trichoderma* (K114, K 132, K150 and K160).

In the present study, all the seed treatments significantly increased the shoot length, fresh weight and shoot dry weight of chickpea. Among the treated treatments, after 40 & 60 days of sowing, *T. viride* @ 5g/kg recorded maximum shoot length at par with Carbendazim @ 2g/kg. Carbendazim + Mancozeb @ 2.5g/kg, Carboxin +

Thiram @2g/kg and Thiophanate methyl + Pyraclostrobin @ 2ml/kg were equally effective in increasing the shoot length of chickpea after 40 & 60 days of sowing. Kumar *et al.* , (2014) showed that Chickpea seeds were treated with different concentrations of *Trichoderma* bioformulation such as 5%, 10%, 20% gm/kg seed followed by treatment with 0.2% Bavistin. As a result, the percentage of seed germination was found to be higher in *T. harzianum* (Th. azad) & *T. viride* (01PP) treated seeds with 5% bioformulation as compared to the other concentrations. Various attributes with their observations include seed germination (92% and 90%), root length (12.38 and 12.19 cm) shoot length (4.97 and 4.32 cm) seedling length (17.38 and 16.50 cm), dry weight (1.19 and 1.88 cm), vigour index I (5197.12 and 1485) and vigour index II (109.48 and 169.20).

Similarly, **maximum shoot fresh weight and dry weight was recorded in *T. viride* @ 5g/kg** at par with Carbendazim @ 2g/kg after 40 and 60 days of sowing. Carbendazim+ Mancozeb @2.5g/kg, Carboxin + Thiram @2g/kg and Thiophanate methyl + Pyraclostrobin @ 2ml/kg were same as in increasing the shoot fresh weight and dry weight of chickpea after 40 & 60 days of sowing. Suresh *et al.* (2011) also reported induction of shoot and root length of chickpea up to 10-15% by *T. viride* and *T. harzianum*.

All the seed treatments significantly increased the root length, fresh and dry weight of root of chickpea after 40 & 60 days of sowing. However, **seed treatment with *T. viride* and fungicides significantly recorded maximum increase in the root length, fresh root weight and dry weight as compared to control. Among the treatments, maximum root length, root fresh weight and dry weight were recorded in *T. viride* @ 5g/kg** at par with Carbendazim @ 2g/ kg and significantly superior to rest of the treatments. Root length, root fresh weight and dry weight was at par in Carbendazim+ Mancozeb @2.5g/kg, Carboxin + Thiram @2g/kg and Thiophanate methyl + Pyraclostrobin @ 2ml/kg after 40 and 60 days of sowing. Improved dry weight of root by three strains of *Trichoderma* (T.h4, T.v7 and T.a9) was also reported by Entesari, (2013).

All the seed treatments with *T. viride* and fungicides significantly increased number of nodules/plant, nodule fresh and dry weight/plant at 40 & 60 days after sowing. Among the treatments, maximum numbers of nodules / plant, nodule fresh and dry weight/plant were obtained in *T. viride* @ 5g/kg at par with Carbendazim @ 2g/kg and significantly superior to rest of the treatments. Carbendazim+ Mancozeb @ 2.5g/kg, Carboxin + Thiram @2g/kg and Thiophanate methyl + Pyraclostrobin @

2ml/kg were same as in increasing the number of nodules/plant, nodule fresh and dry weight after 40 and 60 days of sowing. Patel *et al.* (2012) reported increase in plant height, fresh and dry weight, no. of nodules of chickpea by *Trichoderma* seed treatment.

All the seed treatments significantly increased the number of pods/plant, number of seeds/plant and yield/ha of chickpea as compared to control. Among the treatments, maximum average number of pods/plant, maximum average number of seeds/plant and highest yield/ha were recorded in *T. viride* @ 5g/kg at par with Carbendazim @ 2g/kg after 40 & 60 days of sowing. Carbendazim+ Mancozeb @2.5g/kg, Carboxin + Thiram @2g/kg and Thiophanate methyl + Pyraclostrobin @ 2ml/kg were equally effective in increasing the average number of pods/plant, seeds/plant and yield/ha of chickpea. Increase in yield by *Trichoderma* seed treatment results are in agreement with Pandey *et al.* (2016) who observed disease control and yield enhancement with *T. viride* followed by *T. harzianum*.

Incidence of dry root rot caused by *R. bataticola* was significantly less in all the seed treatments as compared to control. Upon field trials, seed treatment with *T. viride*@ 5g/kg (5.91%) was the best treatment in controlling dry root rot of chickpea at par with Carbendazim @ 2g/kg (6.38%), Carbendazim + Mancozeb @2.5g/kg (7.03%) and significantly superior to rest of the treatments. However, Carboxin + Thiram @ 2g/kg and Thiophanate methyl+ Pyraclostrobin @ 2ml/kg were same as in controlling dry root rot incidence. Lakhra (2017) tested efficacy of biocontrol against *R. bataticola* in *in vivo* and found that minimum disease incidence (23.00 %) was recorded with seed application of *T. viride* @ 4g/kg followed by *T. harzianum* (26.60 %) as compared to control (43.30 %) at 60 days after sowing respectively. Manjunatha *et al.* (2011) evaluated two biocontrol agents individually and in combinations and in integration with carbendazim seed treatment and hexaconazole seed soaking for management of dry root rot of chickpea. Seed and soil application with bioagents was effective for the maximum germination with less root rot incidence as compared to seed treatment with carbendazim alone and seed soaking in hexaconazole alone. Integration of seed and soil application of biocontrol agents resulted in higher germination and reduced mortality due to disease as against check.

Drenching with all the fungicides (30 and 45 DAS) significantly increased shoot length, fresh shoot weight and shoot dry weight of chickpea. The result indicates that after 40 days of sowing, significant increase in shoot length/plant was observed in drenching with fungicides as compared to control. Among the treatments, maximum shoot length was obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800, Fosetyl Al 80% WP a.i. 2400, Fosetyl Al 80% WP a.i. 2000 but it was significantly superior to Fosetyl Al 80% WP a.i. 1600. **Likewise, after 60 days of sowing,** significant increase in shoot length/plant was observed in drenching with fungicides as compared to control. Among the treatments, maximum shoot length was obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800, Fosetyl Al 80% WP a.i. 2400, Fosetyl Al 80% WP a.i. 2000 and Fosetyl Al 80% WP a.i. 1600. [Nagarjuna Reddy \(2010\)](#) tested efficacy of potential biocontrol agent (REB-8) and compatible fungicide (carbendazim) in pot culture against dry root rot of chickpea. The results revealed that treatment (soil application with potential biocontrol agent + soil drenching with fungicide) was superior in reducing per cent disease incidence and increasing plant growth parameters, like root length, shoot length, dry weight of shoot and root when compared to other treatments.

Drenching with all the fungicides (30 and 45 DAS) significantly increased fresh shoot weight of chickpea when observed 40 and 60 days after sowing as compared to control. Among the treatments, Carbendazim 50% WP significantly recorded maximum increase in the fresh shoot weight at par with Fosetyl Al 80% WP a.i. 4800 and Fosetyl Al 80% WP a.i. 2400. **However,** Fosetyl Al 80% WP a.i. 2000 and Fosetyl Al 80% WP a.i. 1600 were equally effective in increasing fresh shoot weight of chickpea when observed 40 and 60 days after sowing.

Drenching with all the fungicides (30 and 45 DAS) significantly increased shoot dry weight of chickpea when observed 40 and 60 days after sowing as compared to control. Among the treatments, Carbendazim 50% WP significantly recorded maximum increase in shoot dry weight at par with Fosetyl Al 80% WP a.i. 4800. However, Fosetyl Al 80% WP a.i. 2400, Fosetyl Al 80% WP a.i. 2000 and Fosetyl Al 80% WP a.i. 1600 were same as in increasing shoot dry weight of chickpea when observed 40 and 60 days after sowing.

Significantly, drenching with all the fungicides (30 and 45 DAS) increased root length, fresh root weight and root dry weight of chickpea. Among the drenched treatments, maximum root length, fresh root weight and root dry weight were

obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 when observed after 40 and 60 days of sowing. Fosetyl Al 80% WP a.i. 2400, Fosetyl Al 80% WP a.i. 2000 and Fosetyl Al 80% WP a.i. 1600 were same as in increasing root length, fresh root weight and root dry weight of chickpea. Pawar *et al.* (2018) reported Thiram + Carbendazim @ (0.3%) was found most effective for increasing the seed germination (81.66%), shoot length (10.77 cm), root length (9.27 cm) and seedling vigour index (1636.46) followed by T4 Thiophanate methyl @ (0.1%) with seed germination (73.33%) shoot length (9.96 cm) root length (8.45 cm) and seedling vigour index (1350.00).

Result shows that drenching with all the fungicides (30 and 45 DAS) significantly increased number of nodules/plant, nodule fresh and dry weight/plant. Significantly, highest number of nodules/plant was found in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 and Fosetyl Al 80% WP a.i. 2400. Fosetyl Al 80% WP a.i. 2000 and Fosetyl Al 80% WP a.i. 1600 were equally effective in increasing number of nodules/plant as compared to control after 40 days of sowing.

Similarly, 60 days after sowing, drenching with all the fungicides (30 and 45 DAS) significantly increased the number of nodules/plant as compared to control. Highest number of nodules/plant was obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800, Fosetyl Al 80% WP a.i. 2400, Fosetyl Al 80% WP a.i. 2000 but it was superior to Fosetyl Al 80% WP a.i. 1600 as compared to control.

Drenching with all the fungicides (30 and 45 DAS) significantly increased nodule fresh weight and nodule dry weight of chickpea when observed 40 and 60 days after sowing as compared to control. Among the treatments, Carbendazim 50% WP significantly recorded maximum increase in the nodule fresh and nodule dry weight at par with Fosetyl Al 80% WP a.i. 4800. However, Fosetyl Al 80% WP a.i. 2400, Fosetyl Al 80% WP a.i. 2000 and Fosetyl Al 80% WP a.i. 1600 were equally effective in increasing nodule fresh and nodule dry weight of chickpea when observed 40 and 60 days after sowing.

Results thus obtained show that drenching with all the fungicides (30 and 45 DAS) significantly increased **yield parameters** of chickpea **viz. number of pods/plant, number of seeds/plant and yield/ha (kg)**. **Significantly, maximum yield parameters [i.e. number of pods/plant, number of seeds/plant and yield/ha (kg)] were recorded in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 as compared to control. However, Fosetyl Al 80% WP a.i. 2400, Fosetyl Al 80% WP a.i. 2000 and Fosetyl Al 80% WP a.i. 1600 were same as in increasing above**

mentioned growth parameters. Incidence of dry root rot caused by *R. bataticola* was significantly less in all the drenching of fungicides as compared to control. Among the drenched treatments, Carbendazim 50% WP was the best treatment in controlling dry root rot of chickpea at par with Fosetyl Al 80% WP a.i. 4800 as compared to control. However, significantly, Fosetyl Al 80% WP a.i. 2400, Fosetyl Al 80% WP a.i. 2000 and Fosetyl Al 80% WP a.i. 1600 were equally effective in controlling dry root rot of chickpea as compared to control. Ebben (1973) The development of *Fusarium* wilt symptoms in carnations grown in soils artificially infested with different amounts of inocula was recorded in four experiments in 1973-76. Initial numbers of plants with symptoms and rates of increase in disease severity were directly related to inoculum levels in each year, but differed between years. Disease increased relatively slowly until 10% of the plants showed symptoms and was then more rapid. Carbendazim drenches decreased disease severity at all inoculum levels, drenches applied soon after planting being more effective than drenches delayed until after symptoms had appeared. Only where initial inocula were low did the decrease in disease with carbendazim drenches prevent loss of flower production. Drenches applied in the 2nd year of the crop did not decrease wilt. Lakhran and Ahir (2018) tested efficacy of fungicides against dry root rot of chickpea under *in vivo*. According to the result, all fungicides found significantly superior over control in reducing per cent disease incidence. Minimum percent disease incidence was recorded in carbendazim (13.33 and 15.38 %) followed by carbendazim + mancozeb (16.66 and 20.00%) while the maximum percentage disease incidence was observed in control (46.60 and 62.50 %) at both of the stage. Singh and Sindhan (1998) observed that carbendazim and carboxin (0.2%) used as seed treatment and soil drench or seed treatment + soil drench significantly reduced the incidence of dry root rot of chickpea caused by *R. bataticola* (*M. phaseolina*) with a corresponding increase in phenolics and carbohydrates in the plants. They examined that application of carbendazim and carboxin with seed treatment + soil drenching was the most effective method in reducing disease incidence followed by soil drenching and seed treatment alone. Sharma and Kumari (2017) reported that Carbendazim was proved most effective in minimizing the disease as well as pathogen population in soil followed by Thiophnate Methyl. Seed treatment with Chlorothalonil was found least effective in minimizing the disease as well as pathogen population in soil. Mahmood *et al.* (2015) result out that comparative study of biological control agents and fungicides showed that both are almost equally

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effective against chickpea wilt disease. Seed treatment with *T. harzanium* followed by Carbendazim drenching was proved most effective by increasing the disease reduction percentage up to 93.75%

CHAPTER-VI

SUMMARY, CONCLUSION AND SUGGESTION FOR FURTHER WORK

Summary

- ✓ The present investigation on application technology of fungicides for the management of dry root rot of chickpea was carried out during *rabi* 2017-18 at Department of Plant Pathology and field of R. A. K. College of Agriculture, Sehore (M.P.), R.V.S.K.V.V. Gwalior (M.P.)
- ✓ A roving survey was conducted during *rabi* 2017-18 around Sehore district of Madhya Pradesh. The highly prone area of dry root rot of chickpea was village Kosmi followed by Nander, Ramgarha, Jamni, Bijori and Kaneriya villages. The least incidence of disease was noticed in Hirapur village as compared to other locations survey. In block-wise seasonal incidence, Rehti was found to be infected with maximum per cent of disease incidence followed by the other blocks viz., Ichhawar and Budni.
- ✓ Dry root rot affected plants were identified based on symptoms like drying and withering of plants and when these plants were pulled out, showed blackening of tap root and devoid of lateral and finer roots and shredding of bark which comes cut in the form of flakes.
- ✓ The pathogen isolated from the root region was purified by single hyphal tip method and maintained on PDA for further studies. Based on morphological characters, the pathogen was identified as *R. bataticola* Butler.
- ✓ Pathogenicity of the fungus was established by inoculation and re-isolation from artificially infected plants. In the present study, soil inoculation method was followed to establish the disease. Several workers found soil inoculation method as the most suitable in establishing the disease caused by *R. bataticola*.
- ✓ Seven fungicides were evaluated in laboratory for their toxicity against *R. bataticola* by poison food technique. Observations on radial growth of *R. bataticola* were recorded after seven days of inoculation. All fungicides completely inhibited mycelial growth of *R. bataticola* except Fosetyl Al 80% WP @ 0.20%.

- ✓ Maximum seed germination (82.34 %) was recorded in *T. viride* @ 5g/kg which was at par with Carboxin +Thiram @ 2g/kg (80.50%), Carbendazim + Mancozeb @ 2.5g/kg (78.35%) , Carbendazim @ 2 g/kg (77.00%) but it was superior over to Thiophanate methyl+ Pyraclostrobin @ 2ml/kg (76.20%).
- ✓ Maximum plant vigour (6519.20) was recorded in *T. viride* @ 5g/kg which was superior to rest of the treatments. Carbendazim @ 2g/kg (5872.27) was at par with Carbendazim + Mancozeb @ 2.5g/kg (5682.48), Carboxin + Thiram @ 2g/kg (5592.99). However, seed treatment with Carbendazim was significantly superior to Thiophanate methyl+ Pyraclostrobin @ 2ml/kg (4917.97)
- ✓ *T. viride* @ 5g/kg recorded maximum shoot length, maximum fresh shoot weight and shoot dry weight at par with Carbendazim @ 2g/kg and superior over rest of all the treatments after 40 and 60 days of sowing.
- ✓ **The maximum** root length, fresh root weight and root dry weight of chickpea **were recorded in** *T. viride* @ 5g/kg at par with Carbendazim @ 2g/kg and Carbendazim + Mancozeb @ 2.5g/kg after 40 and 60 days of sowing.
- ✓ Maximum number of nodules/plant, nodule fresh weight and nodule dry weight was obtained in *T. viride* @ 5g/kg at par with Carbendazim @ 2g/kg and significantly superior to rest of the treatments after 40 and 60 days of sowing.
- ✓ **Maximum average number of pods/plant (46.18) were recorded in** *T. viride* @ 5g/kg **at par with** Carbendazim @ 2g/kg **(43.91)**. Similarly, **maximum** number of seeds/plant **and yield(kg/ha) were recorded in** *T. viride* @ 5g/kg **at par with** Carbendazim @ 2g/kg.
- ✓ Seed treatment with *T. viride* @ 5g/kg was the best treatment (5.91%) for controlling dry root rot of chickpea at par with Carbendazim @ 2g/kg (6.38%), Carbendazim + Mancozeb @ 2.5g/kg (7.03%) and significantly superior to rest of the treatments .In control, where no treatment was done, the incidence of root rot was significantly maximum (33.57%).
- ✓ Maximum shoot length (57.08 cm) was obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 (56.26 cm), Fosetyl Al 80% WP a.i. 2400 (55.00 cm), Fosetyl Al 80% WP a.i. 2000 (54.65 cm) but it was superior to Fosetyl Al 80% WP a.i. 1600 (51.96 cm) after 60 days of sowing.
- ✓ **Drenching of** Carbendazim 50% WP **significantly recorded maximum increase in the fresh shoot weight** at par with Fosetyl Al 80% WP a.i. 4800

and Fosetyl Al 80% WP a.i. 2400 and significantly superior to rest of the treatments as well as control after 40 and 60 days of sowing.

- ✓ **Drenching of Carbendazim 50% WP significantly recorded maximum increase in the shoot dry weight** at par with Fosetyl Al 80% WP a.i. 4800 and Fosetyl Al 80% WP a.i. 2400 and significantly superior to rest of the treatments as well as control after 40 and 60 days of sowing.
- ✓ Maximum root length (20.63 cm) was obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 (19.71 cm), Fosetyl Al 80% WP a.i. 2400 (17.75 cm), Fosetyl Al 80% WP a.i. 2000 (16.90 cm) but it was superior to Fosetyl Al 80% WP a.i. 1600 (15.53 cm) after 60 days of sowing.
- ✓ **Maximum fresh root weight (21.84 g)** was obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 (20.28 g) and significantly superior to rest of the treatments after 60 days of sowing.
- ✓ **Maximum dry root weight (14.30 g)** was obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 (13.52 g) and significantly superior to rest of the treatments after 60 days of sowing.
- ✓ Maximum number of nodules/plant (43.42) was obtained in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 (42.12) and Fosetyl Al 80% WP a.i. 2400 (40.19) and significantly superior to rest of the treatments as compared to control after 60 day of sowing.
- ✓ Maximum nodule fresh weight/plant and nodule dry weight/plant were obtained in Carbendazim 50% WP (0.136 g and 0.047 g respectively) at par with Fosetyl Al 80% WP a.i. 4800 (0.124 g and 0.043 g respectively) and Fosetyl Al 80% WP a.i. 2400 (0.118 g and 0.040 g respectively) and significantly superior to rest of the treatments as compared to control after 60 day of sowing.
- ✓ **Maximum average numbers of pods/plant (44.02) were recorded in Carbendazim 50% WP at par with Fosetyl Al 80% WP a.i. 4800 (42.05)** as compared to control.
- ✓ Maximum number of seeds/plant (41.98) and yield (kg/ha) 1729.60 kg/ha were obtained in Carbendazim 50% WP **at par with** Fosetyl Al 80% WP a.i. 4800.
- ✓ Drenching with Carbendazim 50% WP (5.57%) at 30 and 45 days after sowing was the best treatment for the control of dry root rot of chickpea at par with drenching with Fosetyl Al 80% WP a.i. 4800 (6.15%) ,Fosetyl Al 80% WP

a.i. 2400, Fosetyl AI 80% WP a.i. 2000 and Fosetyl AI 80% WP a.i. 1600 (7.95%,10.69% and 12.35%, respectively). In control, where no treatment was done, the incidence of root rot was significantly maximum (32.27%).

Conclusion

- ✓ The highly prone area of dry root rot of chickpea was Kosmi followed by Nander, Ramgarha, Jamni, Bijori and Kaneriya. The least incidence of disease was noticed in Hirapur as compared to other locations survey. In block-wise seasonal incidence, Rehti was found to be infected with maximum per cent of disease incidence followed by the other blocks viz., Ichhawar and Budni.
- ✓ Seven fungicides were evaluated in laboratory for their toxicity against *R. bataticola* by poison food technique. Observations on radial growth of *R. bataticola* were recorded after seven days of inoculation. All fungicides completely inhibited the mycelial growth of *R. bataticola* except Fosetyl AI @ 0.20%.
- ✓ As an evaluation of different seed treatments for growth promoting ability (GPA) of chickpea, it could be concluded that higher growth attributes were found in seed treated with treatment *T. viride*@ 5g/kg which was at par with Carbendazim @ 2g/kg as compared to control at 40 and 60 DAS.
- ✓ The highest yield/ha (kg) was obtained in *T. viride* @ 5g/kg **with minimum dry root rot incidence at par with** Carbendazim @ 2g/kg as compared to control.
- ✓ As an evaluation of different seed treatments for growth promoting ability (GPA) of chickpea, it could be concluded that higher growth attributes were found in seed treated with treatment Carbendazim 50% WP at par with Fosetyl AI 80% WP a.i. 4800, Fosetyl AI 80% WP a.i. 2400, Fosetyl AI 80% WP a.i. 2000 and Fosetyl AI 80% WP a.i. 1600 as compared to control at 40 and 60 DAS.
- ✓ **The highest yield(kg/ha) was recorded in** Carbendazim 50% WP **with lowest dry root rot incidence at par with** Fosetyl AI 80% WP a.i. 4800 as compared to control.
- ✓ **Seed treatment is only effective till early stage of the chickpea crop, but due to appearance of DRR at later stage of the crop i.e. at podding stage, drenching can reduce the incidence of DRR significantly.**
- ❖ **Suggestions for further work**
 - ✓ Evaluation of combination of seed treatment and drenching needs more attention.
 - ✓ More combination of fungicides should be evaluated against *R. bataticola*.
 - ✓ The various aspects of drenching should be studied in detail.

REFERENCES

- Ahmed, E. M. and Mouselli, M. N. (1990). Wilt and root rot of chickpea in Southern Syria. *Arab J. Pl. Protect.*, **8**: 60-67.
- Ahmed, Q., Abu, M. A. O. and Mohammad, A. (1986). Losses in yield due to *Rhizoctonia* root rot of chickpea in Bihar. *Indian Phytopath.*, **39**: 590-592.
- Ammajamma, R. and Hegde, Y. R. (2009). Efficacy of fungicides against *Rhizoctonia bataticola* causing wilt of *Coleus forskohlii* (Wild) Briq. *Int. J. Plant Prot.*, **2**(1): 31-32.
- Anonymous (2015). Project coordinator's report 2014-15, AICRP on chickpea. ICAR-IJPR Kanpur.
- Ashby, S. C.(1927). *Macrophomina phaseolina* (Maub.) Comb. Nov: The pycnidial stage of *Rhizoctonia bataticola* (Taub.) Butler. *Transactions of the British Mycological Society*,**12**: 141-147.
- Aung, B., May, K. and Maung, T. (1992) . Second survey of chickpea disease in Mayanmar. *Int. chickpea Newsletter*, **27**: 80-81.
- Baki, Abdul A. and Anderson, J.D. (1973). Vigour determination of Soyabean seed by multiple criteria. *Crop Sci.* **13**, 630-633.
- Barnett, H. L. and Barry, B. Hunter (1972). Illustrated genera of imperfect fungi. Burgess publishing company, Minneseta.
- Barnett, H.L. and Hunter, B.B. (1987). Illustrated Genera of Imperfect Fungi. Mc. Millan Pub. Co., NY. pp. 218.
- Beniwal, S. P. S., Ahmed, S. and Goryu, D. (1992). Wilt/root rot disease of chickpea Ethiopia. *Tropical Pest Management*, **38**(1): 48-51.
- Briton-Jones, H. R. (1925). Mycological work in Egypt during the period 1920-1922. *Technology and Science Bulletin*, **49**: 129.
- Butler, E. J. (1918). *Fungi and Diseases in plants*. Thacker, Spink and Co., Calcutta.
- Butler, E. J. (1925). Fungi of India. ICAR, New Delhi.
- Chilkuri, A. and Giri, G. K. (2014). Detection and transmission of seed borne mycoflora in green gram and effect of different fungicides. *Int. J. Advanced Res.*, **2**(5): 1182-1186.
- De Candolle, A. P. (1815). Memories Surles Rhizoctons Medme museum d'histoire. **2**: 209-219.
- Deepa, Sunkad, G., Sharma, M., Mallesh, S. B. Mannur, D. M. and Sreenivas, A. G. (2018). Distribution and Severity of Dry Root Rot of Chickpea Caused by *Rhizoctonia bataticola* in Parts of North Karnataka, India. *Int.J.Curr.Microbiol.App.Sci.* **7**(04): 194-200.
- De-Mooy, C.J. and Burke, D.W.(1990). External infection mechanism of hypocotyls and cotyledons of cowpea seedlings by *Macrophomina phaseolina*. *Pl. Dis.***74**:720

- Deshkar, M.V., Khare M.N. and Joshi, L.K. (1973). Paper towel Method a rapid technique for testing susceptibility to *Rhizoctonia bataticola* (Taub.) But. *JNKVV Research Journal*. **7** : 300-301.
- Dhingra, O. D. and Sinclair, J. B. (1985). Basic Plant Pathology Methods. *CRC Press, Inc. Boca Raton, Florida*. pp. 132-163.
- Dhingra, O.D. and Sinclair, J.B. (1977). An annotated bibliography of *Macrophomina phaseolina*, 1905-1975. Brazil Univ. Stdade Federal de Vicoso University of Illinois Urbana, U.S.A., pp. 224.
- Du Hamel, H. L. (1728). Explication physique d'une maladie que fait petire plantes dans Le getinois et particul erment Le safram. *Hist. Acda. Roy. Sci. Paris*, pp. 100-112.
- El-Araby, M. E, Kurle, J. E. and Stetina, S. R. (2009). First report of charcoal rot (*Macrophomina phaseolina*) on Soyabean in Minnesata. *Pl. Dis.*, **87** (2) : 202.
- Entesari, M., Sharifzadeh, F., Ahmadzadeh, M. and Farhangfar, M. (2013). Seed biopriming with *Trichoderma* species and *Pseudomonas fluorescent* on growth parameters, enzymes activity and nutritional status of soybean. *Int. J. Agronomy and Pl. Production*. **4** (4): 610-619.
- Ghosh, R. Sharma, M., Telangre, R. and Pande, S. (2013). Occurrence and distribution of chickpea diseases in central and southern parts of India. *American J. Pl. Sci.*, **4**: 940-944.
- Goidanich, G. (1947). A revision of the genus *Macrophomina petrak* type species. *Annalidella Sperimentazione Agraria, Nuova Serie*, **1**(3): 449-461.
- Goyal, M. K. and Mehrotra, R. S. (1981). Chemical control of dry root-rot of gram caused by *R. bataticola*. *Acta-Botanica-India*.**9**: 228-232.
- Goyal, S. K. and Jain, M. P. (1998). Storage mycoflora of blackgram (*Vigna mungo* L. Hepper) and their pathogenicity. *Int. J. Trop. Pl. Dis.*,**16**(2): 195-202.
- Grover, R. K. and Sakhuja, P. K. (1981). Some pathological studies on *Rhizoctonia bataticola* leaf blight of mungbean. *Indian Phytopath.*, **34**: 24-29.
- Gupta, Om and Sharma ,Mamta (2015) Dry root rot of chickpea: An overview. *Journal of Food Legumes* **28** (4): 267-276
- Gupta, R. N., Gupta, J. S. and Sharma, S. L. (1983). Studies on wilt and root rot incidence of *Cicer arietinum* of Madhya Pradesh. *Indian Phytopath.*, **36** (1): 82-84.
- Gurha, S. N., Singh, G., Sharma, Y. R. (2003). Diseases of chickpea and their management. India: Ali M, Kumar S, Singh NB, editors. *Chickpea research in India*. Lucknow: Army Printing Press; pp. 195–227.
- Haigh,J.C.(1930). *Macrophomina phaseoli* (Mauubl.) Ashby, the pycnidial stage of *Rhizoctonia*

- bataticola* (Taub.) Butler. *Tropical Agriculturist*, **70**: 77-79.
- Halsted, B. D. (1890). Some fungus diseases of the sweet potato. *New Jersey Agricultural College Experimental Station Bulletin*, **76**: 7-14.
- Holliday, P. and Punithalingam, E. (1970). *Macrophomina phaseolina*. CMI Description of pathogenic fungi and bacteria. No. 275. Farnham Royal: Commonwealth Agricultural Bureaux.
- Hooda, I. and Grover, R. K. (1988). Effect of age, quantity of inoculum and isolates of *Macrophomina phaseolina* on the pathogenesis of mungbean and its control by chemicals. *Indian Phytopathol*, **41**:107–117.
- Ingle, Y.V., Zope, A.V., Wasule, D.L., Dandge M.S. and Patil P.V. (2018) Effect of seed dressing on germination and root rot of soybean. *International Journal of Chemical Studies* **6** (1): 653-655.
- Jagre, A. (2012). Succession of soil borne diseases of chickpea and their management through host plant resistance. M.Sc. (Ag.) Thesis. Deptt. of Plant Pathology, JNKVV, Jabalpur, pp. 79.
- Kadam, A. M. Chavan, S. S., Dhutraj, D. N. and Rewale, K. A. (2018). Survey of dry root rot of chickpea incidence in Marathwada region. *J. Pharmacognosy and Phytochemistry*, **1**: 3004-3008
- Kaiser, W. J., Danesh, D., Okhovat, M. and Mossahebi, H. (1968). Diseases of pulse crops (edible legumes) in Iran. *Pl. Dis Rep.* **52**:687–691.
- Katariya, L. V., Gaur, K., and Panwar, P. K. (2012). Cultural and pathogenic variability among isolates of *Rhizoctonia bataticola* causing dry root rot in chickpea. *J. Mycology and Pl. Path.*, **42**(4): 520-522.
- Kausal, R. P. (2008). Management of root rot in chickpea. *J. Food Legumes*, **21**(3): 178-181.
- Khalid, I. and Ilyas, M. B. (2000). Screening of chickpea germplasm against dry root rot disease (*Macrophomina phaseolina*) in pots/glass house. *Pakistan J. Phytopathol.*, **12**:66-70.
- Khan, R. A., Bhat, T. A. and Kumar, K. (2012). Management of chickpea (*Cicer arietinum* L.) dry root rot caused by *Rhizoctonia bataticola* (Taub.) Butler. *Int. J. Res. in Pharmaceutical and Biomedical Sci.*, **3** (4): 1539-1548.
- Khan, R. A., Towseef, A.B. and K. Krishna. (2012). Management of chickpea (*Cicer arietinum* L.) caused by *Rhizoctonia bataticola* (Taub.) Butler. *Int. J. Res. Pharma Biomedical Sci.* **3**(4): 207-211.

- Konde, S. A., Raut, B. T. and Gade, M. R. (2008). Chemical and biological management of root rot (*Rhizoctonia bataticola*) of soybean. *Ann. Plant Physiol.*, **22**(2): 275-277.
- Kumar , Vipul , Shahid , Mohammad, Srivastava , Mukesh, Singh, Anuradha, Pandey, Sonika and Sharma, Antima (2014) Enhancing Seed Germination and Vigor of Chickpea by Using Potential and Effective Strains of *Trichoderma* Species .*Virolgy and Mycology*.**3** (2):128.
- Kumar, S., Sharma, S., Pathak, D.V. and Beniwal, J. (2011). Integrated management of Jatropa root rot caused by *Rhizoctonia bataticola*. *J. Tropi. Fore. Sci.* **23** (1):35-11.
- Lakhran, L. (2017). Management of dry root rot of chickpea (*Cicer arietinum* L.) caused by *Macrophomina phaseolina* (Tassi.) Goid.(SKNAU)(Doctoral dissertation, SKNAU).
- Lakhran, L. and Ahir, R. R. (2018). *In-vivo* evaluation of different fungicides, plant extracts, bio-control agents and organics amendments for management of dry root rot of chickpea caused by *Macrophomina phaseolina*. *Leg. Res.*, pp.1-6.
- Lambhate, S. S., Choudhari, G. K., Mehetre, S. S., Zanjare, S. R. (2002). *In vitro* evaluations of chemicals against root rot of cotton caused by *Macrophomina phaseolina*. *J. Mah. Agric. Univ.*, **27**(1):202-203.
- Mahmood, Y., Khan, M.A., Javed, N. and Arif, M.J. (2015) Comparative efficacy of fungicides and biological control agents for the management of chickpea wilt caused by *Fusarium oxysporum* f. sp.ciceris *The Journal of Animal and Plant Sciences*, **25** (4):1063-1071
- Malviya, J. P. (2018). Studies on variability and management of *Rhizoctonia bataticola* (Taub.) Butler causing dry root rot of chickpea. M. Sc. (Thesis) Rajmata Vijayaraje Scindia Krishi Vishwavidhyalaya, Gwalior (M.P.).
- Manjunatha, S.V., Naik, M. K., Patil, M. B., Devika Rani, G. S. and Sudha, S. (2011). Prevalence of dry root rot of chickpea in north-eastern Karnataka. *Karnataka J. Agric. Sci.* **24**(3): 404–405.
- Marion H. Ebben (1973). The development of *Fusarium* wilt of carnation in soils with different soil inoculums levels and disease control with Carbendazim drenches .*Acta. Horticulture* pp: 115-121
- Maruti, S. A. S., Sunkad, G. and Amaresh, Y. S. (2017). *In Vitro* efficacy of fungicides and bioagents against dry root rot of pigeonpea caused by *Rhizoctonia bataticola* (Taub.) Butler, *Int. J.Pure App. Biosci.* **5**(6): 1341-1347.
- Mathieson, J. T. and Rush , C. M. (1991). Influence of temperature and five fungicides on *Rhizoctonia* root rot of harq red winter wheat. *Pl. Dis. Res.*, **75** (10): 983 - 986.

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- Mathur, S. B. and Nath, R. (1970). Seed borne fungi of mungbean from India and their significance. *Procter. Int. Seed Test Association*, **31**(1): 225-241.
- Maublanc, A. (1905). *Macrophoma phaseoli*. *Bul. Soc. Myc. France*, **21**: 90.
- Mitra, M. (1931). Report of the imperial mycologist. *Sci Rep of the Agric Res Inst.* 1929–1930:58-71.
- Mittal, R. K. (1997). Effect of sowing dates and disease development in lentil as sole and mixed crop with wheat. *J. Myco. Pl. Pathol.* **27**(2): 203 -209.
- Nagamma, G., Saifulla, M. and Narendrappa, T. (2012). *In vitro* Evaluation of fungicides bioagents and botanicals against *Macrophomina phaseolina* (Tassi) Goid the Causal agent of dry root rot in Chickpea. *Mysore J. Agric. Sci.*, **46**(4):778-783.
- Nagar, S. (2017). Studies on bio-control and plant growth promoting efficacy of *Trichoderma* strains in Chickpea (*Cicer arietinum* L.). M. Sc. (Thesis) Rajmata Vijayaraje Scindia Krishi Vishwavidhyalaya Gwalior (M.P.).
- [Nagarjuna](#) Reddy G. (2010) Integrated management of dry root rot of chickpea and molecular characterization of potential bio-control agents. Acharya N.G.Ranga agricultural University, Hyderabad.
- Narsimhan, R. (1929). A preliminary note on a Fusarium Parasitic on Bengal gram (*Cicer arietinum*) Madras Agricultural Department year Book , pp. 5-11
- Nene, Y. L. and Thapliyal, P. N. (1979). *Fungicides in Plant Disease Control*, Oxford and IBH Publishing House, New Delhi, pp. 163.
- Nene, Y. L., Shelia, V. K. and Sharma, S. B. (1996). A world list of chickpea and pigeonpea pathogens. 5thed. Hyderabad: International Crops Research Institute for Semi-Arid Tropics; pp. 27.
- Padwick, G. W.(1948). Plant protection and food crops of India, plant pest and disease of rice, wheat, sorghum and gram. *Imperial J. Exp. Agri.* **16**:55-64.
- Pandey, G. and Singh, R. B. (1990). Survey of root disease of chickpea in Allhabad region. *Current Nematology*,**1**: 77-78.
- Pandey, R. N., Gohel, N. M. and Jaisani, P. (2016). Management of wilt and root rot of chickpea caused by *Fusarium oxysporum* f. sp. *ciceri* and *Macrophomina phaseolina* through seed biopriming and soil application of bio-agents. *Int. J. Curr. Microbiol. App. Sci.*, **6**(5): 2516-2522.
- Pardeshi, V. F., Reddy,V.G. and Nalwandikar, P. K. (1989). Effects of different fungicides on

- seedling vigour and seed viability in soybean. *J. Maharashtra Agric. Univ.*, **14**(1): 33-36.
- Patel, D. P. Singh, H. B. Shroff, S. and Sahu, J. (2012). Studies on growth promotion activities of *Trichoderma* strains *harzianum* on chickpea. *Adv. Pl. Sci.* **25** (1): 193-195.
- Patel, K. K. and Patel, A. J. (1990). Meterological correlation of charcoal rot of sesamum. *Indian J. Mycol. Pl. Pathol.*, **20**(1): 64-65.
- Pawar, A. D., Ingole, M. N. and Kendre, V. P. (2018). Efficacy of fungicides, bioagents and organic amendment to manage root rot in green gram. *Int. J. Chemical Studies*, **6**(1): 1355-1359
- Peshney, N. L., Gades, R. M. and Thakur, K. G. (1992). Sensitivity and adaptability of *R. bataticola* to different fungicides. *J. Soils and Crops.*, **2**: 35-38.
- Prajapati, R. K., Gangwar, R. K., Srivastava, S.S.L. and Shahid Ahamad. (2002). Efficacy of fungicides, non-target pesticides and bio- agents against the dry root rot of chickpea. *Ann. Pl. Prot.* **10** (1): 154-155.
- Rangaswami, G. and Mahadevan, A. (2008). Diseases of crop plants in India (4th ed). New Delhi, *PHI Learning Private Limited*, pp. 275 – 278.
- Ravichandran, S. and Hegde, Y. R. (2017). Management of dry root rot of chickpea caused by *Rhizoctonia bataticola* through fungicides. *Int. J. Curr. Microbiol. App. Sci.* **6**(7): 1594-1600.
- Reddy, M. V., Baw, V. A., Hein, V. M. and Sethi, S. C. (1991). Survey of chickpea diseases in Myanmar. *Int. Chickpea Newsletter.* **24**: 46-47.
- Reddy, M. V., Raju, R. N. and Pundar, R. P. S. (1990). Additional sources of resistance to wilt and root rot in chickpea. *Int. Chickpea Newsletter*, **22**: 36-38.
- Riker, A.J. and Riker, R.S. (1936). Introduction to research on plant disease. p.1176. John wift Co. Inc. St. Loris.
- Salunke, V. N., Armarkar, S. and Ingle, R. W. (2008). Efficacy of fungicides and antagonistic effect of bioagent against *Rhizoctonia bataticola* isolates. *Ann. Plant Physiol.*, **22** (1):134-137.
- Sarwar, H. A. K. and Raju, D. G. (1985). Topsin M-70 the most effective fungicide for the control of *R. bataticola*, the root and stem rot disease causing pathogen in castor. *Pesticides*, **19**: 56-57.
- Sharma, M. and Pande, S. (2013). Unraveling effects of temperature and soil moisture stress response on development of dry root rot [*Rhizoctonia bataticola* (Taub.)] butler in

- chickpea. *American J Pl. Sci.* **4**:584–589.
- Sharma, M., Ghosh, R. and Pande, S.(2015). Dry root rot (*Rhizoctonia bataticola* (Taub.) Butler): an emerging disease of chickpea—where do we stand? *Arch. Phytopath. Pl. Prot.*, **48**: 1–16.
- Sharma, M., Ghosh, R., Krishnan, R. R., Nagamangala, U. N., Chamarthi, S., Varshney, R. and Pande, S. (2012). Molecular and morphological diversity in *Rhizoctonia bataticola* isolates causing dry root rot of chickpea (*Cicer arietinum* L.) in India. *African J. Biotech.*, **11** (37): 8948-8959.
- Sharma, O. P. and Kumari, M. (2017). Management of dry root rot disease (*Rhizoctonia bataticola*) of chickpea through fungicides. *Int. J. Chemical Studies*, **5**(4):45-47
- Shukla, D. N. and Bhargava, S. N. (1976). Some pathogenic fungi from pulse and oil seed crops. *Proc. Nat. Acad. Sci. India*, **46** (2): 530-531.
- Singh, A. S. and Agarwal, S. N. (2002) .Prevalence of chickpea wilt and dry root rot diseases in Vindhyan plateau zone of Madhya Pradesh. *Indian J. Pulses Res.*, **15**(2):14-19.
- Singh, J. and Kumar, K. (2002). Location, survival, transmission and control of seed borne, *Macrophomina phaseolina* causing dry root rot and leaf blight in urdbean. *Ann. Pl. Prot. Sci.*, **10**(1): 111-113.
- Singh, R. and Sindhan, G. S. (1998). Effect of fungicides on the incidence of dry root rot and biochemical status of chickpea plants. *Pl. Dis. Res.*, **13** (1) : 14-17.
- Singh, S. D. (1999). Soil borne diseases of chickpea. File under plant disease B 10 (field crops). Issued March-1999. 4782000399 M.S.S.D. Singh @ Cgiar. [org/http://grep.icrisat.cgiar.org](http://grep.icrisat.cgiar.org).
- Singh, S. K. and Srivastava, H. P. (1988). Symptoms of *M. phaseolina* infection on mothbean seedlings. *Ann. Arid Zone*, **27**: 151-152.
- Small, W. A. (1924). *Rhizoctonia* causing root diseases in Uganda. *Brit. Mycol. Soc. Trans.* **9**: 152-156.
- Srinivas, P.(2016).Studies on dry root rot (*Rhizoctonia bataticola* (taub.) butler) of chickpea (*Cicer arietinum*). Ph.D.,Thesis, Professor Jayashankar Telangana State Agricultural University, India.
- Srivastava, A., Singh, S. N. and Agarwal, S. C. (2002). Prevalence of chickpea wilt and dry root disease in Vindhyan plateau zone of Madhya Pradesh. *Indian J. Pulses Res.*, **15**:192.
- Suresh M., Kumar C. V. Sameer and Sudhakar R. (2011) Efficacy of Native Isolates of *Trichoderma* sp. against Chickpea Wilt in Southern Telangana Zone of Andhra Pradesh

- Tancic, Sonja, Skrobonja, Jelica, Lalosevic, Mirjana, Jevtic, Radivoje and Vidic, Milos (2013) Impact of *Trichoderma* spp. on Soybean Seed Germination and Potential Antagonistic Effect on *Sclerotinia sclerotiorum* estic. *Phytomed.* (Belgrade), **28** (3), 181–185
- Taubenhaus, J. J. (1913). The black rots of the sweet potato. *Phytopathology*, **3**: 159-165.
- Taya, R. S., Tripathi, M. H. and Pawan, M. S. (1990). Influence of soil type, soil moisture and fertilizers on the severity of chickpea dry root rot caused by *Rhizoctonia bataticola* (Taub) Butler. *Indian J. Pl. Path.*, **18**: 133-136.
- Thawale, S., Kakade, S. U. and Joshi, S. (2010). Effect of seed soaking treatments on quality parameters of soybean. *Int. J. Agril. Sci.*, **6**(1): 35-38.
- Tripathi, N. N. and Sharma, B. K. (1983). Incidence of chickpea dry root rot (*Rhizoctonia bataticola*) in Southern Haryana. *Int. Chickpea Newslet.*, **8**: 22-23.
- Udit Narain (1986). Some new diseases of sunflower from India. *Indian Phytopath.*, **36**: 196.
- Verma , R. K. (1976). Chemotherapeutic activity of five systemic fungicides in gram seedling against three soil borne pathogens. M. Sc. (Ag). Thesis J. N. Agri. University, Jabalpur pp. 78.
- Vijay-Mohan, S. M., Prasad, M., Barnwal, M. K., Kudada, N. (2006). Fungicidal management of dry root rot disease and yield of chickpea. *J. Appl. Biol.*, **16**:42–44.
- Westerlund, F. V., Jr-Cambell, R. N. and Kimble, K. A. (1974). Fungal root rots and wilt of chickpea in California. *Phytopathology*, 664:432–436.
- Wyllie, T. D. (1993). Charcoal Rot. In: Compendium of soybean diseases, Sinclair, J. B. And P.A. Backman (Eds.) 3rd Edn., APS Press, ST. Paul, M.N., pp: 30-33.

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Plate 1: Dry root rot infected root of chickpea plant, exhibiting blackening and lack of secondary roots

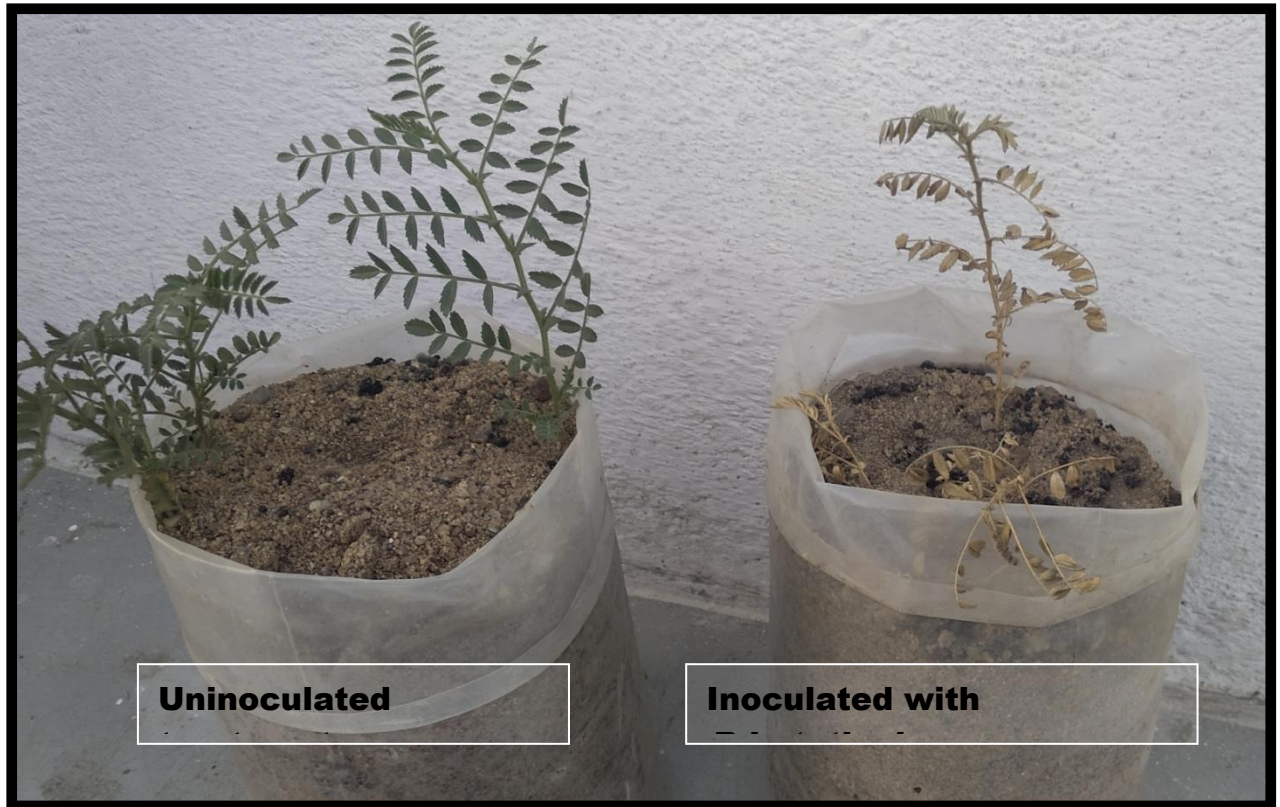


Plate 2: Pathogenicity test on susceptible chickpea variety L 550

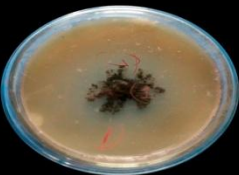
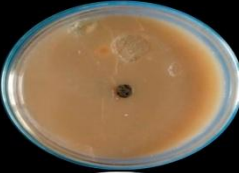
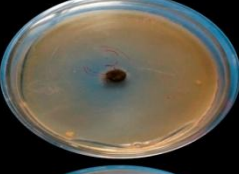
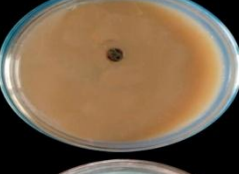
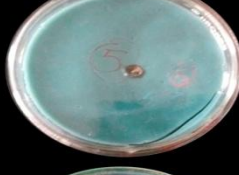
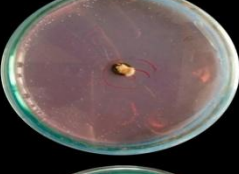
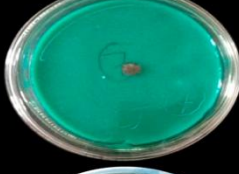
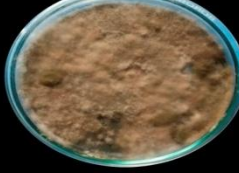
Fungicides	Radial growth (mm)
Fosetyl AI @ 0.20%	
Fosetyl AI @ 0.25%	
Fosetyl AI @ 0.30%	
Carbendazim	
Mancozeb+carbendazim	
Thiram+carboxin	
Thiophanate methyl + Pyraclostrobin	
Control	

Plate 3 Mycelial radial growth of *Rhizoctonia bataticola* in fungicides amended medium